Derivations for *Mostly Harmless Econometrics* *(2008)*

Derivations assisted by LLMs*

Contents

*https://www.mostlyharmlesseconometrics.com/

1 Introduction

This document presents detailed derivations based on *Mostly Harmless Econometrics* (2008). Each section corresponds to a chapter of the book, and each subsection is organized around a specific derivation. Since Chapter 1 contains no formal derivations, the document begins with Chapter 2.

1.1 Conceptual Map Across Chapters (Figure 1)

Figure 1 summarizes the conceptual relationship among the main chapters.

2 Chapter 2 Derivations: The Experimental Ideal

2.1 Potential Outcomes Setup

Let $d_i \in \{0, 1\}$ denote treatment status. In the hospitalization example,

$$d_i = 1$$

means individual i goes to the hospital, and

$$d_i = 0$$

means individual i does not go to the hospital.

There are two potential outcomes:

$$y_{1i} = \text{health outcome if individual } i \text{ receives treatment,}$$

and

$$y_{0i} = \text{health outcome if individual } i \text{ does not receive treatment.}$$

The individual-level causal effect is

$$\boxed{y_{1i} - y_{0i}.}$$

The fundamental problem of causal inference is that for each individual, we observe only one of these two potential outcomes.

2.2 Observed Outcome Equation

The observed outcome is

$$y_i = \begin{cases} y_{1i}, & \text{if } d_i = 1, \\ y_{0i}, & \text{if } d_i = 0. \end{cases}$$

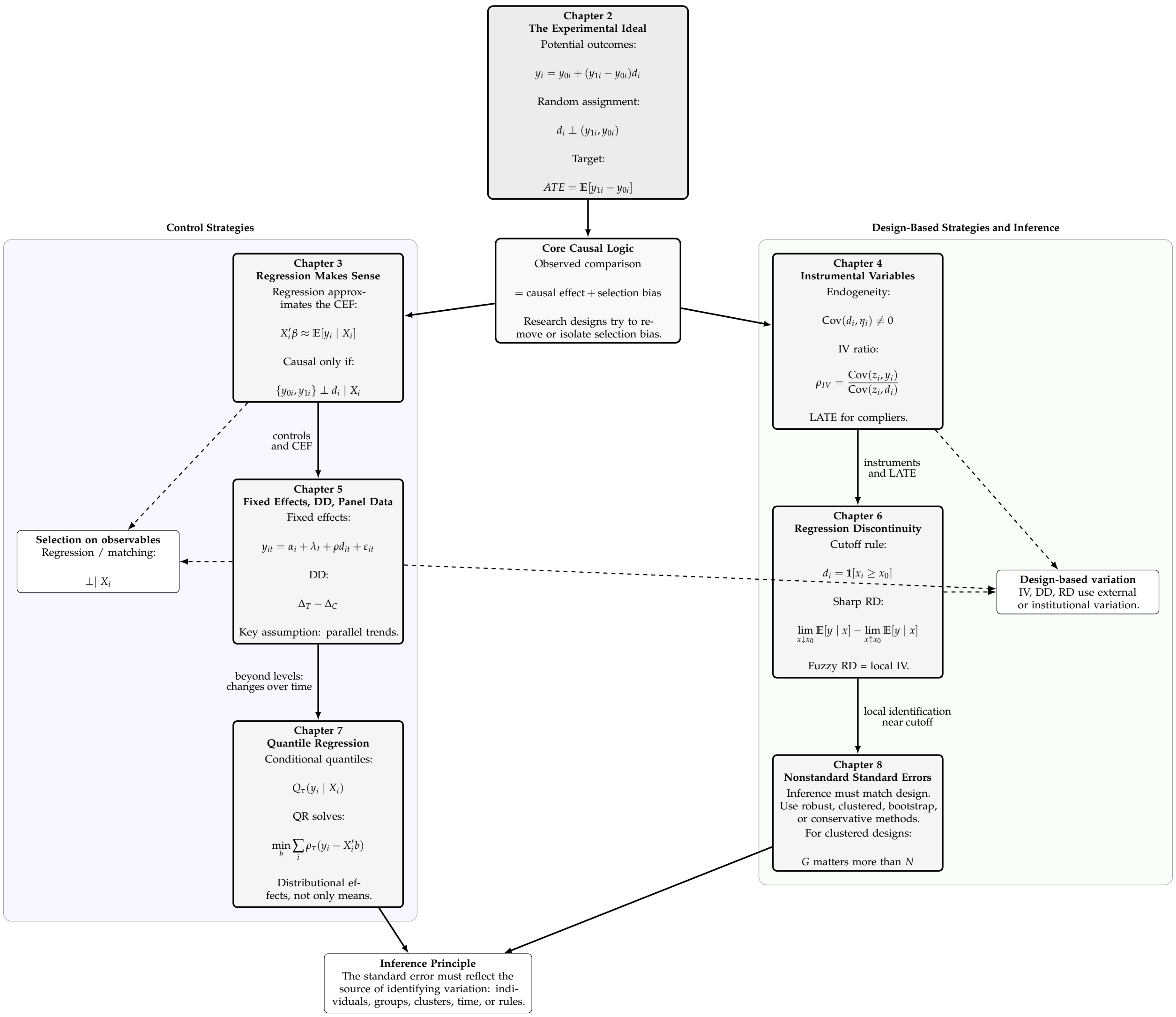

Figure 1. Conceptual map of Chapters 2–8. The sequence starts from the experimental ideal, then moves to regression as CEF approximation, instrumental variables, panel and difference-in-differences designs, regression discontinuity, quantile regression, and nonstandard inference.

This can be written compactly as

$$\boxed{y_i = y_{0i} + (y_{1i} - y_{0i})d_i.}$$

To verify this, consider two cases.

If $d_i = 1$, then

$$y_i = y_{0i} + (y_{1i} - y_{0i}) = y_{1i}.$$

If $d_i = 0$, then

$$y_i = y_{0i} + (y_{1i} - y_{0i})0 = y_{0i}.$$

Therefore, the compact expression reproduces the observed outcome in both treatment states.

2.3 *Average Treatment Effect*

The average treatment effect is

$$\boxed{ATE = \mathbb{E}[y_{1i} - y_{0i}].}$$

Using linearity of expectation,

$$ATE = \mathbb{E}[y_{1i}] - \mathbb{E}[y_{0i}].$$

This is the average causal effect in the full population. It asks:

> What would be the average effect if everyone were treated instead of untreated?

2.4 *Average Treatment Effect on the Treated*

The average treatment effect on the treated is

$$\boxed{ATT = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].}$$

Using linearity of conditional expectation,

$$ATT = \mathbb{E}[y_{1i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 1].$$

This asks:

What was the average effect for those who actually received treatment?

The first term,

$$\mathbb{E}[y_{1i} \mid d_i = 1],$$

is observed because treated individuals reveal y_{1i}. The second term,

$$\mathbb{E}[y_{0i} \mid d_i = 1],$$

is counterfactual because we do not observe what treated individuals would have experienced without treatment.

2.5 Naive Difference in Means

The naive comparison between treated and untreated individuals is

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].$$

Since treated individuals reveal y_{1i},

$$\mathbb{E}[y_i \mid d_i = 1] = \mathbb{E}[y_{1i} \mid d_i = 1].$$

Since untreated individuals reveal y_{0i},

$$\mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{0i} \mid d_i = 0].$$

Therefore,

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{1i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].}$$

This is observable, but it is generally not equal to the causal effect.

2.6 Selection Bias Decomposition

Start from the naive difference:

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].$$

Using potential outcomes,

$$= \mathbb{E}[y_{1i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].$$

Add and subtract $\mathbb{E}[y_{0i} \mid d_i = 1]$:

$$= \mathbb{E}[y_{1i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 1] + \mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].$$

Group the terms:

$$= [\mathbb{E}[y_{1i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 1]] + [\mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0]] .$$

Thus,

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = ATT + \text{selection bias.}}$$

where

$$\boxed{ATT = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1]}$$

and

$$\boxed{\text{selection bias} = \mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].}$$

2.7 *Interpretation of Selection Bias*

Selection bias is

$$\mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].$$

It compares untreated potential outcomes for those who were treated and those who were not treated.

In the hospitalization example, people who go to the hospital are likely to be sicker even without hospitalization. If worse health corresponds to a larger numerical health score, then

$$\mathbb{E}[y_{0i} \mid d_i = 1] > \mathbb{E}[y_{0i} \mid d_i = 0].$$

Therefore, the naive comparison may exaggerate the harmful appearance of hospitalization.

More generally:

$$\boxed{\text{selection bias arises when treated and untreated groups differ in their untreated potential outcomes.}}$$

2.8 *Random Assignment*

Random assignment means that treatment status is independent of potential outcomes:

$$\boxed{d_i \perp (y_{1i}, y_{0i}).}$$

This implies

$$\mathbb{E}[y_{1i} \mid d_i = 1] = \mathbb{E}[y_{1i} \mid d_i = 0] = \mathbb{E}[y_{1i}],$$

and

$$\mathbb{E}[y_{0i} \mid d_i = 1] = \mathbb{E}[y_{0i} \mid d_i = 0] = \mathbb{E}[y_{0i}].$$

Therefore, random assignment makes treated and untreated groups comparable in terms of potential outcomes.

2.9 Random Assignment Eliminates Selection Bias

Selection bias is

$$\mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].$$

Under random assignment,

$$\mathbb{E}[y_{0i} \mid d_i = 1] = \mathbb{E}[y_{0i} \mid d_i = 0].$$

Therefore,

$$\boxed{\text{selection bias} = 0.}$$

Thus,

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = ATT.$$

2.10 Random Assignment Identifies the Average Treatment Effect

Under random assignment,

$$d_i \perp (y_{1i}, y_{0i}).$$

Start from the observed difference:

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].$$

Using potential outcomes,

$$= \mathbb{E}[y_{1i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].$$

By random assignment,

$$\mathbb{E}[y_{1i} \mid d_i = 1] = \mathbb{E}[y_{1i}],$$

and

$$\mathbb{E}[y_{0i} \mid d_i = 0] = \mathbb{E}[y_{0i}].$$

Therefore,

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{1i}] - \mathbb{E}[y_{0i}].$$

Hence,

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{1i} - y_{0i}] = ATE.}$$

2.11 Random Assignment Makes ATT Equal to ATE

The treatment effect on the treated is

$$ATT = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].$$

Under random assignment,

$$d_i \perp (y_{1i}, y_{0i}).$$

Therefore,

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1] = \mathbb{E}[y_{1i} - y_{0i}].$$

Thus,

$$\boxed{ATT = ATE.}$$

In a randomized experiment, the treated group is statistically representative of the population with respect to potential outcomes.

2.12 The Experimental Estimand

In a randomized experiment, the population experimental estimand is

$$\boxed{\delta = \mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].}$$

Under random assignment,

$$\delta = ATE.$$

The corresponding sample estimator is

$$\boxed{\widehat{\delta} = \bar{y}_1 - \bar{y}_0,}$$

where

$$\bar{y}_1 = \frac{1}{N_1} \sum_{i:d_i=1} y_i,$$

and

$$\bar{y}_0 = \frac{1}{N_0} \sum_{i:d_i=0} y_i.$$

2.13 *Sampling Variance of the Difference in Means*

Let

$$N_1 = \sum_i d_i$$

be the number of treated observations, and

$$N_0 = \sum_i (1 - d_i)$$

be the number of untreated observations.

The difference-in-means estimator is

$$\widehat{\delta} = \bar{y}_1 - \bar{y}_0.$$

Assuming independent observations,

$$\mathrm{Var}(\widehat{\delta}) = \mathrm{Var}(\bar{y}_1) + \mathrm{Var}(\bar{y}_0).$$

If

$$\mathrm{Var}(y_i \mid d_i = 1) = \sigma_1^2$$

and

$$\mathrm{Var}(y_i \mid d_i = 0) = \sigma_0^2,$$

then

$$\mathrm{Var}(\bar{y}_1) = \frac{\sigma_1^2}{N_1},$$

and

$$\mathrm{Var}(\bar{y}_0) = \frac{\sigma_0^2}{N_0}.$$

Thus,

$$\boxed{\mathrm{Var}(\widehat{\delta}) = \frac{\sigma_1^2}{N_1} + \frac{\sigma_0^2}{N_0}.}$$

A sample analog is

$$\boxed{\widehat{\mathrm{Var}}(\widehat{\delta}) = \frac{s_1^2}{N_1} + \frac{s_0^2}{N_0},}$$

where s_1^2 and s_0^2 are sample variances within treatment and control groups.

2.14 Regression with a Binary Treatment

Consider the regression

$$\boxed{y_i = \alpha + \rho d_i + \eta_i.}$$

The OLS slope coefficient on a binary variable is the difference in conditional means:

$$\boxed{\rho = \mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].}$$

Derivation

The conditional expectation when $d_i = 0$ is

$$\mathbb{E}[y_i \mid d_i = 0] = \alpha + \rho \cdot 0 + \mathbb{E}[\eta_i \mid d_i = 0].$$

In a population regression with an intercept and a binary regressor, the fitted conditional mean for the control group is α, so

$$\alpha = \mathbb{E}[y_i \mid d_i = 0].$$

Similarly, for $d_i = 1$,

$$\mathbb{E}[y_i \mid d_i = 1] = \alpha + \rho.$$

Subtract:

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = (\alpha + \rho) - \alpha = \rho.$$

Therefore,

$$\boxed{\rho = \mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].}$$

2.15 Constant Treatment Effects Regression Model

Suppose the treatment effect is the same for every individual:

$$y_{1i} - y_{0i} = \rho.$$

Then

$$y_{1i} = y_{0i} + \rho.$$

Using the observed outcome equation,

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i,$$

we get

$$y_i = y_{0i} + \rho d_i.$$

Define

$$\alpha = \mathbb{E}[y_{0i}],$$

and

$$\eta_i = y_{0i} - \mathbb{E}[y_{0i}].$$

Then

$$y_{0i} = \alpha + \eta_i.$$

Substitute:

$$y_i = \alpha + \rho d_i + \eta_i.$$

Therefore,

$$\boxed{y_i = \alpha + \rho d_i + \eta_i.}$$

This is the regression representation of the constant-effects causal model.

2.16 Selection Bias as Error-Regressor Correlation

In the constant-effects model,

$$y_i = \alpha + \rho d_i + \eta_i,$$

where

$$\eta_i = y_{0i} - \mathbb{E}[y_{0i}].$$

Taking conditional expectations:

$$\mathbb{E}[y_i \mid d_i = 1] = \alpha + \rho + \mathbb{E}[\eta_i \mid d_i = 1],$$

and

$$\mathbb{E}[y_i \mid d_i = 0] = \alpha + \mathbb{E}[\eta_i \mid d_i = 0].$$

Subtract:

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \rho + \mathbb{E}[\eta_i \mid d_i = 1] - \mathbb{E}[\eta_i \mid d_i = 0].$$

Since

$$\eta_i = y_{0i} - \mathbb{E}[y_{0i}],$$

we have

$$\mathbb{E}[\eta_i \mid d_i = 1] - \mathbb{E}[\eta_i \mid d_i = 0] = \mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0].$$

Therefore,

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \rho + \text{selection bias.}}$$

Thus, in regression language,

$$\boxed{\text{selection bias is correlation between } d_i \text{ and } \eta_i.}$$

2.17 *Random Assignment and Regression Identification*

Under random assignment,

$$d_i \perp y_{0i}.$$

Since

$$\eta_i = y_{0i} - \mathbb{E}[y_{0i}],$$

it follows that

$$d_i \perp \eta_i.$$

Therefore,

$$\mathbb{E}[\eta_i \mid d_i = 1] = \mathbb{E}[\eta_i \mid d_i = 0] = 0.$$

The observed difference becomes

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \rho.$$

Thus, in a randomized experiment,

$$\boxed{\text{the regression coefficient on treatment identifies the causal effect.}}$$

2.18 *Multiple Treatment Groups in the STAR Experiment*

The STAR experiment assigned students to three class types:

$$\text{small class,}$$

$$\text{regular class,}$$

and

$$\text{regular class with aide.}$$

Let regular class be the omitted reference category. Define

$$S_i = \mathbf{1}[\text{small class}],$$

and

$$A_i = \mathbf{1}[\text{regular class with aide}].$$

A regression for test scores is

$$\boxed{y_i = \alpha + \rho_S S_i + \rho_A A_i + \eta_i.}$$

The conditional means are:

$$\mathbb{E}[y_i \mid S_i = 0, A_i = 0] = \alpha,$$

$$\mathbb{E}[y_i \mid S_i = 1, A_i = 0] = \alpha + \rho_S,$$

and

$$\mathbb{E}[y_i \mid S_i = 0, A_i = 1] = \alpha + \rho_A.$$

Therefore,

$$\boxed{\rho_S = \mathbb{E}[y_i \mid \text{small class}] - \mathbb{E}[y_i \mid \text{regular class}],}$$

and

$$\boxed{\rho_A = \mathbb{E}[y_i \mid \text{regular/aide class}] - \mathbb{E}[y_i \mid \text{regular class}].}$$

2.19 Regression with Covariates

A regression with covariates is

$$\boxed{y_i = \alpha + \rho d_i + X_i'\gamma + \eta_i.}$$

Here X_i includes predetermined student characteristics such as race, age, or free-lunch status.

If treatment is randomly assigned, then

$$d_i \perp X_i.$$

Therefore, adding X_i is not necessary to remove selection bias. However, adding X_i can improve precision if X_i explains outcome variation.

2.20 Why Randomly Balanced Covariates Do Not Change the Treatment Coefficient Much

Suppose the short regression is

$$y_i = \alpha_s + \rho_s d_i + u_i,$$

and the long regression is

$$y_i = \alpha + \rho d_i + X_i'\gamma + \eta_i.$$

The omitted variables bias formula gives

$$\rho_s = \rho + \gamma'\delta_{Xd},$$

where δ_{Xd} is the vector of coefficients from regressing X_i on d_i.

Under random assignment,

$$d_i \perp X_i,$$

so

$$\delta_{Xd} = 0.$$

Therefore,

$$\boxed{\rho_s = \rho.}$$

Thus, adding balanced covariates should not systematically change the treatment-effect estimate.

2.21 Why Covariates Can Improve Precision

Consider the long regression

$$y_i = \alpha + \rho d_i + X_i'\gamma + \eta_i.$$

Suppose X_i is uncorrelated with d_i, but X_i strongly predicts y_i. Then including X_i reduces residual variance:

$$\text{Var}(\eta_i) < \text{Var}(u_i),$$

where u_i is the residual from the short regression without X_i.

The approximate variance of the treatment coefficient is proportional to residual variance:

$$\text{Var}(\widehat{\rho}) \propto \sigma_\eta^2.$$

Therefore,

$$\boxed{\text{balanced covariates can reduce standard errors even if they do not change the point estimate.}}$$

2.22 *Conditional Random Assignment*

In the STAR experiment, random assignment was conducted within schools. This means treatment is randomly assigned conditional on school:

$$d_i \perp (y_{1i}, y_{0i}) \mid \text{school}_i.$$

Let G_i be a vector of school indicators. A regression that respects the design is

$$\boxed{y_i = \alpha + \rho d_i + G_i'\lambda + \eta_i.}$$

The school fixed effects $G_i'\lambda$ control for permanent differences across schools.

Under within-school random assignment,

$$\mathbb{E}[\eta_i \mid d_i, G_i] = \mathbb{E}[\eta_i \mid G_i].$$

Therefore, after controlling for school fixed effects,

$$\boxed{\rho \text{ identifies the within-school randomized treatment effect.}}$$

2.23 *Fixed Effects as Blocked Experimental Controls*

Suppose there are schools indexed by g. Within each school, treatment is randomly assigned. Let

$$\bar{y}_{1g}$$

be the mean outcome for treated students in school g, and let

$$\bar{y}_{0g}$$

be the mean outcome for control students in school g.

The school-specific treatment-control difference is

$$\widehat{\rho}_g = \bar{y}_{1g} - \bar{y}_{0g}.$$

A fixed-effects regression combines these school-specific differences using regression weights:

$$\boxed{\widehat{\rho}_{FE} = \sum_g \omega_g \widehat{\rho}_g,}$$

where the weights satisfy

$$\omega_g \geq 0, \qquad \sum_g \omega_g = 1.$$

The weights are larger for schools with more within-school treatment variation and larger sample sizes.

Thus, school fixed effects estimate a weighted average of within-school experimental contrasts.

2.24 *Testing Randomization Balance*

A common diagnostic in randomized experiments is to check whether predetermined characteristics are balanced across treatment groups.

Let X_i be a predetermined characteristic. For a binary treatment, one can estimate

$$\boxed{X_i = \alpha + \pi d_i + v_i.}$$

The null hypothesis of balance is

$$\boxed{H_0 : \pi = 0.}$$

For multiple treatment groups, as in STAR, one can estimate

$$X_i = \alpha + \pi_S S_i + \pi_A A_i + v_i,$$

and test

$$\boxed{H_0 : \pi_S = \pi_A = 0.}$$

Failing to reject this null supports the claim that randomization produced comparable treatment groups.

2.25 *Joint Balance Test*

Suppose there are K predetermined covariates:

$$X_{1i}, X_{2i}, \ldots, X_{Ki}.$$

One way to test balance is to test equality of covariate means across treatment groups. For a single covariate X_{ki}, the null is

$$\mathbb{E}[X_{ki} \mid d_i = 1] = \mathbb{E}[X_{ki} \mid d_i = 0].$$

For all covariates jointly, the null is

$$\boxed{\mathbb{E}[X_i \mid d_i = 1] = \mathbb{E}[X_i \mid d_i = 0].}$$

In regression form, for binary treatment:

$$d_i = \alpha + X_i'\pi + v_i.$$

The joint balance null is

$$\boxed{H_0 : \pi = 0.}$$

If treatment assignment is genuinely random, predetermined characteristics should not predict treatment status.

2.26 Attrition as a Threat to Experimental Validity

Let $R_i \in \{0, 1\}$ indicate whether outcome y_i is observed:

$$R_i = 1 \quad \text{if } y_i \text{ is observed,}$$

and

$$R_i = 0 \quad \text{if } y_i \text{ is missing.}$$

Even if treatment is initially randomized,

$$d_i \perp (y_{1i}, y_{0i}),$$

the analysis sample may be selected if

$$R_i$$

depends on treatment and potential outcomes.

The observed experimental contrast is

$$\mathbb{E}[y_i \mid d_i = 1, R_i = 1] - \mathbb{E}[y_i \mid d_i = 0, R_i = 1].$$

This identifies the causal effect only if

$$\boxed{R_i \perp (y_{1i}, y_{0i}) \mid d_i}$$

or if attrition is unrelated to potential outcomes within treatment groups.

If attrition differs across treatment arms and is related to outcomes, then

$$\boxed{\text{Random assignment in the original sample does not guarantee random assignment in the observed sample.}}$$

2.27 *Treatment-Control Difference with Heterogeneous Effects*

Suppose treatment effects are heterogeneous:

$$\tau_i = y_{1i} - y_{0i}.$$

The observed outcome is

$$y_i = y_{0i} + \tau_i d_i.$$

Under random assignment,

$$d_i \perp (y_{0i}, \tau_i).$$

The difference in means is

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].$$

Using the observed outcome equation,

$$\mathbb{E}[y_i \mid d_i = 1] = \mathbb{E}[y_{0i} + \tau_i \mid d_i = 1],$$

and

$$\mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{0i} \mid d_i = 0].$$

By random assignment,

$$\mathbb{E}[y_{0i} \mid d_i = 1] = \mathbb{E}[y_{0i} \mid d_i = 0],$$

and

$$\mathbb{E}[\tau_i \mid d_i = 1] = \mathbb{E}[\tau_i].$$

Therefore,

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[\tau_i] = ATE.}$$

Thus, random assignment identifies the average treatment effect even when treatment effects are heterogeneous.

2.28 Treatment-Control Difference with Unequal Assignment Probabilities

Suppose treatment is randomly assigned, but not necessarily with probability one-half:

$$\mathbb{P}(d_i = 1) = p, \qquad 0 < p < 1.$$

Random assignment means

$$d_i \perp (y_{1i}, y_{0i}).$$

Therefore,

$$\mathbb{E}[y_i \mid d_i = 1] = \mathbb{E}[y_{1i}],$$

and

$$\mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{0i}].$$

Thus,

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = ATE}$$

regardless of the value of p, as long as both treatment and control groups have positive probability:

$$0 < p < 1.$$

2.29 Why the Experimental Ideal Matters for Observational Research

In an ideal randomized experiment,

$$d_i \perp (y_{1i}, y_{0i}).$$

This makes

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0]$$

a causal comparison.

In observational research, treatment is usually not randomly assigned:

$$d_i \not\perp (y_{1i}, y_{0i}).$$

Therefore, the naive comparison generally equals

$$ATT + \text{selection bias}.$$

The experimental ideal gives the benchmark:

$$\boxed{\text{A credible research design should make treated and untreated units comparable in potential outcomes.}}$$

Regression, matching, instrumental variables, difference-in-differences, and regression discontinuity can all be understood as attempts to approximate this experimental ideal.

2.30 *Summary of Chapter 2*

Chapter 2 introduces the experimental ideal as the foundation of causal inference. The key objects are potential outcomes:

$$y_{1i} \quad \text{and} \quad y_{0i}.$$

The observed outcome is

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i.$$

The naive difference in means can be decomposed as

$$\boxed{\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = ATT + [\mathbb{E}[y_{0i} \mid d_i = 1] - \mathbb{E}[y_{0i} \mid d_i = 0]]\,.}$$

The second term is selection bias.

Random assignment gives

$$d_i \perp (y_{1i}, y_{0i}),$$

which eliminates selection bias and identifies

$$\boxed{ATE = \mathbb{E}[y_{1i} - y_{0i}].}$$

Regression analysis of experiments is a convenient way to estimate treatment-control differences:

$$y_i = \alpha + \rho d_i + \eta_i.$$

When treatment is randomly assigned,

$$\boxed{\rho \text{ has a causal interpretation.}}$$

Covariates and fixed effects are not required for unbiasedness when randomization is valid, but they can improve precision and account for conditional random assignment.

3 Chapter 3 Derivations: Making Regression Make Sense

3.1 Conditional Expectation Function, CEF

Let y_i be an outcome and let X_i be a vector of covariates. The conditional expectation function is defined as

$$\mathbb{E}[y_i \mid X_i].$$

For a particular value $X_i = x$, if y_i is continuously distributed with conditional density $f_y(t \mid X_i = x)$, then

$$\mathbb{E}[y_i \mid X_i = x] = \int t f_y(t \mid X_i = x)\,\mathrm{d}t.$$

If y_i is discrete, then

$$\mathbb{E}[y_i \mid X_i = x] = \sum_t t f_y(t \mid X_i = x).$$

The CEF summarizes the average value of y_i among observations with the same covariate value $X_i = x$.

3.2 Law of Iterated Expectations

The law of iterated expectations states that

$$\mathbb{E}[y_i] = \mathbb{E}\{\mathbb{E}[y_i \mid X_i]\}.$$

Derivation

Suppose (X_i, y_i) has joint density $f_{xy}(u,t)$. Let $g_X(u)$ be the marginal density of X_i, and let $f_y(t \mid X_i = u)$ be the conditional density of y_i given $X_i = u$. Start from the outer expectation:

$$\mathbb{E}\{\mathbb{E}[y_i \mid X_i]\} = \int \mathbb{E}[y_i \mid X_i = u] g_X(u)\,\mathrm{d}u.$$

Substitute the definition of the conditional expectation:

$$= \int \left[\int t f_y(t \mid X_i = u)\,\mathrm{d}t\right] g_X(u)\,\mathrm{d}u.$$

Move the marginal density inside:

$$= \int \int t f_y(t \mid X_i = u) g_X(u)\,\mathrm{d}t\,\mathrm{d}u.$$

Using

$$f_y(t \mid X_i = u) g_X(u) = f_{xy}(u,t),$$

we have

$$= \int \int t f_{xy}(u, t)\, \mathrm{d}u\, \mathrm{d}t.$$

Integrating out u,

$$= \int t \left[\int f_{xy}(u, t)\, \mathrm{d}u \right] \mathrm{d}t.$$

The inner integral is the marginal density of y_i, denoted $g_y(t)$. Hence

$$= \int t g_y(t)\, \mathrm{d}t = \mathbb{E}[y_i].$$

Therefore,

$$\boxed{\mathbb{E}[y_i] = \mathbb{E}\{\mathbb{E}[y_i \mid X_i]\}.}$$

3.3 CEF Decomposition Property

Define the residual

$$\varepsilon_i = y_i - \mathbb{E}[y_i \mid X_i].$$

Then

$$y_i = \mathbb{E}[y_i \mid X_i] + \varepsilon_i.$$

Step 1: Conditional Mean Zero

We show that

$$\mathbb{E}[\varepsilon_i \mid X_i] = 0.$$

By definition,

$$\mathbb{E}[\varepsilon_i \mid X_i] = \mathbb{E}[y_i - \mathbb{E}[y_i \mid X_i] \mid X_i].$$

Using linearity of conditional expectation,

$$= \mathbb{E}[y_i \mid X_i] - \mathbb{E}[\mathbb{E}[y_i \mid X_i] \mid X_i].$$

Because $\mathbb{E}[y_i \mid X_i]$ is already a function of X_i,

$$\mathbb{E}[\mathbb{E}[y_i \mid X_i] \mid X_i] = \mathbb{E}[y_i \mid X_i].$$

Thus

$$\mathbb{E}[\varepsilon_i \mid X_i] = \mathbb{E}[y_i \mid X_i] - \mathbb{E}[y_i \mid X_i] = 0.$$

Therefore,

$$\boxed{\mathbb{E}[\varepsilon_i \mid X_i] = 0.}$$

Step 2: Orthogonality to Any Function of X_i

Let $h(X_i)$ be any function of X_i. Then

$$\mathbb{E}[h(X_i)\varepsilon_i] = \mathbb{E}\{\mathbb{E}[h(X_i)\varepsilon_i \mid X_i]\}.$$

Since $h(X_i)$ is known once X_i is known,

$$\mathbb{E}[h(X_i)\varepsilon_i \mid X_i] = h(X_i)\mathbb{E}[\varepsilon_i \mid X_i].$$

Using $\mathbb{E}[\varepsilon_i \mid X_i] = 0$,

$$\mathbb{E}[h(X_i)\varepsilon_i] = \mathbb{E}[h(X_i) \cdot 0] = 0.$$

Therefore,

$$\boxed{\mathbb{E}[h(X_i)\varepsilon_i] = 0}$$

for every function $h(X_i)$.

3.4 CEF Prediction Property

The CEF solves the minimum mean squared error prediction problem:

$$\mathbb{E}[y_i \mid X_i] = \arg\min_{m(X_i)} \mathbb{E}[(y_i - m(X_i))^2].$$

Derivation

Let

$$\mu(X_i) = \mathbb{E}[y_i \mid X_i].$$

Then

$$y_i - m(X_i) = [y_i - \mu(X_i)] + [\mu(X_i) - m(X_i)].$$

Squaring both sides,

$$(y_i - m(X_i))^2 = [y_i - \mu(X_i)]^2 + 2[y_i - \mu(X_i)][\mu(X_i) - m(X_i)] + [\mu(X_i) - m(X_i)]^2.$$

Taking expectations,

$$\mathbb{E}[(y_i - m(X_i))^2] = \mathbb{E}[(y_i - \mu(X_i))^2] + 2\mathbb{E}\{[y_i - \mu(X_i)][\mu(X_i) - m(X_i)]\} + \mathbb{E}[(\mu(X_i) - m(X_i))^2].$$

The first term does not depend on $m(X_i)$. For the middle term, define

$$\varepsilon_i = y_i - \mu(X_i).$$

Then $\mu(X_i) - m(X_i)$ is a function of X_i. By the CEF decomposition property,

$$\mathbb{E}\{[y_i - \mu(X_i)][\mu(X_i) - m(X_i)]\} = 0.$$

Therefore,

$$\mathbb{E}[(y_i - m(X_i))^2] = \mathbb{E}[\varepsilon_i^2] + \mathbb{E}[(\mu(X_i) - m(X_i))^2].$$

The first term is fixed. The second term is minimized when

$$m(X_i) = \mu(X_i) = \mathbb{E}[y_i \mid X_i].$$

Thus,

$$\boxed{\mathbb{E}[y_i \mid X_i] = \arg\min_{m(X_i)} \mathbb{E}[(y_i - m(X_i))^2].}$$

3.5 ANOVA Variance Decomposition

The ANOVA theorem states that

$$\mathrm{Var}(y_i) = \mathrm{Var}(\mathbb{E}[y_i \mid X_i]) + \mathbb{E}[\mathrm{Var}(y_i \mid X_i)].$$

Derivation

Let

$$\mu_i = \mathbb{E}[y_i \mid X_i], \qquad \varepsilon_i = y_i - \mu_i.$$

Then

$$y_i = \mu_i + \varepsilon_i.$$

Taking variance,

$$\mathrm{Var}(y_i) = \mathrm{Var}(\mu_i + \varepsilon_i).$$

Expanding,

$$\mathrm{Var}(y_i) = \mathrm{Var}(\mu_i) + \mathrm{Var}(\varepsilon_i) + 2\,\mathrm{Cov}(\mu_i, \varepsilon_i).$$

Because μ_i is a function of X_i, and ε_i is orthogonal to every function of X_i,

$$\mathrm{Cov}(\mu_i, \varepsilon_i) = 0.$$

Therefore,

$$\mathrm{Var}(y_i) = \mathrm{Var}(\mu_i) + \mathrm{Var}(\varepsilon_i).$$

Since $\mathbb{E}[\varepsilon_i] = 0$,

$$\text{Var}(\varepsilon_i) = \mathbb{E}[\varepsilon_i^2].$$

Using iterated expectations,

$$\mathbb{E}[\varepsilon_i^2] = \mathbb{E}\{\mathbb{E}[\varepsilon_i^2 \mid X_i]\}.$$

But

$$\varepsilon_i = y_i - \mathbb{E}[y_i \mid X_i],$$

so

$$\mathbb{E}[\varepsilon_i^2 \mid X_i] = \text{Var}(y_i \mid X_i).$$

Hence,

$$\text{Var}(\varepsilon_i) = \mathbb{E}[\text{Var}(y_i \mid X_i)].$$

Therefore,

$$\boxed{\text{Var}(y_i) = \text{Var}(\mathbb{E}[y_i \mid X_i]) + \mathbb{E}[\text{Var}(y_i \mid X_i)].}$$

3.6 Population Linear Regression

The population regression coefficient vector is defined by

$$\beta = \arg\min_b \mathbb{E}[(y_i - X_i'b)^2].$$

Derivation

Define the population objective function:

$$Q(b) = \mathbb{E}[(y_i - X_i'b)^2].$$

Differentiate with respect to b:

$$\frac{\partial Q(b)}{\partial b} = \mathbb{E}[-2X_i(y_i - X_i'b)].$$

The first-order condition is

$$\mathbb{E}[X_i(y_i - X_i'b)] = 0.$$

Expanding,

$$\mathbb{E}[X_i y_i] - \mathbb{E}[X_i X_i']b = 0.$$

Thus

$$\mathbb{E}[X_i X_i']b = \mathbb{E}[X_i y_i].$$

If $\mathbb{E}[X_i X_i']$ is invertible, then

$$\boxed{\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i y_i].}$$

Define the population regression residual as

$$e_i = y_i - X_i'\beta.$$

By construction,

$$\boxed{\mathbb{E}[X_i e_i] = 0.}$$

This orthogonality condition follows from the definition of β; it is not an extra economic assumption.

3.7 Bivariate Regression with an Intercept

Consider the population regression

$$y_i = \alpha + \beta x_i + e_i.$$

The population least-squares conditions are

$$\mathbb{E}[e_i] = 0, \qquad \mathbb{E}[x_i e_i] = 0.$$

Derivation of the Intercept

Since

$$e_i = y_i - \alpha - \beta x_i,$$

the condition $\mathbb{E}[e_i] = 0$ gives

$$\mathbb{E}[y_i - \alpha - \beta x_i] = 0.$$

Thus

$$\mathbb{E}[y_i] - \alpha - \beta\mathbb{E}[x_i] = 0,$$

so

$$\boxed{\alpha = \mathbb{E}[y_i] - \beta\mathbb{E}[x_i].}$$

Derivation of the Slope

The second condition is

$$\mathbb{E}[x_i(y_i - \alpha - \beta x_i)] = 0.$$

Substitute $\alpha = \mathbb{E}[y_i] - \beta\mathbb{E}[x_i]$:

$$\mathbb{E}[x_i y_i] - \mathbb{E}[x_i]\mathbb{E}[y_i] + \beta\mathbb{E}[x_i]\mathbb{E}[x_i] - \beta\mathbb{E}[x_i^2] = 0.$$

Rearranging,

$$\mathbb{E}[x_i y_i] - \mathbb{E}[x_i]\mathbb{E}[y_i] = \beta\left[\mathbb{E}[x_i^2] - (\mathbb{E}[x_i])^2\right].$$

The left-hand side is $\text{Cov}(y_i, x_i)$, and the bracketed term is $\text{Var}(x_i)$. Hence

$$\boxed{\beta = \frac{\text{Cov}(y_i, x_i)}{\text{Var}(x_i)}.}$$

3.8 Regression Anatomy and the Frisch-Waugh-Lovell Logic

Suppose the long regression is

$$y_i = \beta_k x_{ki} + Z_i'\gamma + e_i,$$

where Z_i contains all other controls. Let $\widetilde{x}_{ki}$ be the residual from the population regression of x_{ki} on Z_i:

$$x_{ki} = Z_i'\pi + \widetilde{x}_{ki}.$$

By construction,

$$\mathbb{E}[Z_i \widetilde{x}_{ki}] = 0.$$

We want to show

$$\boxed{\beta_k = \frac{\text{Cov}(y_i, \widetilde{x}_{ki})}{\text{Var}(\widetilde{x}_{ki})}.}$$

Derivation

Take the covariance of y_i with $\widetilde{x}_{ki}$:

$$\text{Cov}(y_i, \widetilde{x}_{ki}) = \text{Cov}(\beta_k x_{ki} + Z_i'\gamma + e_i, \widetilde{x}_{ki}).$$

Using linearity,

$$= \beta_k \text{Cov}(x_{ki}, \widetilde{x}_{ki}) + \text{Cov}(Z_i'\gamma, \widetilde{x}_{ki}) + \text{Cov}(e_i, \widetilde{x}_{ki}).$$

The second term is zero because $\widetilde{x}_{ki}$ is orthogonal to Z_i:

$$\text{Cov}(Z_i'\gamma, \widetilde{x}_{ki}) = 0.$$

The third term is zero because e_i is orthogonal to all regressors, and $\widetilde{x}_{ki}$ is a linear combination of regressors:

$$\mathrm{Cov}(e_i, \widetilde{x}_{ki}) = 0.$$

Since

$$x_{ki} = Z_i'\pi + \widetilde{x}_{ki},$$

we have

$$\mathrm{Cov}(x_{ki}, \widetilde{x}_{ki}) = \mathrm{Cov}(Z_i'\pi + \widetilde{x}_{ki}, \widetilde{x}_{ki}) = \mathrm{Var}(\widetilde{x}_{ki}).$$

Therefore,

$$\mathrm{Cov}(y_i, \widetilde{x}_{ki}) = \beta_k \,\mathrm{Var}(\widetilde{x}_{ki}),$$

and hence

$$\boxed{\beta_k = \frac{\mathrm{Cov}(y_i, \widetilde{x}_{ki})}{\mathrm{Var}(\widetilde{x}_{ki})}.}$$

3.9 Linear CEF Theorem

If the CEF is linear,

$$\mathbb{E}[y_i \mid X_i] = X_i'\beta^*,$$

then the population regression coefficient equals β^*.

Derivation

The population regression coefficient is

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i y_i].$$

Using iterated expectations,

$$\mathbb{E}[X_i y_i] = \mathbb{E}\{\mathbb{E}[X_i y_i \mid X_i]\}.$$

Since X_i is known conditional on X_i,

$$\mathbb{E}[X_i y_i \mid X_i] = X_i \mathbb{E}[y_i \mid X_i].$$

Therefore,

$$\mathbb{E}[X_i y_i] = \mathbb{E}[X_i \mathbb{E}[y_i \mid X_i]].$$

If

$$\mathbb{E}[y_i \mid X_i] = X_i'\beta^*,$$

then

$$\mathbb{E}[X_i y_i] = \mathbb{E}[X_i X_i' \beta^*] = \mathbb{E}[X_i X_i']\beta^*.$$

Thus

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i X_i']\beta^* = \beta^*.$$

Therefore,

$$\boxed{\text{If the CEF is linear, the regression function is the CEF.}}$$

3.10 Best Linear Predictor Theorem

The population regression function $X_i'\beta$ is the best linear predictor of y_i given X_i. This follows directly because

$$\beta = \arg\min_b \mathbb{E}[(y_i - X_i'b)^2].$$

Therefore, among all linear functions $X_i'b$, the best one in the minimum mean squared error sense is $X_i'\beta$.

3.11 Regression-CEF Theorem

The regression function is the best linear approximation to the CEF:

$$\beta = \arg\min_b \mathbb{E}\left[(\mathbb{E}[y_i \mid X_i] - X_i'b)^2\right].$$

Derivation

Let

$$\mu_i = \mathbb{E}[y_i \mid X_i].$$

Write

$$y_i - X_i'b = (y_i - \mu_i) + (\mu_i - X_i'b).$$

Squaring,

$$(y_i - X_i'b)^2 = (y_i - \mu_i)^2 + (\mu_i - X_i'b)^2 + 2(y_i - \mu_i)(\mu_i - X_i'b).$$

Taking expectations,

$$\mathbb{E}[(y_i - X_i'b)^2] = \mathbb{E}[(y_i - \mu_i)^2] + \mathbb{E}[(\mu_i - X_i'b)^2] + 2\mathbb{E}[(y_i - \mu_i)(\mu_i - X_i'b)].$$

The first term does not depend on b. The third term is zero because $y_i - \mu_i$ is orthogonal to every function of X_i, and $\mu_i - X_i'b$ is a function of X_i. Hence,

$$\arg\min_b \mathbb{E}[(y_i - X_i'b)^2] = \arg\min_b \mathbb{E}[(\mu_i - X_i'b)^2].$$

Therefore,

$$\boxed{\text{Regression gives the best linear approximation to the CEF.}}$$

3.12 *Grouped-Data Regression*

The population coefficient is

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i y_i].$$

Using iterated expectations,

$$\mathbb{E}[X_i y_i] = \mathbb{E}[X_i \mathbb{E}[y_i \mid X_i]].$$

Therefore,

$$\boxed{\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i \mathbb{E}[y_i \mid X_i]].}$$

If X_i is discrete and takes values u, then

$$\mathbb{E}[X_i X_i'] = \sum_u uu'\mathbb{P}(X_i = u),$$

and

$$\mathbb{E}[X_i \mathbb{E}[y_i \mid X_i]] = \sum_u u\mathbb{E}[y_i \mid X_i = u]\mathbb{P}(X_i = u).$$

Thus, β can be obtained by regressing the cell mean $\mathbb{E}[y_i \mid X_i = u]$ on u, weighting each cell by $\mathbb{P}(X_i = u)$. In sample form, this means weighted least squares on grouped means, with group size as the weight.

3.13 *OLS Estimator and Asymptotic Distribution*

The population coefficient is

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i y_i].$$

The sample analog is

$$\widehat{\beta} = \left(\sum_i X_i X_i'\right)^{-1} \sum_i X_i y_i.$$

Equivalently,

$$\widehat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i y_i \right).$$

Derivation

Let

$$y_i = X_i'\beta + e_i.$$

Substitute into $\widehat{\beta}$:

$$\widehat{\beta} = \left(\sum_i X_i X_i' \right)^{-1} \sum_i X_i (X_i'\beta + e_i).$$

Expanding,

$$= \left(\sum_i X_i X_i' \right)^{-1} \left[\sum_i X_i X_i'\beta + \sum_i X_i e_i \right].$$

Therefore,

$$\widehat{\beta} = \beta + \left(\sum_i X_i X_i' \right)^{-1} \sum_i X_i e_i.$$

Hence

$$\widehat{\beta} - \beta = \left(\sum_i X_i X_i' \right)^{-1} \sum_i X_i e_i.$$

Multiplying by $\sqrt{N}$,

$$\sqrt{N}(\widehat{\beta} - \beta) = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_i X_i e_i \right).$$

By the law of large numbers,

$$\frac{1}{N} \sum_i X_i X_i' \xrightarrow{p} \mathbb{E}[X_i X_i'] = Q.$$

By the central limit theorem,

$$\frac{1}{\sqrt{N}} \sum_i X_i e_i \xrightarrow{d} \mathcal{N}(0, \Omega),$$

where

$$\Omega = \mathbb{E}[X_i X_i' e_i^2].$$

By Slutsky's theorem,

$$\boxed{\sqrt{N}(\widehat{\beta}-\beta) \xrightarrow{d} \mathcal{N}(0, Q^{-1}\Omega Q^{-1}).}$$

Thus the asymptotic variance is

$$\boxed{Q^{-1}\Omega Q^{-1} = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i X_i' e_i^2]\mathbb{E}[X_i X_i']^{-1}.}$$

3.14 Robust Standard Error Formula

The asymptotic covariance matrix is

$$\mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i X_i' e_i^2]\mathbb{E}[X_i X_i']^{-1}.$$

In practice, estimate e_i by

$$\widehat{e_i} = y_i - X_i'\widehat{\beta}.$$

Estimate $\mathbb{E}[X_i X_i']$ by

$$\frac{1}{N}\sum_i X_i X_i',$$

and estimate $\mathbb{E}[X_i X_i' e_i^2]$ by

$$\frac{1}{N}\sum_i X_i X_i' \widehat{e}_i^2.$$

Therefore, the robust covariance estimator is

$$\boxed{\widehat{V}_{robust}(\widehat{\beta}) = \left(\sum_i X_i X_i'\right)^{-1}\left(\sum_i X_i X_i' \widehat{e}_i^2\right)\left(\sum_i X_i X_i'\right)^{-1}.}$$

Finite-sample corrections may multiply this expression by factors such as $N/(N-k)$, depending on the software.

3.15 Homoskedastic Simplification

Assume

$$\mathbb{E}[e_i^2 \mid X_i] = \sigma^2.$$

Then

$$\mathbb{E}[X_i X_i' e_i^2] = \mathbb{E}\{X_i X_i' \mathbb{E}[e_i^2 \mid X_i]\}.$$

Substituting the homoskedasticity assumption,

$$= \mathbb{E}[X_i X_i' \sigma^2] = \sigma^2 \mathbb{E}[X_i X_i'].$$

Therefore,

$$Q^{-1}\Omega Q^{-1} = \mathbb{E}[X_i X_i']^{-1} \left[\sigma^2 \mathbb{E}[X_i X_i']\right] \mathbb{E}[X_i X_i']^{-1}.$$

Canceling,

$$= \sigma^2 \mathbb{E}[X_i X_i']^{-1}.$$

Thus,

$$\boxed{V_{homoskedastic}(\widehat{\beta}) = \sigma^2 \mathbb{E}[X_i X_i']^{-1}.}$$

3.16 Why a Nonlinear CEF Generates Heteroskedastic Regression Residuals

Let

$$\mu_i = \mathbb{E}[y_i \mid X_i].$$

The regression residual is

$$e_i = y_i - X_i'\beta.$$

Write

$$e_i = (y_i - \mu_i) + (\mu_i - X_i'\beta).$$

Then

$$e_i^2 = (y_i - \mu_i)^2 + (\mu_i - X_i'\beta)^2 + 2(y_i - \mu_i)(\mu_i - X_i'\beta).$$

Taking conditional expectation given X_i,

$$\mathbb{E}[e_i^2 \mid X_i] = \mathbb{E}[(y_i - \mu_i)^2 \mid X_i] + (\mu_i - X_i'\beta)^2 + 2(\mu_i - X_i'\beta)\mathbb{E}[y_i - \mu_i \mid X_i].$$

The final term is zero because

$$\mathbb{E}[y_i - \mu_i \mid X_i] = 0.$$

Thus,

$$\boxed{\mathbb{E}[e_i^2 \mid X_i] = \mathrm{Var}(y_i \mid X_i) + (\mathbb{E}[y_i \mid X_i] - X_i'\beta)^2.}$$

Therefore, even if $\mathrm{Var}(y_i \mid X_i)$ is constant, the regression residual variance can vary with X_i when the linear regression does not perfectly fit the CEF.

3.17 Linear Probability Model Residual Variance

Suppose

$$y_i \in \{0, 1\}.$$

Then, conditional on X_i,

$$y_i \sim Bernoulli(p_i),$$

where

$$p_i = \mathbb{P}(y_i = 1 \mid X_i).$$

For a Bernoulli variable,

$$\mathrm{Var}(y_i \mid X_i) = p_i(1 - p_i).$$

If the linear probability model is saturated, then

$$p_i = \mathbb{E}[y_i \mid X_i] = X_i'\beta.$$

Therefore,

$$\boxed{\mathrm{Var}(y_i \mid X_i) = X_i'\beta(1 - X_i'\beta).}$$

This generally depends on X_i, so the linear probability model is generally heteroskedastic.

3.18 Saturated Model with One Discrete Regressor

Suppose schooling s_i takes values

$$0, 1, \ldots, \tau.$$

Define

$$d_{ji} = \mathbf{1}[s_i = j].$$

A saturated model is

$$y_i = \beta_0 + \beta_1 d_{1i} + \cdots + \beta_\tau d_{\tau i} + \varepsilon_i.$$

For $s_i = 0$, all dummy variables equal zero, so

$$\mathbb{E}[y_i \mid s_i = 0] = \beta_0.$$

For $s_i = j$,

$$\mathbb{E}[y_i \mid s_i = j] = \beta_0 + \beta_j.$$

Therefore,

$$\boxed{\beta_j = \mathbb{E}[y_i \mid s_i = j] - \mathbb{E}[y_i \mid s_i = 0].}$$

A saturated regression exactly fits the CEF for a discrete regressor.

3.19 Saturated Model with Two Dummy Variables

Let

$$x_{1i} = 1$$

if individual i is a college graduate, and let

$$x_{2i} = 1$$

if individual i is female. A saturated model is

$$y_i = \alpha + \beta x_{1i} + \gamma x_{2i} + \delta(x_{1i}x_{2i}) + \varepsilon_i.$$

The four conditional means are

$$\mathbb{E}[y_i \mid x_{1i} = 0, x_{2i} = 0] = \alpha,$$

$$\mathbb{E}[y_i \mid x_{1i} = 1, x_{2i} = 0] = \alpha + \beta,$$

$$\mathbb{E}[y_i \mid x_{1i} = 0, x_{2i} = 1] = \alpha + \gamma,$$

and

$$\mathbb{E}[y_i \mid x_{1i} = 1, x_{2i} = 1] = \alpha + \beta + \gamma + \delta.$$

Thus,

$$\beta = \mathbb{E}[y_i \mid 1, 0] - \mathbb{E}[y_i \mid 0, 0],$$

$$\gamma = \mathbb{E}[y_i \mid 0, 1] - \mathbb{E}[y_i \mid 0, 0],$$

and

$$\delta = [\mathbb{E}[y_i \mid 1, 1] - \mathbb{E}[y_i \mid 0, 1]] - [\mathbb{E}[y_i \mid 1, 0] - \mathbb{E}[y_i \mid 0, 0]] \,.$$

Therefore, δ is a difference-in-differences interaction effect.

3.20 Selection Bias Decomposition

Let $c_i \in \{0, 1\}$ indicate college attendance. Define potential outcomes as

$$y_{1i} = \text{earnings if college,} \qquad y_{0i} = \text{earnings if no college.}$$

The observed outcome is

$$y_i = y_{0i} + (y_{1i} - y_{0i})c_i.$$

The observed difference is

$$\mathbb{E}[y_i \mid c_i = 1] - \mathbb{E}[y_i \mid c_i = 0].$$

When $c_i = 1$, $y_i = y_{1i}$. When $c_i = 0$, $y_i = y_{0i}$. Hence,

$$\mathbb{E}[y_i \mid c_i = 1] - \mathbb{E}[y_i \mid c_i = 0] = \mathbb{E}[y_{1i} \mid c_i = 1] - \mathbb{E}[y_{0i} \mid c_i = 0].$$

Add and subtract $\mathbb{E}[y_{0i} \mid c_i = 1]$:

$$= \mathbb{E}[y_{1i} \mid c_i = 1] - \mathbb{E}[y_{0i} \mid c_i = 1] + \mathbb{E}[y_{0i} \mid c_i = 1] - \mathbb{E}[y_{0i} \mid c_i = 0].$$

Therefore,

$$\boxed{\mathbb{E}[y_i \mid c_i = 1] - \mathbb{E}[y_i \mid c_i = 0] = \mathbb{E}[y_{1i} - y_{0i} \mid c_i = 1] + \left[\mathbb{E}[y_{0i} \mid c_i = 1] - \mathbb{E}[y_{0i} \mid c_i = 0]\right].}$$

The first term is the treatment effect on the treated. The second term is selection bias.

3.21 Conditional Independence Assumption

The conditional independence assumption says

$$\{y_{0i}, y_{1i}\} \perp c_i \mid X_i.$$

Then

$$\mathbb{E}[y_{0i} \mid X_i, c_i = 1] = \mathbb{E}[y_{0i} \mid X_i, c_i = 0] = \mathbb{E}[y_{0i} \mid X_i],$$

and similarly for y_{1i}.

Consider

$$\mathbb{E}[y_i \mid X_i, c_i = 1] - \mathbb{E}[y_i \mid X_i, c_i = 0].$$

This equals

$$\mathbb{E}[y_{1i} \mid X_i, c_i = 1] - \mathbb{E}[y_{0i} \mid X_i, c_i = 0].$$

By the conditional independence assumption,

$$= \mathbb{E}[y_{1i} \mid X_i] - \mathbb{E}[y_{0i} \mid X_i].$$

Thus,

$$\boxed{\mathbb{E}[y_i \mid X_i, c_i = 1] - \mathbb{E}[y_i \mid X_i, c_i = 0] = \mathbb{E}[y_{1i} - y_{0i} \mid X_i].}$$

Therefore, conditional comparisons have a causal interpretation under the conditional independence assumption.

3.22 Average Treatment Effect from Conditional Effects

The conditional causal effect is

$$\mathbb{E}[y_{1i} - y_{0i} \mid X_i].$$

The population average treatment effect is

$$\mathbb{E}[y_{1i} - y_{0i}].$$

Using iterated expectations,

$$\mathbb{E}[y_{1i} - y_{0i}] = \mathbb{E}\{\mathbb{E}[y_{1i} - y_{0i} \mid X_i]\}.$$

Under conditional independence,

$$\mathbb{E}[y_{1i} - y_{0i} \mid X_i] = \mathbb{E}[y_i \mid X_i, c_i = 1] - \mathbb{E}[y_i \mid X_i, c_i = 0].$$

Therefore,

$$\boxed{ATE = \mathbb{E}\left[\mathbb{E}[y_i \mid X_i, c_i = 1] - \mathbb{E}[y_i \mid X_i, c_i = 0]\right].}$$

3.23 Constant-Effects Causal Model and Regression

Suppose individual potential earnings are

$$f_i(s) = \alpha + \rho s + \eta_i.$$

Observed schooling is s_i, so

$$y_i = f_i(s_i) = \alpha + \rho s_i + \eta_i.$$

Suppose

$$\eta_i = X_i'\gamma + v_i,$$

where

$$\mathbb{E}[\eta_i \mid X_i] = X_i'\gamma.$$

Then

$$y_i = \alpha + \rho s_i + X_i'\gamma + v_i.$$

If conditional independence holds, then

$$\mathbb{E}[\eta_i \mid X_i, s_i] = \mathbb{E}[\eta_i \mid X_i].$$

Therefore,

$$\mathbb{E}[v_i \mid X_i, s_i] = \mathbb{E}[\eta_i - X_i'\gamma \mid X_i, s_i] = \mathbb{E}[\eta_i \mid X_i, s_i] - X_i'\gamma = 0.$$

Thus, v_i is orthogonal to s_i and X_i, so the regression coefficient on schooling identifies ρ. Hence,

$$\boxed{\text{Under constant effects and conditional independence, regression identifies the causal effect.}}$$

3.24 Omitted Variables Bias Formula

Consider the long regression

$$y_i = \alpha + \rho s_i + A_i'\gamma + \varepsilon_i.$$

The short regression omits A_i:

$$y_i = \alpha_s + \rho_s s_i + u_i.$$

The short-regression slope is

$$\rho_s = \frac{\mathrm{Cov}(y_i, s_i)}{\mathrm{Var}(s_i)}.$$

Substitute the long regression into the numerator:

$$\mathrm{Cov}(y_i, s_i) = \mathrm{Cov}(\alpha + \rho s_i + A_i'\gamma + \varepsilon_i, s_i).$$

The constant has zero covariance, so

$$= \rho \,\mathrm{Var}(s_i) + \mathrm{Cov}(A_i'\gamma, s_i) + \mathrm{Cov}(\varepsilon_i, s_i).$$

In the long regression, ε_i is orthogonal to s_i, so

$$\mathrm{Cov}(\varepsilon_i, s_i) = 0.$$

Also,

$$\mathrm{Cov}(A_i'\gamma, s_i) = \gamma' \,\mathrm{Cov}(A_i, s_i).$$

Thus,

$$\rho_s = \frac{\rho \,\mathrm{Var}(s_i) + \gamma' \,\mathrm{Cov}(A_i, s_i)}{\mathrm{Var}(s_i)}.$$

Therefore,

$$\boxed{\rho_s = \rho + \gamma' \frac{\text{Cov}(A_i, s_i)}{\text{Var}(s_i)}.}$$

Let

$$\delta_{As} = \frac{\text{Cov}(A_i, s_i)}{\text{Var}(s_i)}$$

be the coefficient from regressing A_i on s_i. Then

$$\boxed{\rho_s = \rho + \gamma' \delta_{As}.}$$

In words,

$$\boxed{\text{short coefficient} = \text{long coefficient} + \text{effect of omitted variable} \times \text{relationship between omitted and inc}}$$

3.25 Bad Control Derivation

Let c_i be college attendance and w_i be white-collar occupation. Potential earnings are

$$y_i = c_i y_{1i} + (1 - c_i) y_{0i}.$$

Potential occupation is

$$w_i = c_i w_{1i} + (1 - c_i) w_{0i}.$$

Suppose college is randomly assigned. Then the unconditional comparison can estimate

$$\mathbb{E}[y_{1i} - y_{0i}].$$

However, consider conditioning on white-collar status:

$$\mathbb{E}[y_i \mid w_i = 1, c_i = 1] - \mathbb{E}[y_i \mid w_i = 1, c_i = 0].$$

When $c_i = 1$, $w_i = w_{1i}$ and $y_i = y_{1i}$. When $c_i = 0$, $w_i = w_{0i}$ and $y_i = y_{0i}$. Hence,

$$= \mathbb{E}[y_{1i} \mid w_{1i} = 1, c_i = 1] - \mathbb{E}[y_{0i} \mid w_{0i} = 1, c_i = 0].$$

Because c_i is randomly assigned,

$$= \mathbb{E}[y_{1i} \mid w_{1i} = 1] - \mathbb{E}[y_{0i} \mid w_{0i} = 1].$$

Add and subtract $\mathbb{E}[y_{0i} \mid w_{1i} = 1]$:

$$= \mathbb{E}[y_{1i} - y_{0i} \mid w_{1i} = 1] + \left[\mathbb{E}[y_{0i} \mid w_{1i} = 1] - \mathbb{E}[y_{0i} \mid w_{0i} = 1]\right].$$

Thus,

$$\boxed{\begin{array}{c}\text{conditional comparison} = \text{causal effect for those who would be white-collar if treated} \\ +\text{selection bias.}\end{array}}$$

The problem is that w_i is itself affected by treatment. It is an outcome, not a pre-treatment control.

3.26 Proxy-Control Bad Control Derivation

Suppose the desired long regression is

$$y_i = \alpha + \rho s_i + \gamma a_i + \varepsilon_i,$$

where a_i is true ability. However, a_i is unobserved. Instead, we observe late ability a_i^l, determined by

$$a_i^l = \pi_0 + \pi_1 s_i + \pi_2 a_i.$$

Solve for a_i:

$$a_i = \frac{a_i^l - \pi_0 - \pi_1 s_i}{\pi_2}.$$

Substitute into the true equation:

$$y_i = \alpha + \rho s_i + \gamma \left(\frac{a_i^l - \pi_0 - \pi_1 s_i}{\pi_2} \right) + \varepsilon_i.$$

Expanding,

$$y_i = \alpha - \frac{\gamma \pi_0}{\pi_2} + \left(\rho - \frac{\gamma \pi_1}{\pi_2} \right) s_i + \frac{\gamma}{\pi_2} a_i^l + \varepsilon_i.$$

Thus, the coefficient on schooling when controlling for late ability is

$$\boxed{\rho - \frac{\gamma \pi_1}{\pi_2}.}$$

If

$$\gamma > 0, \qquad \pi_1 > 0, \qquad \pi_2 > 0,$$

then

$$\rho - \frac{\gamma\pi_1}{\pi_2} < \rho.$$

Thus, controlling for late ability can understate the true schooling effect because late ability is partly caused by schooling.

3.27 Matching Estimand: Treatment on the Treated

Let treatment be $d_i \in \{0, 1\}$, and let potential outcomes be y_{1i} and y_{0i}. The treatment effect on the treated is

$$TOT = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].$$

Using iterated expectations,

$$TOT = \mathbb{E}\{\mathbb{E}[y_{1i} - y_{0i} \mid X_i, d_i = 1] \mid d_i = 1\}.$$

Expand:

$$= \mathbb{E}\{\mathbb{E}[y_{1i} \mid X_i, d_i = 1] - \mathbb{E}[y_{0i} \mid X_i, d_i = 1] \mid d_i = 1\}.$$

The first term is observed among treated:

$$\mathbb{E}[y_{1i} \mid X_i, d_i = 1] = \mathbb{E}[y_i \mid X_i, d_i = 1].$$

The second term is counterfactual. Under conditional independence,

$$\mathbb{E}[y_{0i} \mid X_i, d_i = 1] = \mathbb{E}[y_{0i} \mid X_i, d_i = 0] = \mathbb{E}[y_i \mid X_i, d_i = 0].$$

Therefore,

$$\boxed{TOT = \mathbb{E}\left[\mathbb{E}[y_i \mid X_i, d_i = 1] - \mathbb{E}[y_i \mid X_i, d_i = 0] \mid d_i = 1\right].}$$

If X_i is discrete,

$$\boxed{TOT = \sum_x \delta_x \mathbb{P}(X_i = x \mid d_i = 1),}$$

where

$$\delta_x = \mathbb{E}[y_i \mid X_i = x, d_i = 1] - \mathbb{E}[y_i \mid X_i = x, d_i = 0].$$

3.28 Average Treatment Effect Matching Estimand

The average treatment effect is

$$ATE = \mathbb{E}[y_{1i} - y_{0i}].$$

Using iterated expectations,

$$ATE = \mathbb{E}\{\mathbb{E}[y_{1i} - y_{0i} \mid X_i]\}.$$

Under conditional independence,

$$\mathbb{E}[y_{1i} - y_{0i} \mid X_i] = \mathbb{E}[y_i \mid X_i, d_i = 1] - \mathbb{E}[y_i \mid X_i, d_i = 0].$$

Thus,

$$\boxed{ATE = \mathbb{E}[\mathbb{E}[y_i \mid X_i, d_i = 1] - \mathbb{E}[y_i \mid X_i, d_i = 0]].}$$

For discrete X_i,

$$\boxed{ATE = \sum_x \delta_x \mathbb{P}(X_i = x).}$$

The difference between TOT and ATE is the weighting distribution:

$$TOT : \mathbb{P}(X_i = x \mid d_i = 1), \qquad ATE : \mathbb{P}(X_i = x).$$

3.29 Regression as Weighted Matching

Consider the regression

$$y_i = \sum_x d_{ix}\beta_x + \delta_R d_i + \varepsilon_i,$$

where $d_{ix} = \mathbf{1}[X_i = x]$. This regression is saturated in X_i, but treatment has one additive coefficient.

By regression anatomy,

$$\delta_R = \frac{\text{Cov}(y_i, \widetilde{d}_i)}{\text{Var}(\widetilde{d}_i)},$$

where

$$\widetilde{d}_i = d_i - \mathbb{E}[d_i \mid X_i].$$

Let

$$p(X_i) = \mathbb{E}[d_i \mid X_i].$$

Then

$$\widetilde{d}_i = d_i - p(X_i).$$

Thus,

$$\delta_R = \frac{\mathbb{E}[(d_i - p(X_i))y_i]}{\mathbb{E}[(d_i - p(X_i))^2]}.$$

Use iterated expectations:

$$\mathbb{E}[(d_i - p(X_i))y_i] = \mathbb{E}\{(d_i - p(X_i))\mathbb{E}[y_i \mid d_i, X_i]\}.$$

The conditional mean can be written as

$$\mathbb{E}[y_i \mid d_i, X_i] = \mathbb{E}[y_i \mid d_i = 0, X_i] + \delta_X d_i,$$

where

$$\delta_X = \mathbb{E}[y_i \mid d_i = 1, X_i] - \mathbb{E}[y_i \mid d_i = 0, X_i].$$

Substitute:

$$\mathbb{E}[(d_i - p)y_i] = \mathbb{E}[(d_i - p)\mathbb{E}[y_i \mid d_i = 0, X_i]] + \mathbb{E}[(d_i - p)d_i\delta_X].$$

The first term is zero because $\mathbb{E}[y_i \mid d_i = 0, X_i]$ is a function of X_i, and $d_i - p(X_i)$ has conditional mean zero. Therefore,

$$\mathbb{E}[(d_i - p)y_i] = \mathbb{E}[(d_i - p)d_i\delta_X].$$

Since d_i is binary,

$$\mathbb{E}[(d_i - p)d_i \mid X_i] = p(X_i)(1 - p(X_i)).$$

Also,

$$\mathbb{E}[(d_i - p)^2 \mid X_i] = p(X_i)(1 - p(X_i)).$$

Therefore,

$$\boxed{\delta_R = \frac{\mathbb{E}[p(X_i)(1 - p(X_i))\delta_X]}{\mathbb{E}[p(X_i)(1 - p(X_i))]}.}$$

Regression weights each covariate-specific treatment effect by

$$p(X_i)(1 - p(X_i)).$$

This weight is largest when $p(X_i) = 1/2$.

3.30 Continuous Treatment Average Derivative Formula

Let s_i be a continuous treatment and

$$h(s) = \mathbb{E}[y_i \mid s_i = s].$$

The bivariate regression slope is

$$\beta = \frac{\mathrm{Cov}(y_i, s_i)}{\mathrm{Var}(s_i)} = \frac{\mathbb{E}[h(s_i)(s_i - \mathbb{E}[s_i])]}{\mathbb{E}[s_i(s_i - \mathbb{E}[s_i])]}.$$

Let $\mu_s = \mathbb{E}[s_i]$. The numerator is

$$\mathbb{E}[h(s_i)(s_i - \mu_s)].$$

Assume

$$h(s) = \kappa + \int_{-\infty}^{s} h'(t)\,\mathrm{d}t.$$

Substitute:

$$\mathbb{E}[h(s_i)(s_i - \mu_s)] = \mathbb{E}\left[\left(\kappa + \int_{-\infty}^{s_i} h'(t)\,\mathrm{d}t\right)(s_i - \mu_s)\right].$$

The constant term disappears because

$$\mathbb{E}[\kappa(s_i - \mu_s)] = 0.$$

Thus,

$$= \mathbb{E}\left[\left(\int_{-\infty}^{s_i} h'(t)\,\mathrm{d}t\right)(s_i - \mu_s)\right].$$

Write the expectation as an integral:

$$= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{s} h'(t)\,\mathrm{d}t\right](s - \mu_s)g(s)\,\mathrm{d}s.$$

Reverse the order of integration:

$$= \int_{-\infty}^{\infty} h'(t)\left[\int_{t}^{\infty}(s - \mu_s)g(s)\,\mathrm{d}s\right]\mathrm{d}t.$$

Define

$$\mu_t = [\mathbb{E}[s_i \mid s_i \geq t] - \mathbb{E}[s_i \mid s_i < t]]\,\mathbb{P}(s_i \geq t)[1 - \mathbb{P}(s_i \geq t)].$$

Then

$$\int_{t}^{\infty}(s - \mu_s)g(s)\,\mathrm{d}s = \mu_t.$$

So

$$\mathbb{E}[h(s_i)(s_i - \mu_s)] = \int h'(t)\mu_t\,\mathrm{d}t.$$

Similarly,

$$\operatorname{Var}(s_i) = \mathbb{E}[s_i(s_i - \mu_s)] = \int \mu_t \, \mathrm{d}t.$$

Therefore,

$$\boxed{\beta = \frac{\int h'(t)\mu_t \, \mathrm{d}t}{\int \mu_t \, \mathrm{d}t}.}$$

So the regression slope is a weighted average derivative of the CEF.

3.31 Normal Regressor Special Case

Suppose

$$s_i \sim \mathcal{N}(\mu_s, \sigma_s^2),$$

and define

$$z_i = \frac{s_i - \mu_s}{\sigma_s}.$$

Then

$$z_i \sim \mathcal{N}(0, 1).$$

For

$$t^* = \frac{t - \mu_s}{\sigma_s},$$

the truncated normal formulas are

$$\mathbb{E}[z_i \mid z_i > t^*] = \frac{\phi(t^*)}{1 - \Phi(t^*)},$$

and

$$\mathbb{E}[z_i \mid z_i < t^*] = -\frac{\phi(t^*)}{\Phi(t^*)}.$$

Therefore,

$$\mathbb{E}[s_i \mid s_i \geq t] = \mu_s + \sigma_s \frac{\phi(t^*)}{1 - \Phi(t^*)},$$

and

$$\mathbb{E}[s_i \mid s_i < t] = \mu_s - \sigma_s \frac{\phi(t^*)}{\Phi(t^*)}.$$

The difference is

$$\sigma_s \left[\frac{\phi(t^*)}{1 - \Phi(t^*)} + \frac{\phi(t^*)}{\Phi(t^*)} \right].$$

Multiplying by

$$\mathbb{P}(s_i \geq t)[1 - \mathbb{P}(s_i \geq t)] = [1 - \Phi(t^*)]\Phi(t^*),$$

we get

$$\mu_t = \sigma_s \left[\frac{\phi(t^*)}{1 - \Phi(t^*)} + \frac{\phi(t^*)}{\Phi(t^*)} \right] [1 - \Phi(t^*)]\Phi(t^*).$$

Simplifying,

$$\mu_t = \sigma_s \phi(t^*)[\Phi(t^*) + 1 - \Phi(t^*)] = \sigma_s \phi(t^*).$$

Thus,

$$\int h'(t)\mu_t \, \mathrm{d}t = \int h'(t)\sigma_s \phi \left(\frac{t - \mu_s}{\sigma_s} \right) \mathrm{d}t.$$

Let

$$t = \mu_s + \sigma_s z, \qquad \mathrm{d}t = \sigma_s \, \mathrm{d}z.$$

Then

$$= \sigma_s^2 \int h'(\mu_s + \sigma_s z)\phi(z) \, \mathrm{d}z = \sigma_s^2 \mathbb{E}[h'(s_i)].$$

Also,

$$\int \mu_t \, \mathrm{d}t = \sigma_s^2.$$

Therefore,

$$\boxed{\beta = \mathbb{E}[h'(s_i)].}$$

When the regressor is normally distributed, the bivariate regression slope equals the unweighted average derivative of the CEF.

3.32 Propensity Score Theorem

Let

$$p(X_i) = \mathbb{P}(d_i = 1 \mid X_i) = \mathbb{E}[d_i \mid X_i].$$

The theorem says that if

$$\{y_{0i}, y_{1i}\} \perp d_i \mid X_i,$$

then

$$\{y_{0i}, y_{1i}\} \perp d_i \mid p(X_i).$$

Derivation

It is enough to show that

$$\mathbb{P}(d_i = 1 \mid y_{ji}, p(X_i))$$

does not depend on y_{ji}, for $j = 0, 1$. Start with

$$\mathbb{P}(d_i = 1 \mid y_{ji}, p(X_i)) = \mathbb{E}[d_i \mid y_{ji}, p(X_i)].$$

Use iterated expectations:

$$= \mathbb{E}\{\mathbb{E}[d_i \mid y_{ji}, p(X_i), X_i] \mid y_{ji}, p(X_i)\}.$$

Since $p(X_i)$ is a function of X_i, conditioning on both $p(X_i)$ and X_i is equivalent to conditioning on X_i:

$$= \mathbb{E}\{\mathbb{E}[d_i \mid y_{ji}, X_i] \mid y_{ji}, p(X_i)\}.$$

By conditional independence,

$$\mathbb{E}[d_i \mid y_{ji}, X_i] = \mathbb{E}[d_i \mid X_i].$$

Therefore,

$$= \mathbb{E}\{\mathbb{E}[d_i \mid X_i] \mid y_{ji}, p(X_i)\}.$$

But

$$\mathbb{E}[d_i \mid X_i] = p(X_i).$$

Thus,

$$= \mathbb{E}[p(X_i) \mid y_{ji}, p(X_i)] = p(X_i).$$

Therefore,

$$\boxed{\mathbb{P}(d_i = 1 \mid y_{ji}, p(X_i)) = p(X_i).}$$

Hence d_i is independent of potential outcomes conditional on the propensity score.

3.33 Inverse Probability Weighting for the ATE

Under conditional independence,

$$\mathbb{E}\left[\frac{y_i d_i}{p(X_i)}\right] = \mathbb{E}[y_{1i}].$$

Derivation

Using consistency,

$$d_i y_i = d_i y_{1i}.$$

Then

$$\mathbb{E}\left[\frac{d_i y_i}{p(X_i)}\right] = \mathbb{E}\left[\frac{d_i y_{1i}}{p(X_i)}\right].$$

Use iterated expectations:

$$= \mathbb{E}\left[\mathbb{E}\left[\frac{d_i y_{1i}}{p(X_i)} \mid X_i\right]\right].$$

Since $p(X_i)$ is fixed given X_i,

$$= \mathbb{E}\left[\frac{1}{p(X_i)}\mathbb{E}[d_i y_{1i} \mid X_i]\right].$$

By conditional independence,

$$\mathbb{E}[d_i y_{1i} \mid X_i] = \mathbb{E}[d_i \mid X_i]\mathbb{E}[y_{1i} \mid X_i].$$

Therefore,

$$= \mathbb{E}\left[\frac{1}{p(X_i)}p(X_i)\mathbb{E}[y_{1i} \mid X_i]\right] = \mathbb{E}[\mathbb{E}[y_{1i} \mid X_i]] = \mathbb{E}[y_{1i}].$$

Similarly,

$$\mathbb{E}\left[\frac{(1-d_i)y_i}{1-p(X_i)}\right] = \mathbb{E}[y_{0i}].$$

Thus,

$$\boxed{ATE = \mathbb{E}\left[\frac{d_i y_i}{p(X_i)} - \frac{(1-d_i)y_i}{1-p(X_i)}\right].}$$

This can also be written as

$$\boxed{ATE = \mathbb{E}\left[\frac{(d_i - p(X_i))y_i}{p(X_i)(1-p(X_i))}\right].}$$

3.34 Inverse Probability Weighting for Treatment on the Treated

The treatment effect on the treated is

$$TOT = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].$$

The corresponding inverse probability representation is

$$\boxed{TOT = \mathbb{E}\left[\frac{(d_i - p(X_i))y_i}{(1-p(X_i))\mathbb{P}(d_i = 1)}\right].}$$

Derivation

Take conditional expectation given X_i:

$$\mathbb{E}\left[\frac{(d_i - p)y_i}{1-p} \mid X_i\right] = \frac{1}{1-p}\mathbb{E}[(d_i - p)y_i \mid X_i].$$

Now

$$\mathbb{E}[(d_i - p)y_i \mid X_i] = \mathbb{E}[d_i y_i \mid X_i] - p\mathbb{E}[y_i \mid X_i].$$

Under conditional independence,

$$\mathbb{E}[d_i y_i \mid X_i] = p\mathbb{E}[y_{1i} \mid X_i],$$

and

$$\mathbb{E}[y_i \mid X_i] = p\mathbb{E}[y_{1i} \mid X_i] + (1-p)\mathbb{E}[y_{0i} \mid X_i].$$

Therefore,

$$\mathbb{E}[(d_i - p)y_i \mid X_i] = p\mathbb{E}[y_{1i} \mid X_i] - p[p\mathbb{E}[y_{1i} \mid X_i] + (1-p)\mathbb{E}[y_{0i} \mid X_i]].$$

Simplifying,

$$= p(1-p)\left[\mathbb{E}[y_{1i} \mid X_i] - \mathbb{E}[y_{0i} \mid X_i]\right].$$

Divide by $1 - p$:

$$= p\left[\mathbb{E}[y_{1i} \mid X_i] - \mathbb{E}[y_{0i} \mid X_i]\right].$$

Taking expectations over X_i,

$$\mathbb{E}\left[\frac{(d_i - p)y_i}{1 - p}\right] = \mathbb{E}[p(X_i)\delta(X_i)],$$

where

$$\delta(X_i) = \mathbb{E}[y_{1i} - y_{0i} \mid X_i].$$

Since

$$\mathbb{P}(d_i = 1) = \mathbb{E}[p(X_i)],$$

we have

$$\mathbb{E}[p(X_i)\delta(X_i)] = \mathbb{P}(d_i = 1)\mathbb{E}[\delta(X_i) \mid d_i = 1].$$

Therefore,

$$\boxed{TOT = \frac{\mathbb{E}\left[\frac{(d_i - p(X_i))y_i}{1 - p(X_i)}\right]}{\mathbb{P}(d_i = 1)}.}$$

3.35 Regression as a Propensity-Score Weighted Estimand

The saturated regression coefficient is

$$\delta_R = \frac{\mathbb{E}[(d_i - p(X_i))y_i]}{\mathbb{E}[p(X_i)(1 - p(X_i))]}.$$

This belongs to the weighted class

$$\mathbb{E}\left\{g(X_i)\left[\frac{d_i y_i}{p(X_i)} - \frac{(1-d_i)y_i}{1-p(X_i)}\right]\right\}.$$

For regression, choose

$$g(X_i) = \frac{p(X_i)(1-p(X_i))}{\mathbb{E}[p(X_i)(1-p(X_i))]}.$$

Then

$$g(X_i)\left[\frac{d_i y_i}{p(X_i)} - \frac{(1-d_i)y_i}{1-p(X_i)}\right] = \frac{p(1-p)}{\mathbb{E}[p(1-p)]}\left[\frac{d_i y_i}{p} - \frac{(1-d_i)y_i}{1-p}\right].$$

Simplifying,

$$= \frac{1}{\mathbb{E}[p(1-p)]}\left[(1-p)d_i y_i - p(1-d_i)y_i\right].$$

Expanding,

$$= \frac{1}{\mathbb{E}[p(1-p)]}\left[d_i y_i - p d_i y_i - p y_i + p d_i y_i\right].$$

Canceling terms,

$$= \frac{(d_i - p)y_i}{\mathbb{E}[p(1-p)]}.$$

Taking expectations gives

$$\boxed{\delta_R = \frac{\mathbb{E}[(d_i - p(X_i))y_i]}{\mathbb{E}[p(X_i)(1 - p(X_i))]}.}$$

3.36 Limited Dependent Variables: Binary Outcome

Suppose $y_i \in \{0, 1\}$, and treatment d_i is randomly assigned. Then

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0] = \mathbb{E}[y_{1i} - y_{0i}].$$

Because y_i is binary,

$$\mathbb{E}[y_i \mid d_i = 1] = \mathbb{P}(y_i = 1 \mid d_i = 1),$$

and

$$\mathbb{E}[y_i \mid d_i = 0] = \mathbb{P}(y_i = 1 \mid d_i = 0).$$

Thus,

$$\boxed{\mathbb{E}[y_{1i} - y_{0i}] = \mathbb{P}(y_i = 1 \mid d_i = 1) - \mathbb{P}(y_i = 1 \mid d_i = 0).}$$

For a binary outcome, the average causal effect is a difference in probabilities.

3.37 Probit with a Binary Treatment

Suppose the latent outcome is

$$y_i^* = \beta_0^* + \beta_1^* d_i - \nu_i,$$

and

$$y_i = \mathbf{1}[y_i^* > 0].$$

If

$$\nu_i \sim \mathcal{N}(0, \sigma^2),$$

then

$$\mathbb{P}(y_i = 1 \mid d_i) = \mathbb{P}(\nu_i < \beta_0^* + \beta_1^* d_i).$$

Standardizing,

$$= \Phi\left(\frac{\beta_0^* + \beta_1^* d_i}{\sigma}\right).$$

Thus,

$$\mathbb{E}[y_i \mid d_i] = \Phi\left(\frac{\beta_0^* + \beta_1^* d_i}{\sigma}\right).$$

Since d_i is binary,

$$\mathbb{E}[y_i \mid d_i] = \Phi\left(\frac{\beta_0^*}{\sigma}\right) + \left[\Phi\left(\frac{\beta_0^* + \beta_1^*}{\sigma}\right) - \Phi\left(\frac{\beta_0^*}{\sigma}\right)\right] d_i.$$

Therefore, the causal effect on probability is

$$\boxed{\Phi\left(\frac{\beta_0^* + \beta_1^*}{\sigma}\right) - \Phi\left(\frac{\beta_0^*}{\sigma}\right).}$$

For a binary treatment with no other covariates, OLS and Probit marginal effects recover the same conditional mean difference.

3.38 Nonnegative Outcome Decomposition

Suppose $y_i \geq 0$, such as medical expenditure. Then

$$\mathbb{E}[y_i \mid d_i] = \mathbb{E}[y_i \mid y_i > 0, d_i]\mathbb{P}(y_i > 0 \mid d_i).$$

The treatment-control difference is

$$\mathbb{E}[y_i \mid d_i = 1] - \mathbb{E}[y_i \mid d_i = 0].$$

Substituting,

$$= \mathbb{E}[y_i \mid y_i > 0, d_i = 1]\mathbb{P}(y_i > 0 \mid d_i = 1) - \mathbb{E}[y_i \mid y_i > 0, d_i = 0]\mathbb{P}(y_i > 0 \mid d_i = 0).$$

Add and subtract

$$\mathbb{E}[y_i \mid y_i > 0, d_i = 1]\mathbb{P}(y_i > 0 \mid d_i = 0).$$

Then

$$\begin{aligned} &= \mathbb{E}[y_i \mid y_i > 0, d_i = 1]\left[\mathbb{P}(y_i > 0 \mid d_i = 1) - \mathbb{P}(y_i > 0 \mid d_i = 0)\right] \\ &+\mathbb{P}(y_i > 0 \mid d_i = 0)\left[\mathbb{E}[y_i \mid y_i > 0, d_i = 1] - \mathbb{E}[y_i \mid y_i > 0, d_i = 0]\right]. \end{aligned}$$

Thus,

$$\boxed{\text{overall mean effect} = \text{participation effect} + \text{conditional-on-positive effect.}}$$

The second term generally does not have a causal interpretation because $y_i > 0$ is itself an outcome affected by treatment.

3.39 Conditional-on-Positive Selection Bias

The conditional-on-positive comparison is

$$\mathbb{E}[y_i \mid y_i > 0, d_i = 1] - \mathbb{E}[y_i \mid y_i > 0, d_i = 0].$$

Using potential outcomes,

$$= \mathbb{E}[y_{1i} \mid y_{1i} > 0] - \mathbb{E}[y_{0i} \mid y_{0i} > 0].$$

Add and subtract $\mathbb{E}[y_{0i} \mid y_{1i} > 0]$:

$$= \mathbb{E}[y_{1i} \mid y_{1i} > 0] - \mathbb{E}[y_{0i} \mid y_{1i} > 0] + \mathbb{E}[y_{0i} \mid y_{1i} > 0] - \mathbb{E}[y_{0i} \mid y_{0i} > 0].$$

Therefore,

$$\boxed{COP = \mathbb{E}[y_{1i} - y_{0i} \mid y_{1i} > 0] + \left[\mathbb{E}[y_{0i} \mid y_{1i} > 0] - \mathbb{E}[y_{0i} \mid y_{0i} > 0]\right].}$$

The second term is selection bias.

3.40 Tobit Expected Value Formula

Suppose

$$y_i = \max(0, y_i^*),$$

where

$$y_i^* = m_i + u_i, \qquad u_i \sim \mathcal{N}(0, \sigma^2).$$

Here

$$m_i = X_i'\beta.$$

Then

$$\mathbb{E}[y_i \mid X_i] = \mathbb{E}[y_i^* \mathbf{1}[y_i^* > 0] \mid X_i].$$

This is

$$\mathbb{E}[(m_i + u_i)\mathbf{1}[u_i > -m_i]].$$

Split the expression:

$$= m_i \mathbb{P}(u_i > -m_i) + \mathbb{E}[u_i \mathbf{1}[u_i > -m_i]].$$

Now

$$\mathbb{P}(u_i > -m_i) = \Phi\left(\frac{m_i}{\sigma}\right).$$

For a normal random variable,

$$\mathbb{E}[u_i \mathbf{1}[u_i > -m_i]] = \sigma\phi\left(\frac{m_i}{\sigma}\right).$$

Therefore,

$$\boxed{\mathbb{E}[y_i \mid X_i] = \Phi\left(\frac{m_i}{\sigma}\right) m_i + \sigma\phi\left(\frac{m_i}{\sigma}\right).}$$

If

$$m_i = X_i'\beta_0 + \beta_1 d_i,$$

then

$$\mathbb{E}[y_i \mid X_i, d_i] = \Phi\left(\frac{X_i'\beta_0 + \beta_1 d_i}{\sigma}\right)(X_i'\beta_0 + \beta_1 d_i) + \sigma\phi\left(\frac{X_i'\beta_0 + \beta_1 d_i}{\sigma}\right).$$

3.41 Tobit Marginal Effect

Let

$$m_i = X_i'\beta.$$

The Tobit conditional mean is

$$\mathbb{E}[y_i \mid X_i] = m_i \Phi\left(\frac{m_i}{\sigma}\right) + \sigma\phi\left(\frac{m_i}{\sigma}\right).$$

Let

$$z_i = \frac{m_i}{\sigma}.$$

Differentiate with respect to a covariate x_{ki}. Since

$$\frac{\partial m_i}{\partial x_{ki}} = \beta_k,$$

we compute

$$\frac{\partial}{\partial x_{ki}}\left[m_i\Phi(z_i) + \sigma\phi(z_i)\right].$$

For the first term,

$$\frac{\partial}{\partial x_{ki}}[m_i\Phi(z_i)] = \beta_k\Phi(z_i) + m_i\phi(z_i)\frac{\partial z_i}{\partial x_{ki}}.$$

Since

$$\frac{\partial z_i}{\partial x_{ki}} = \frac{\beta_k}{\sigma},$$

this becomes

$$\beta_k\Phi(z_i) + m_i\phi(z_i)\frac{\beta_k}{\sigma}.$$

For the second term,

$$\frac{\partial}{\partial x_{ki}}[\sigma\phi(z_i)] = \sigma[-z_i\phi(z_i)]\frac{\beta_k}{\sigma}.$$

Thus,

$$= -z_i\phi(z_i)\beta_k.$$

Because

$$z_i = \frac{m_i}{\sigma},$$

this is

$$-\frac{m_i}{\sigma}\phi(z_i)\beta_k.$$

This term cancels with

$$m_i\phi(z_i)\frac{\beta_k}{\sigma}.$$

Therefore,

$$\boxed{\frac{\partial\mathbb{E}[y_i \mid X_i]}{\partial x_{ki}} = \beta_k\Phi\left(\frac{m_i}{\sigma}\right).}$$

The average marginal effect is

$$\boxed{\mathbb{E}\left[\Phi\left(\frac{X_i'\beta}{\sigma}\right)\right]\beta_k.}$$

3.42 Probit Marginal Effect with Covariates

For Probit,

$$\mathbb{E}[y_i \mid X_i, d_i] = \Phi\left(\frac{X_i'\beta_0 + \beta_1 d_i}{\sigma}\right).$$

For a continuous regressor d_i, the derivative is

$$\frac{\partial \mathbb{E}[y_i \mid X_i, d_i]}{\partial d_i} = \phi\left(\frac{X_i'\beta_0 + \beta_1 d_i}{\sigma}\right)\frac{\beta_1}{\sigma}.$$

Thus, the average marginal effect is

$$\boxed{\mathbb{E}\left[\phi\left(\frac{X_i'\beta_0 + \beta_1 d_i}{\sigma}\right)\right]\frac{\beta_1}{\sigma}.}$$

For a binary treatment, the finite-difference marginal effect is

$$\boxed{\mathbb{E}\left[\Phi\left(\frac{X_i'\beta_0 + \beta_1}{\sigma}\right) - \Phi\left(\frac{X_i'\beta_0}{\sigma}\right)\right].}$$

3.43 Regression to the Mean

Let x_i be parent height and y_i be child height. The bivariate regression is

$$\mathbb{E}[y_i \mid x_i] = \alpha + \beta x_i.$$

The slope is

$$\beta = \frac{\mathrm{Cov}(y_i, x_i)}{\mathrm{Var}(x_i)}.$$

If height is stationary across generations, then

$$\mathbb{E}[y_i] = \mathbb{E}[x_i] = \mu,$$

and

$$\mathrm{Var}(y_i) = \mathrm{Var}(x_i).$$

Therefore,

$$\beta = \frac{\mathrm{Cov}(y_i, x_i)}{\sqrt{\mathrm{Var}(y_i)}\sqrt{\mathrm{Var}(x_i)}} = \rho_{xy},$$

where ρ_{xy} is the correlation coefficient. The intercept is

$$\alpha = \mathbb{E}[y_i] - \beta\mathbb{E}[x_i] = \mu - \rho_{xy}\mu = \mu(1 - \rho_{xy}).$$

Thus,

$$\mathbb{E}[y_i \mid x_i] = \mu(1 - \rho_{xy}) + \rho_{xy}x_i.$$

Rewrite:

$$\boxed{\mathbb{E}[y_i \mid x_i] = \mu + \rho_{xy}(x_i - \mu).}$$

If

$$0 < \rho_{xy} < 1,$$

then the predicted child height lies between parent height and the population mean. For tall parents,

$$x_i > \mu,$$

but

$$\mathbb{E}[y_i \mid x_i] - \mu = \rho_{xy}(x_i - \mu) < x_i - \mu.$$

Therefore, children of tall parents are expected to be tall, but less extremely tall than their parents. Similarly, children of short parents are expected to be short, but less extremely short. This is regression to the mean.

3.44 Brief Summary of Chapter 3

Chapter 3 explains why regression is central in applied econometrics. The main idea is that regression is not only a mechanical tool for fitting lines; it is a disciplined way to summarize the conditional expectation function,

$$\mathbb{E}[y_i \mid X_i].$$

The chapter begins from the CEF because the CEF provides the best mean-squared-error prediction of y_i given X_i. Linear regression is then interpreted as the best linear approximation to this CEF:

$$X_i'\beta \approx \mathbb{E}[y_i \mid X_i].$$

Thus, even when the true CEF is nonlinear, regression can still be useful as a compact summary of the relationship between outcomes and covariates.

The chapter then separates two distinct questions. The first is a statistical question:

What population object does regression estimate?

The second is a causal question:

When can the regression coefficient be interpreted causally?

The answer to the first question comes from the population regression formula,

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i y_i],$$

and from the regression-CEF theorem. The answer to the second question depends on the conditional independence assumption:

$$\{y_{0i}, y_{1i}\} \perp d_i \mid X_i.$$

When this assumption holds, conditional comparisons across treatment groups can be interpreted as causal effects.

A central warning of the chapter is that adding controls is not automatically good. The omitted variables bias formula,

$$\beta^{short} = \beta^{long} + \text{effect of omitted variable} \times \text{relationship between omitted and included variables},$$

explains why coefficients change when controls are added. But some controls are bad controls: variables affected by the treatment should not be controlled for if the goal is to estimate the total causal effect.

The chapter also shows that regression and matching are closely related. Matching estimates treatment effects by comparing treated and untreated observations with the same covariates. Regression does something similar, but it uses a particular weighting scheme. With a binary treatment d_i, saturated regression estimates a weighted average of covariate-specific treatment effects:

$$\delta_R = \frac{\mathbb{E}[p(X_i)(1 - p(X_i))\delta(X_i)]}{\mathbb{E}[p(X_i)(1 - p(X_i))]},$$

where

$$p(X_i) = \mathbb{P}(d_i = 1 \mid X_i)$$

is the propensity score. This shows that regression gives more weight to covariate cells where there is substantial variation in treatment status.

The propensity score theorem gives another way to reduce the dimension of the control problem. If treatment is conditionally independent of potential outcomes given X_i, then it is also conditionally independent given the scalar propensity score:

$$\{y_{0i}, y_{1i}\} \perp d_i \mid p(X_i).$$

This connects regression, matching, and inverse-probability weighting as related control strategies rather than completely separate methods.

Finally, the chapter discusses limited dependent variables. The main message is practical: linear regression often remains useful even when the outcome is binary, censored, or nonnegative. For binary outcomes, the linear probability model estimates changes in probabilities. For nonnegative outcomes, the chapter warns against conditioning on positive outcomes because this creates a bad-control problem. Nonlinear models such as Probit and Tobit can be useful for curve-fitting, but their coefficients must be converted into marginal effects, and these marginal effects are often close to OLS estimates in practice.

Overall, Chapter 3 teaches that regression should be understood as a tool for approximating conditional expectations and summarizing causal comparisons when the identifying assumptions are credible. The main conceptual chain is:

$$\begin{gathered}\text{CEF} \longrightarrow \text{linear approximation} \longrightarrow \text{regression coefficient} \\ \longrightarrow \text{causal interpretation only under identifying assumptions.}\end{gathered}$$

4 Chapter 4 Derivations: Instrumental Variables in Action

4.1 Constant-Effects IV Setup

Suppose the potential outcome function is

$$y_{si} \equiv f_i(s),$$

and assume the constant-effects causal model

$$f_i(s) = \pi_0 + \pi_1 s + \eta_i.$$

When individual i's observed schooling is s_i, the observed outcome is

$$y_i = f_i(s_i) = \pi_0 + \pi_1 s_i + \eta_i.$$

In the chapter's notation, write

$$y_i = \alpha + \rho s_i + \eta_i.$$

The problem is that

$$\text{Cov}(s_i, \eta_i) \neq 0,$$

so OLS does not identify ρ.

Suppose we have an instrument z_i such that

$$\text{Cov}(z_i, \eta_i) = 0$$

and

$$\text{Cov}(z_i, s_i) \neq 0.$$

The first condition is the exclusion or exogeneity condition. The second is the first-stage relevance condition.

4.2 Deriving the Basic IV Covariance Ratio

Start from the structural equation

$$y_i = \alpha + \rho s_i + \eta_i.$$

Take covariance with z_i:

$$\text{Cov}(y_i, z_i) = \text{Cov}(\alpha + \rho s_i + \eta_i, z_i).$$

The covariance of a constant with z_i is zero, so

$$\text{Cov}(y_i, z_i) = \rho\,\text{Cov}(s_i, z_i) + \text{Cov}(\eta_i, z_i).$$

By the exclusion condition,

$$\text{Cov}(\eta_i, z_i) = 0.$$

Therefore,

$$\text{Cov}(y_i, z_i) = \rho\,\text{Cov}(s_i, z_i).$$

If

$$\mathrm{Cov}(s_i, z_i) \neq 0,$$

then

$$\boxed{\rho = \frac{\mathrm{Cov}(y_i, z_i)}{\mathrm{Cov}(s_i, z_i)}.}$$

Equivalently,

$$\boxed{\rho = \frac{\mathrm{Cov}(y_i, z_i) / \mathrm{Var}(z_i)}{\mathrm{Cov}(s_i, z_i) / \mathrm{Var}(z_i)}.}$$

The numerator is the reduced-form regression coefficient of y_i on z_i. The denominator is the first-stage regression coefficient of s_i on z_i.

4.3 *First Stage and Reduced Form*

The first-stage equation is

$$s_i = X_i' \pi_{10} + \pi_{11} z_i + \xi_{1i}.$$

The reduced-form equation is

$$y_i = X_i' \pi_{20} + \pi_{21} z_i + \xi_{2i}.$$

The first-stage coefficient is π_{11}, and the reduced-form coefficient is π_{21}.

If the structural model is

$$y_i = X_i' \alpha + \rho s_i + \eta_i,$$

then IV identifies

$$\boxed{\rho = \frac{\pi_{21}}{\pi_{11}}.}$$

4.4 *Deriving the Covariate-Adjusted IV Ratio*

Let $\widetilde{z}_i$ be the residual from the population regression of z_i on X_i. Therefore,

$$\mathbb{E}[X_i \widetilde{z}_i] = 0.$$

The structural equation is

$$y_i = X_i' \alpha + \rho s_i + \eta_i.$$

Take covariance with $\widetilde{z}_i$:

$$\mathrm{Cov}(y_i, \widetilde{z}_i) = \mathrm{Cov}(X_i' \alpha + \rho s_i + \eta_i, \widetilde{z}_i).$$

Using linearity,

$$= \operatorname{Cov}(X_i'\alpha, \widetilde{z}_i) + \rho \operatorname{Cov}(s_i, \widetilde{z}_i) + \operatorname{Cov}(\eta_i, \widetilde{z}_i).$$

Because $\widetilde{z}_i$ is orthogonal to X_i,

$$\operatorname{Cov}(X_i'\alpha, \widetilde{z}_i) = 0.$$

By the IV exogeneity condition,

$$\operatorname{Cov}(\eta_i, \widetilde{z}_i) = 0.$$

Hence

$$\operatorname{Cov}(y_i, \widetilde{z}_i) = \rho \operatorname{Cov}(s_i, \widetilde{z}_i).$$

Therefore,

$$\boxed{\rho = \frac{\operatorname{Cov}(y_i, \widetilde{z}_i)}{\operatorname{Cov}(s_i, \widetilde{z}_i)}.}$$

This is the covariate-adjusted IV estimand.

4.5 Deriving the Reduced Form from the Structural Equation and First Stage

The structural equation is

$$y_i = X_i'\alpha + \rho s_i + \eta_i.$$

The first stage is

$$s_i = X_i'\pi_{10} + \pi_{11} z_i + \xi_{1i}.$$

Substitute the first stage into the structural equation:

$$y_i = X_i'\alpha + \rho(X_i'\pi_{10} + \pi_{11} z_i + \xi_{1i}) + \eta_i.$$

Expand:

$$y_i = X_i'\alpha + \rho X_i'\pi_{10} + \rho\pi_{11} z_i + \rho\xi_{1i} + \eta_i.$$

Collect terms:

$$y_i = X_i'(\alpha + \rho\pi_{10}) + (\rho\pi_{11}) z_i + (\rho\xi_{1i} + \eta_i).$$

Define

$$\pi_{20} = \alpha + \rho\pi_{10},$$

$$\pi_{21} = \rho\pi_{11},$$

and

$$\xi_{2i} = \rho\xi_{1i} + \eta_i.$$

Then the reduced form is

$$\boxed{y_i = X_i'\pi_{20} + \pi_{21}z_i + \xi_{2i}.}$$

Since

$$\pi_{21} = \rho\pi_{11},$$

we obtain

$$\boxed{\rho = \frac{\pi_{21}}{\pi_{11}}.}$$

4.6 Population Fitted-Value Representation of 2SLS

The first stage is

$$s_i = X_i'\pi_{10} + \pi_{11}z_i + \xi_{1i}.$$

The population fitted value is

$$s_i^* = X_i'\pi_{10} + \pi_{11}z_i.$$

The structural equation is

$$y_i = X_i'\alpha + \rho s_i + \eta_i.$$

Since

$$s_i = s_i^* + \xi_{1i},$$

we have

$$y_i = X_i'\alpha + \rho(s_i^* + \xi_{1i}) + \eta_i.$$

Thus

$$y_i = X_i'\alpha + \rho s_i^* + (\rho\xi_{1i} + \eta_i).$$

This gives the population second-stage equation

$$\boxed{y_i = X_i'\alpha + \rho s_i^* + \xi_{2i},}$$

where

$$\xi_{2i} = \rho\xi_{1i} + \eta_i.$$

Because s_i^* is a function of X_i and z_i, and both X_i and z_i are exogenous, s_i^* is uncorrelated with ξ_{2i}. Therefore, regressing y_i on X_i and s_i^* identifies ρ.

4.7 Sample 2SLS Estimator

In practice, the population first-stage fitted value s_i^* is unknown. Estimate the first stage by OLS:

$$s_i = X_i'\pi_{10} + \pi_{11}z_i + \xi_{1i}.$$

Let the fitted value be

$$\widehat{s}_i = X_i'\widehat{\pi}_{10} + \widehat{\pi}_{11}z_i.$$

The manual second stage is

$$y_i = X_i'\alpha + \rho\widehat{s}_i + \left[\eta_i + \rho(s_i - \widehat{s}_i)\right].$$

Thus, 2SLS can be computed in two steps:

$$\boxed{\text{First stage: regress } s_i \text{ on } X_i \text{ and } z_i.}$$

$$\boxed{\text{Second stage: regress } y_i \text{ on } X_i \text{ and } \widehat{s}_i.}$$

The coefficient on $\widehat{s}_i$ is the 2SLS estimator of ρ.

4.8 Why 2SLS is an IV Estimator

Let

$$\widehat{s}_i$$

be the fitted value from the first stage, and let

$$\widehat{s}_i^{\perp}$$

be the residual from regressing $\widehat{s}_i$ on X_i.

By regression anatomy, the coefficient on $\widehat{s}_i$ in the second stage equals

$$\widehat{\rho}_{2SLS} = \frac{\widehat{\mathrm{Cov}}(y_i, \widehat{s}_i^{\perp})}{\widehat{\mathrm{Cov}}(s_i, \widehat{s}_i^{\perp})}.$$

Since $\widehat{s}_i$ is generated by the instruments and covariates, the residual $\widehat{s}_i^{\perp}$ is the part of the endogenous variable predicted by the excluded instrument after removing covariates. Thus, 2SLS uses only the variation in s_i generated by the instrument.

4.9 Single-Instrument Equivalence of 2SLS and ILS

With one instrument z_i, after partialling out covariates X_i, we have

$$\widehat{s}_i^{\perp} = \widehat{\pi}_{11}\widetilde{z}_i,$$

where $\widetilde{z}_i$ is the residual from regressing z_i on X_i.

Then

$$\widehat{\rho}_{2SLS} = \frac{\widehat{\text{Cov}}(y_i, \widehat{s}_i^{\perp})}{\widehat{\text{Cov}}(s_i, \widehat{s}_i^{\perp})}.$$

Substitute

$$\widehat{s}_i^{\perp} = \widehat{\pi}_{11}\widetilde{z}_i :$$

$$\widehat{\rho}_{2SLS} = \frac{\widehat{\text{Cov}}(y_i, \widehat{\pi}_{11}\widetilde{z}_i)}{\widehat{\text{Cov}}(s_i, \widehat{\pi}_{11}\widetilde{z}_i)}.$$

Cancel $\widehat{\pi}_{11}$:

$$\widehat{\rho}_{2SLS} = \frac{\widehat{\text{Cov}}(y_i, \widetilde{z}_i)}{\widehat{\text{Cov}}(s_i, \widetilde{z}_i)}.$$

This equals the ratio of the reduced-form coefficient to the first-stage coefficient:

$$\boxed{\widehat{\rho}_{2SLS} = \frac{\widehat{\pi}_{21}}{\widehat{\pi}_{11}}.}$$

Thus, with one endogenous variable and one instrument, 2SLS equals indirect least squares.

4.10 Multiple Instruments and First-Stage Fitted Values

Suppose there are three excluded instruments:

$$z_{1i}, \quad z_{2i}, \quad z_{3i}.$$

The first stage is

$$s_i = X_i'\pi_{10} + \pi_{11}z_{1i} + \pi_{12}z_{2i} + \pi_{13}z_{3i} + \xi_{1i}.$$

The fitted value is

$$\widehat{s}_i = X_i'\widehat{\pi}_{10} + \widehat{\pi}_{11}z_{1i} + \widehat{\pi}_{12}z_{2i} + \widehat{\pi}_{13}z_{3i}.$$

The second stage is

$$y_i = X_i'\alpha + \rho\widehat{s}_i + \text{second-stage error}.$$

Thus, 2SLS combines multiple instruments into a single fitted value. This fitted value is the best linear prediction of s_i from the instruments and covariates.

4.11 Wald Estimator with a Binary Instrument

Suppose $z_i \in \{0, 1\}$, with

$$\mathbb{P}(z_i = 1) = p.$$

The no-covariate causal model is

$$y_i = \alpha + \rho s_i + \eta_i.$$

Assume

$$\mathbb{E}[\eta_i \mid z_i] = 0.$$

Then

$$\mathbb{E}[y_i \mid z_i] = \alpha + \rho \mathbb{E}[s_i \mid z_i].$$

Evaluate at $z_i = 1$:

$$\mathbb{E}[y_i \mid z_i = 1] = \alpha + \rho \mathbb{E}[s_i \mid z_i = 1].$$

Evaluate at $z_i = 0$:

$$\mathbb{E}[y_i \mid z_i = 0] = \alpha + \rho \mathbb{E}[s_i \mid z_i = 0].$$

Subtract:

$$\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] = \rho \left(\mathbb{E}[s_i \mid z_i = 1] - \mathbb{E}[s_i \mid z_i = 0] \right).$$

If

$$\mathbb{E}[s_i \mid z_i = 1] - \mathbb{E}[s_i \mid z_i = 0] \neq 0,$$

then

$$\boxed{\rho = \frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[s_i \mid z_i = 1] - \mathbb{E}[s_i \mid z_i = 0]}.}$$

This is the Wald estimand.

4.12 Covariance Formula for a Binary Instrument

For binary z_i, with $\mathbb{P}(z_i = 1) = p$, show that

$$\mathrm{Cov}(y_i, z_i) = \left[\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] \right] p(1 - p).$$

Derivation

By definition,

$$\mathrm{Cov}(y_i, z_i) = \mathbb{E}[y_i z_i] - \mathbb{E}[y_i]\mathbb{E}[z_i].$$

Since z_i is binary,

$$\mathbb{E}[z_i] = p.$$

Also,

$$\mathbb{E}[y_i z_i] = \mathbb{E}[y_i \mid z_i = 1]\mathbb{P}(z_i = 1) = p\mathbb{E}[y_i \mid z_i = 1].$$

The unconditional mean is

$$\mathbb{E}[y_i] = p\mathbb{E}[y_i \mid z_i = 1] + (1 - p)\mathbb{E}[y_i \mid z_i = 0].$$

Therefore,

$$\mathrm{Cov}(y_i, z_i) = p\mathbb{E}[y_i \mid z_i = 1] - p\left[p\mathbb{E}[y_i \mid z_i = 1] + (1 - p)\mathbb{E}[y_i \mid z_i = 0]\right].$$

Expand:

$$= p(1 - p)\mathbb{E}[y_i \mid z_i = 1] - p(1 - p)\mathbb{E}[y_i \mid z_i = 0].$$

Thus,

$$\boxed{\mathrm{Cov}(y_i, z_i) = p(1 - p)\left[\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]\right].}$$

The same logic gives

$$\mathrm{Cov}(s_i, z_i) = p(1 - p)\left[\mathbb{E}[s_i \mid z_i = 1] - \mathbb{E}[s_i \mid z_i = 0]\right].$$

Therefore,

$$\frac{\mathrm{Cov}(y_i, z_i)}{\mathrm{Cov}(s_i, z_i)} = \frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[s_i \mid z_i = 1] - \mathbb{E}[s_i \mid z_i = 0]}.$$

4.13 Grouped Data and 2SLS

Suppose the instrument r_i takes values

$$j = 1, \dots, J.$$

Let

$$\bar{y}_j = \mathbb{E}[y_i \mid r_i = j],$$

and

$$\bar{d}_j = \mathbb{E}[d_i \mid r_i = j].$$

In the constant-effects model

$$y_i = \alpha + \rho d_i + \eta_i,$$

with

$$\mathbb{E}[\eta_i \mid r_i] = 0,$$

take conditional expectation given $r_i = j$:

$$\mathbb{E}[y_i \mid r_i = j] = \alpha + \rho \mathbb{E}[d_i \mid r_i = j].$$

Therefore,

$$\boxed{\bar{y}_j = \alpha + \rho \bar{d}_j.}$$

Thus, the slope from a regression of $\bar{y}_j$ on $\bar{d}_j$ identifies ρ.

4.14 Why Grouped-Data IV Equals 2SLS with Dummy Instruments

Let the instruments be a full set of group dummies:

$$Z_i = \{\mathbf{1}[r_i = 1], \ldots, \mathbf{1}[r_i = J-1]\}.$$

The first stage regresses d_i on these group dummies and a constant. Because the model is saturated in the instrument groups, the fitted value for every observation in group j is

$$\widehat{d_i} = \bar{d}_j.$$

The second stage regresses y_i on $\widehat{d_i}$. Since $\widehat{d_i}$ is constant within group, this is equivalent to a weighted regression of group means:

$$\bar{y}_j = \alpha + \rho \bar{d}_j + \bar{\eta}_j,$$

weighted by group size n_j. Hence,

$$\boxed{\text{2SLS with group-dummy instruments equals weighted grouped-data IV.}}$$

4.15 2SLS Limiting Distribution

Let

$$V_i = \begin{bmatrix} X_i \\ s_i^* \end{bmatrix},$$

where s_i^* is the population first-stage fitted value. The second-stage population equation is

$$y_i = V_i'\theta + \eta_i,$$

where

$$\theta = \begin{bmatrix} \alpha \\ \rho \end{bmatrix}.$$

The 2SLS estimator can be written approximately as

$$\widehat{\theta}_{2SLS} = \left(\sum_i V_i V_i' \right)^{-1} \sum_i V_i y_i.$$

Substitute

$$y_i = V_i'\theta + \eta_i :$$

$$\widehat{\theta}_{2SLS} = \left(\sum_i V_i V_i' \right)^{-1} \sum_i V_i (V_i'\theta + \eta_i).$$

Expand:

$$= \theta + \left(\sum_i V_i V_i' \right)^{-1} \sum_i V_i \eta_i.$$

Therefore,

$$\widehat{\theta}_{2SLS} - \theta = \left(\sum_i V_i V_i' \right)^{-1} \sum_i V_i \eta_i.$$

Multiply by $\sqrt{N}$:

$$\sqrt{N}(\widehat{\theta}_{2SLS} - \theta) = \left(\frac{1}{N} \sum_i V_i V_i' \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_i V_i \eta_i \right).$$

By the law of large numbers,

$$\frac{1}{N} \sum_i V_i V_i' \xrightarrow{p} \mathbb{E}[V_i V_i'].$$

By the central limit theorem,

$$\frac{1}{\sqrt{N}}\sum_i V_i\eta_i \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[V_iV_i'\eta_i^2]).$$

Therefore,

$$\boxed{\sqrt{N}(\widehat{\theta}_{2SLS}-\theta) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[V_iV_i']^{-1}\mathbb{E}[V_iV_i'\eta_i^2]\mathbb{E}[V_iV_i']^{-1}\right).}$$

4.16 Robust 2SLS Covariance Matrix

The asymptotic covariance matrix of 2SLS is

$$\mathbb{E}[V_iV_i']^{-1}\mathbb{E}[V_iV_i'\eta_i^2]\mathbb{E}[V_iV_i']^{-1}.$$

The sample analog is

$$\boxed{\widehat{V}_{robust}(\widehat{\theta}_{2SLS}) = \left(\sum_i \widehat{V}_i\widehat{V}_i'\right)^{-1}\left(\sum_i \widehat{V}_i\widehat{V}_i'\widehat{\eta}_i^2\right)\left(\sum_i \widehat{V}_i\widehat{V}_i'\right)^{-1}.}$$

Here

$$\widehat{V}_i = \begin{bmatrix} X_i \\ \widehat{s}_i \end{bmatrix}.$$

The correct 2SLS residual is

$$\widehat{\eta}_i = y_i - X_i'\widehat{\alpha} - \widehat{\rho}s_i,$$

not

$$y_i - X_i'\widehat{\alpha} - \widehat{\rho}\widehat{s}_i.$$

This is why manually running the second stage by OLS gives incorrect standard errors unless the residual variance is corrected.

4.17 Homoskedastic 2SLS Covariance Matrix

If

$$\mathbb{E}[\eta_i^2 \mid X_i, z_i] = \sigma_\eta^2,$$

then

$$\mathbb{E}[V_iV_i'\eta_i^2] = \mathbb{E}[V_iV_i'\mathbb{E}[\eta_i^2 \mid X_i, z_i]] = \sigma_\eta^2\mathbb{E}[V_iV_i'].$$

Therefore,

$$\mathbb{E}[V_iV_i']^{-1}\mathbb{E}[V_iV_i'\eta_i^2]\mathbb{E}[V_iV_i']^{-1} = \sigma_\eta^2\mathbb{E}[V_iV_i']^{-1}.$$

Thus,

$$\boxed{V_{homoskedastic}(\widehat{\theta}_{2SLS}) = \sigma_\eta^2 \mathbb{E}[V_i V_i']^{-1}.}$$

4.18 GMM Moment Condition for IV

Let

$$Z_i = \begin{bmatrix} X_i \\ z_{1i} \\ \vdots \\ z_{qi} \end{bmatrix}, \qquad W_i = \begin{bmatrix} X_i \\ s_i \end{bmatrix}.$$

The structural equation is

$$y_i = W_i'\theta + \eta_i,$$

where

$$\eta_i(\theta) = y_i - W_i'\theta.$$

The IV moment condition is

$$\boxed{\mathbb{E}[Z_i \eta_i(\theta)] = 0.}$$

The sample analog is

$$m_N(\theta) = \frac{1}{N} \sum_i Z_i \eta_i(\theta).$$

If there are more instruments than endogenous regressors, this vector has more moments than unknown parameters. We choose θ to make the sample moment vector as close to zero as possible.

4.19 GMM Minimand for IV

By the central limit theorem,

$$\sqrt{N} m_N(\theta)$$

has asymptotic covariance matrix

$$\Omega = \mathbb{E}[Z_i Z_i' \eta_i^2].$$

The efficient GMM minimand is

$$\boxed{J_N(\theta) = N m_N(\theta)' \Omega^{-1} m_N(\theta).}$$

The GMM estimator solves

$$\boxed{\widehat{\theta}_{GMM} = \arg\min_\theta J_N(\theta).}$$

4.20 Showing Homoskedastic GMM Gives 2SLS

Under homoskedasticity,

$$\Omega = \mathbb{E}[Z_i Z_i' \eta_i^2] = \sigma_\eta^2 \mathbb{E}[Z_i Z_i'].$$

Thus,

$$\Omega^{-1} = \frac{1}{\sigma_\eta^2} \mathbb{E}[Z_i Z_i']^{-1}.$$

Using matrices, the sample moment is

$$m_N(\theta) = \frac{1}{N} Z'(y - W\theta).$$

The minimand becomes proportional to

$$(y - W\theta)' Z (Z'Z)^{-1} Z'(y - W\theta).$$

Define the projection matrix

$$P_Z = Z(Z'Z)^{-1} Z'.$$

Then the minimand is

$$(y - W\theta)' P_Z (y - W\theta).$$

Differentiate with respect to θ:

$$-2W' P_Z (y - W\theta) = 0.$$

Therefore,

$$W' P_Z y = W' P_Z W\theta.$$

If $W' P_Z W$ is invertible,

$$\boxed{\widehat{\theta}_{2SLS} = (W' P_Z W)^{-1} W' P_Z y.}$$

Hence, under homoskedasticity, efficient GMM based on the IV moment condition gives 2SLS.

4.21 Grouped-Data 2SLS Minimand

Suppose instruments are mutually exclusive group dummies. Let group j have size n_j, group mean outcome $\bar{y}_j$, and group mean regressor vector $\bar{W}_j$. The 2SLS minimand becomes

$$\boxed{\widehat{J}_N(\theta) = \frac{1}{\sigma_\eta^2} \sum_j n_j \left(\bar{y}_j - \theta' \bar{W}_j\right)^2.}$$

Without homoskedasticity, the efficient grouped-data minimand is

$$\boxed{\widehat{J}_N(\theta) = \sum_j \frac{n_j}{\sigma_j^2} \left(\bar{y}_j - \theta' \bar{W}_j\right)^2,}$$

where

$$\sigma_j^2 = \mathrm{Var}(\eta_i \mid \text{group } j).$$

Thus, overidentification with dummy instruments can be interpreted as the goodness-of-fit of the grouped-data IV line.

4.22 Overidentification Test Statistic

The minimized GMM criterion is

$$J_N(\widehat{\theta}).$$

Under the null hypothesis that all instruments are valid,

$$\mathbb{E}[Z_i \eta_i] = 0,$$

the minimized criterion has an asymptotic chi-square distribution:

$$\boxed{J_N(\widehat{\theta}) \xrightarrow{d} \chi^2_{q-r},}$$

where q is the number of moment conditions and r is the number of parameters estimated.

In the one-endogenous-variable case with q excluded instruments, the degrees of freedom are

$$q - 1.$$

The test asks whether the instruments tell a mutually consistent story about the same structural parameter.

4.23 Two-Sample IV Moment Equation

The IV moment equation is

$$\frac{Z'y}{N} = \frac{Z'W}{N}\theta + \frac{Z'\eta}{N}.$$

Two-sample IV uses one sample to estimate the reduced-form moments and another sample to estimate the first-stage moments.

Let sample 1 contain Z_1, y_1, and let sample 2 contain Z_2, W_2. Then the two-sample moment equation is

$$\boxed{\frac{Z_1' y_1}{N_1} = \frac{Z_2' W_2}{N_2}\theta + \text{sampling error.}}$$

If both samples are drawn from the same population, then

$$\operatorname{plim} \frac{Z_1' W_1}{N_1} = \operatorname{plim} \frac{Z_2' W_2}{N_2}.$$

Thus, using sample 2 for the first-stage moment matrix remains consistent.

4.24 Split-Sample IV

The first-stage estimate in sample 2 is

$$\widehat{\Pi}_2 = (Z_2' Z_2)^{-1} Z_2' W_2.$$

Carry these first-stage coefficients to sample 1:

$$\widehat{W}_{12} = Z_1 \widehat{\Pi}_2 = Z_1 (Z_2' Z_2)^{-1} Z_2' W_2.$$

Then regress y_1 on $\widehat{W}_{12}$. This gives the split-sample IV estimator.

The key idea is that the first-stage fitted values used in sample 1 are estimated from a separate sample, reducing mechanical overfitting between the endogenous variable and the second-stage error.

4.25 Potential Outcomes with an Instrument

Let treatment be binary:

$$d_i \in \{0, 1\}.$$

Let the instrument be binary:

$$z_i \in \{0, 1\}.$$

Potential outcomes are indexed by both treatment and instrument:

$$y_i(d, z).$$

Potential treatment assignments are

$$d_{1i} = \text{treatment if } z_i = 1,$$

and

$$d_{0i} = \text{treatment if } z_i = 0.$$

Observed treatment is

$$\boxed{d_i = d_{0i} + (d_{1i} - d_{0i})z_i.}$$

Observed outcome is

$$y_i = y_i(d_i, z_i).$$

4.26 Independence Implies Causal Reduced Form and First Stage

Assume instrument independence:

$$\{y_i(d,z) \text{ for all } d, z;\ d_{1i}, d_{0i}\} \perp z_i.$$

Then

$$\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[y_i(d_{1i}, 1)] - \mathbb{E}[y_i(d_{0i}, 0)].$$

Thus,

$$\boxed{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[y_i(d_{1i}, 1) - y_i(d_{0i}, 0)].}$$

This is the causal effect of the instrument on the outcome.

Similarly,

$$\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0] = \mathbb{E}[d_{1i}] - \mathbb{E}[d_{0i}] = \boxed{\mathbb{E}[d_{1i} - d_{0i}].}$$

So the first stage is the average causal effect of the instrument on treatment.

4.27 Exclusion Restriction and Observed Outcomes

The exclusion restriction says that the instrument affects the outcome only through treatment:

$$y_i(d, 0) = y_i(d, 1) \equiv y_{di}, \qquad d = 0, 1.$$

Therefore,

$$y_{1i} = y_i(1, 1) = y_i(1, 0),$$

and

$$y_{0i} = y_i(0, 1) = y_i(0, 0).$$

The observed outcome can be written as

$$\boxed{y_i = y_{0i} + (y_{1i} - y_{0i})d_i.}$$

Let

$$\rho_i = y_{1i} - y_{0i}.$$

Then

$$\boxed{y_i = y_{0i} + \rho_i d_i.}$$

4.28 LATE Theorem

Assume:

$$\text{Independence: } \{y_{1i}, y_{0i}, d_{1i}, d_{0i}\} \perp z_i.$$

$$\text{Exclusion: } y_i(d, 0) = y_i(d, 1).$$

$$\text{First stage: } \mathbb{E}[d_{1i} - d_{0i}] \neq 0.$$

$$\text{Monotonicity: } d_{1i} \geq d_{0i} \quad \text{for all } i.$$

Then the Wald estimand identifies the local average treatment effect:

$$\boxed{\frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]} = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}].}$$

Proof

Using exclusion,

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i.$$

When $z_i = 1$, observed treatment is d_{1i}. Hence,

$$\mathbb{E}[y_i \mid z_i = 1] = \mathbb{E}[y_{0i} + (y_{1i} - y_{0i})d_{1i} \mid z_i = 1].$$

By independence,

$$= \mathbb{E}[y_{0i} + (y_{1i} - y_{0i})d_{1i}].$$

Similarly,

$$\mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[y_{0i} + (y_{1i} - y_{0i})d_{0i}].$$

Subtract:

$$\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[(y_{1i} - y_{0i})(d_{1i} - d_{0i})].$$

Under monotonicity,

$$d_{1i} - d_{0i} \in \{0, 1\}.$$

Therefore,

$$\mathbb{E}[(y_{1i} - y_{0i})(d_{1i} - d_{0i})] = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}]\mathbb{P}(d_{1i} > d_{0i}).$$

The denominator is

$$\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0] = \mathbb{E}[d_{1i} - d_{0i}] = \mathbb{P}(d_{1i} > d_{0i}).$$

Taking the ratio gives

$$\boxed{\frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]} = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}].}$$

4.29 Why Monotonicity Matters

Without monotonicity, some individuals may be defiers:

$$d_{1i} < d_{0i}.$$

The reduced form is

$$\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[(y_{1i} - y_{0i})(d_{1i} - d_{0i})].$$

If both compliers and defiers exist, then

$$\mathbb{E}[(y_{1i} - y_{0i})(d_{1i} - d_{0i})]$$

equals

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}]\mathbb{P}(d_{1i} > d_{0i}) - \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} < d_{0i}]\mathbb{P}(d_{1i} < d_{0i}).$$

Thus,

$$\boxed{\text{complier effects and defier effects can cancel in the reduced form.}}$$

Monotonicity rules out this cancellation and ensures that IV estimates a weighted average of treatment effects with nonnegative weights.

4.30 Latent-Index Model and Monotonicity

Suppose treatment is determined by

$$d_i = \begin{cases} 1, & \text{if } \gamma_0 + \gamma_1 z_i > v_i, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$d_{0i} = \mathbf{1}[\gamma_0 > v_i],$$

and

$$d_{1i} = \mathbf{1}[\gamma_0 + \gamma_1 > v_i].$$

If

$$\gamma_1 > 0,$$

then

$$d_{1i} \geq d_{0i}$$

for every i. Therefore, monotonicity is automatically satisfied.

The complier group is

$$d_{1i} > d_{0i} \iff \gamma_0 + \gamma_1 > v_i > \gamma_0.$$

Thus,

$$\boxed{LATE = \mathbb{E}[y_{1i} - y_{0i} \mid \gamma_0 + \gamma_1 > v_i > \gamma_0].}$$

4.31 Compliers, Always-Takers, and Never-Takers

Under monotonicity, the population can be partitioned into three groups:

$$\text{Compliers: } d_{1i} = 1, \quad d_{0i} = 0.$$

$$\text{Always-takers: } d_{1i} = 1, \quad d_{0i} = 1.$$

$$\text{Never-takers: } d_{1i} = 0, \quad d_{0i} = 0.$$

The LATE estimand identifies the average treatment effect for compliers:

$$\boxed{LATE = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}].}$$

4.32 *Effect on the Treated as a Weighted Average*

The treated population consists of always-takers and compliers whose instrument is switched on.

The effect on the treated is

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].$$

Decompose by groups:

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1] = \mathbb{E}[y_{1i} - y_{0i} \mid d_{0i} = 1]\mathbb{P}(d_{0i} = 1 \mid d_i = 1)$$

$$+\mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}, z_i = 1]\mathbb{P}(d_{1i} > d_{0i}, z_i = 1 \mid d_i = 1).$$

By independence,

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}, z_i = 1] = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}].$$

Therefore,

$$\boxed{\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1] = ATT_{AT} \cdot w_{AT} + LATE \cdot w_C.}$$

Thus, the average treatment effect on the treated is generally not equal to LATE unless all treated units are compliers.

4.33 *Effect on the Non-Treated as a Weighted Average*

The non-treated population consists of never-takers and compliers whose instrument is switched off.

The effect on the non-treated is

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 0].$$

Decompose by groups:

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 0] = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} = 0]\mathbb{P}(d_{1i} = 0 \mid d_i = 0)$$

$$+\mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}, z_i = 0]\mathbb{P}(d_{1i} > d_{0i}, z_i = 0 \mid d_i = 0).$$

By independence,

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}, z_i = 0] = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}].$$

Therefore,

$$\boxed{\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 0] = ATU_{NT} \cdot w_{NT} + LATE \cdot w_C.}$$

4.34 Bloom Result for One-Sided Noncompliance

Suppose z_i is random assignment to treatment, d_i is actual treatment, and there is one-sided noncompliance:

$$\mathbb{E}[d_i \mid z_i = 0] = 0.$$

Then there are no always-takers. Under the LATE assumptions,

$$\boxed{\frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[d_i \mid z_i = 1]} = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].}$$

Proof

Since

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i,$$

we have

$$\mathbb{E}[y_i \mid z_i = 1] = \mathbb{E}[y_{0i} + (y_{1i} - y_{0i})d_i \mid z_i = 1].$$

Also, since $\mathbb{E}[d_i \mid z_i = 0] = 0$,

$$\mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[y_{0i} \mid z_i = 0].$$

By independence,

$$\mathbb{E}[y_{0i} \mid z_i = 1] = \mathbb{E}[y_{0i} \mid z_i = 0].$$

Therefore,

$$\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0] = \mathbb{E}[(y_{1i} - y_{0i})d_i \mid z_i = 1].$$

This equals

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1, z_i = 1]\mathbb{P}(d_i = 1 \mid z_i = 1).$$

With one-sided noncompliance, $d_i = 1$ implies $z_i = 1$, so

$$\mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1, z_i = 1] = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].$$

Divide by $\mathbb{E}[d_i \mid z_i = 1] = \mathbb{P}(d_i = 1 \mid z_i = 1)$:

$$\boxed{\frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[d_i \mid z_i = 1]} = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].}$$

4.35 Size of the Complier Group

Under monotonicity,

$$d_{1i} - d_{0i} = 1$$

for compliers and zero otherwise. Therefore,

$$\mathbb{P}(d_{1i} > d_{0i}) = \mathbb{E}[d_{1i} - d_{0i}].$$

By independence,

$$\mathbb{E}[d_{1i}] = \mathbb{E}[d_i \mid z_i = 1],$$

and

$$\mathbb{E}[d_{0i}] = \mathbb{E}[d_i \mid z_i = 0].$$

Hence,

$$\boxed{\mathbb{P}(d_{1i} > d_{0i}) = \mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0].}$$

The first stage is the share of compliers in the population.

4.36 Share of Treated Units Who Are Compliers

We want

$$\mathbb{P}(d_{1i} > d_{0i} \mid d_i = 1).$$

Using Bayes' rule,

$$\mathbb{P}(d_{1i} > d_{0i} \mid d_i = 1) = \frac{\mathbb{P}(d_i = 1 \mid d_{1i} > d_{0i})\mathbb{P}(d_{1i} > d_{0i})}{\mathbb{P}(d_i = 1)}.$$

For compliers,

$$d_i = 1 \iff z_i = 1.$$

Therefore,

$$\mathbb{P}(d_i = 1 \mid d_{1i} > d_{0i}) = \mathbb{P}(z_i = 1 \mid d_{1i} > d_{0i}).$$

By independence,

$$\mathbb{P}(z_i = 1 \mid d_{1i} > d_{0i}) = \mathbb{P}(z_i = 1).$$

Thus,

$$\boxed{\mathbb{P}(d_{1i} > d_{0i} \mid d_i = 1) = \frac{\mathbb{P}(z_i = 1)\left[\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]\right]}{\mathbb{P}(d_i = 1)}.}$$

4.37 Share of Non-Treated Units Who Are Compliers

Similarly,

$$\mathbb{P}(d_{1i} > d_{0i} \mid d_i = 0) = \frac{\mathbb{P}(d_i = 0 \mid d_{1i} > d_{0i})\mathbb{P}(d_{1i} > d_{0i})}{\mathbb{P}(d_i = 0)}.$$

For compliers,

$$d_i = 0 \iff z_i = 0.$$

Therefore,

$$\mathbb{P}(d_i = 0 \mid d_{1i} > d_{0i}) = \mathbb{P}(z_i = 0).$$

By independence,

$$\mathbb{P}(z_i = 0 \mid d_{1i} > d_{0i}) = \mathbb{P}(z_i = 0).$$

Thus,

$$\boxed{\mathbb{P}(d_{1i} > d_{0i} \mid d_i = 0) = \frac{\mathbb{P}(z_i = 0)\left[\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]\right]}{\mathbb{P}(d_i = 0)}.}$$

4.38 Complier Characteristics Ratio

Let x_i be a binary characteristic. We want the relative likelihood that a complier has characteristic $x_i = 1$:

$$\frac{\mathbb{P}(x_i = 1 \mid d_{1i} > d_{0i})}{\mathbb{P}(x_i = 1)}.$$

By Bayes' rule,

$$\mathbb{P}(x_i = 1 \mid d_{1i} > d_{0i}) = \frac{\mathbb{P}(d_{1i} > d_{0i} \mid x_i = 1)\mathbb{P}(x_i = 1)}{\mathbb{P}(d_{1i} > d_{0i})}.$$

Divide by $\mathbb{P}(x_i = 1)$:

$$\frac{\mathbb{P}(x_i = 1 \mid d_{1i} > d_{0i})}{\mathbb{P}(x_i = 1)} = \frac{\mathbb{P}(d_{1i} > d_{0i} \mid x_i = 1)}{\mathbb{P}(d_{1i} > d_{0i})}.$$

Under monotonicity and independence,

$$\mathbb{P}(d_{1i} > d_{0i} \mid x_i = 1) = \mathbb{E}[d_i \mid z_i = 1, x_i = 1] - \mathbb{E}[d_i \mid z_i = 0, x_i = 1],$$

and

$$\mathbb{P}(d_{1i} > d_{0i}) = \mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0].$$

Therefore,

$$\boxed{\frac{\mathbb{P}(x_i = 1 \mid d_{1i} > d_{0i})}{\mathbb{P}(x_i = 1)} = \frac{\mathbb{E}[d_i \mid z_i = 1, x_i = 1] - \mathbb{E}[d_i \mid z_i = 0, x_i = 1]}{\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]}.}$$

4.39 Multiple Instruments as a Weighted Average of Wald Estimands

Suppose there are two mutually exclusive instruments z_{1i} and z_{2i}. Define the two just-identified IV estimands:

$$\rho_1 = \frac{\operatorname{Cov}(y_i, z_{1i})}{\operatorname{Cov}(d_i, z_{1i})},$$

and

$$\rho_2 = \frac{\operatorname{Cov}(y_i, z_{2i})}{\operatorname{Cov}(d_i, z_{2i})}.$$

The population first-stage fitted value from both instruments is

$$d_i^* = \pi_{11} z_{1i} + \pi_{12} z_{2i}.$$

The 2SLS estimand is

$$\rho_{2SLS} = \frac{\operatorname{Cov}(y_i, d_i^*)}{\operatorname{Cov}(d_i, d_i^*)}.$$

Substitute d_i^*:

$$\rho_{2SLS} = \frac{\pi_{11} \operatorname{Cov}(y_i, z_{1i}) + \pi_{12} \operatorname{Cov}(y_i, z_{2i})}{\pi_{11} \operatorname{Cov}(d_i, z_{1i}) + \pi_{12} \operatorname{Cov}(d_i, z_{2i})}.$$

Use

$$\operatorname{Cov}(y_i, z_{ji}) = \rho_j \operatorname{Cov}(d_i, z_{ji}), \qquad j = 1, 2.$$

Then

$$\rho_{2SLS} = \frac{\pi_{11}\rho_1 \operatorname{Cov}(d_i, z_{1i}) + \pi_{12}\rho_2 \operatorname{Cov}(d_i, z_{2i})}{\pi_{11} \operatorname{Cov}(d_i, z_{1i}) + \pi_{12} \operatorname{Cov}(d_i, z_{2i})}.$$

Define

$$\omega = \frac{\pi_{11} \operatorname{Cov}(d_i, z_{1i})}{\pi_{11} \operatorname{Cov}(d_i, z_{1i}) + \pi_{12} \operatorname{Cov}(d_i, z_{2i})}.$$

Then

$$1 - \omega = \frac{\pi_{12} \operatorname{Cov}(d_i, z_{2i})}{\pi_{11} \operatorname{Cov}(d_i, z_{1i}) + \pi_{12} \operatorname{Cov}(d_i, z_{2i})}.$$

Therefore,

$$\boxed{\rho_{2SLS} = \omega\rho_1 + (1 - \omega)\rho_2.}$$

Thus, 2SLS with multiple instruments is a weighted average of just-identified IV estimands.

4.40 IV with Covariates Under Heterogeneous Effects

Assume conditional independence:

$$\{y_{1i}, y_{0i}, d_{1i}, d_{0i}\} \perp z_i \mid X_i.$$

For each covariate value $X_i = x$, define the covariate-specific LATE:

$$\lambda(x) = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}, X_i = x].$$

A saturated first stage is

$$d_i = \pi_X + \pi_{1X} z_i + \xi_i,$$

where π_X is a full set of covariate-cell indicators and π_{1X} allows a separate first-stage effect for each covariate cell.

A saturated second stage is

$$y_i = \alpha_X + \rho_c d_i + \eta_i.$$

The resulting 2SLS estimand is

$$\boxed{\rho_c = \mathbb{E}[\omega(X_i)\lambda(X_i)].}$$

The weights are

$$\boxed{\omega(X_i) = \frac{\operatorname{Var}\{\mathbb{E}[d_i \mid X_i, z_i] \mid X_i\}}{\mathbb{E}\left[\operatorname{Var}\{\mathbb{E}[d_i \mid X_i, z_i] \mid X_i\}\right]}.}$$

Thus, 2SLS with saturated covariates estimates a weighted average of covariate-specific LATEs.

4.41 Abadie Kappa Weighting

Define

$$p(X_i) = \mathbb{P}(z_i = 1 \mid X_i).$$

Abadie's kappa weight is

$$\boxed{\kappa_i = 1 - \frac{d_i(1 - z_i)}{1 - p(X_i)} - \frac{(1 - d_i)z_i}{p(X_i)}.}$$

For any function $g(y_i, d_i, X_i)$,

$$\boxed{\mathbb{E}[g(y_i, d_i, X_i) \mid d_{1i} > d_{0i}] = \frac{\mathbb{E}[\kappa_i g(y_i, d_i, X_i)]}{\mathbb{E}[\kappa_i]}.}$$

The intuition is that the negative terms remove always-takers and never-takers, leaving the complier population.

4.42 Kappa-Weighted Least Squares

Suppose we want the best linear approximation to the complier causal response function:

$$\mathbb{E}[y_i \mid d_i, X_i, d_{1i} > d_{0i}].$$

Define (α_a, β_a) by

$$(\alpha_a, \beta_a) = \arg\min_{a,b} \mathbb{E}\left[\left(\mathbb{E}[y_i \mid d_i, X_i, d_{1i} > d_{0i}] - ad_i - X_i'b\right)^2 \mid d_{1i} > d_{0i}\right].$$

Using Abadie's kappa theorem, this can be estimated by minimizing

$$\boxed{(\alpha_a, \beta_a) = \arg\min_{a,b} \mathbb{E}\left[\kappa_i (y_i - ad_i - X_i'b)^2\right].}$$

Thus, kappa-weighted least squares approximates the causal response function for compliers.

4.43 Average Causal Response with Multivalued Treatment

Let treatment intensity s_i take values

$$0, 1, \ldots, \bar{s}.$$

Let potential outcomes be

$$Y_{si} = f_i(s).$$

Let s_{1i} be treatment intensity when $z_i = 1$, and s_{0i} be treatment intensity when $z_i = 0$.

Assume:

$$\{Y_{0i}, Y_{1i}, \ldots, Y_{\bar{s}i}, s_{1i}, s_{0i}\} \perp z_i,$$

$$\mathbb{E}[s_{1i} - s_{0i}] \neq 0,$$

and monotonicity:

$$s_{1i} \geq s_{0i} \quad \text{for all } i.$$

Then the Wald estimand is

$$\boxed{\frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[s_i \mid z_i = 1] - \mathbb{E}[s_i \mid z_i = 0]} = \sum_{s=1}^{\bar{s}} \omega_s \mathbb{E}[Y_{si} - Y_{s-1,i} \mid s_{1i} \geq s > s_{0i}].}$$

The weights are

$$\boxed{\omega_s = \frac{\mathbb{P}(s_{1i} \geq s > s_{0i})}{\sum_{j=1}^{\bar{s}} \mathbb{P}(s_{1i} \geq j > s_{0i})}.}$$

The weights are nonnegative and sum to one.

4.44 CDF Representation of Average Causal Response Weights

Under monotonicity,

$$\mathbb{P}(s_{1i} \geq s > s_{0i}) = \mathbb{P}(s_{0i} < s) - \mathbb{P}(s_{1i} < s).$$

By independence,

$$\mathbb{P}(s_{0i} < s) = \mathbb{P}(s_i < s \mid z_i = 0),$$

and

$$\mathbb{P}(s_{1i} < s) = \mathbb{P}(s_i < s \mid z_i = 1).$$

Therefore,

$$\boxed{\mathbb{P}(s_{1i} \geq s > s_{0i}) = \mathbb{P}(s_i < s \mid z_i = 0) - \mathbb{P}(s_i < s \mid z_i = 1).}$$

Thus, the ACR weights can be estimated by comparing the treatment-intensity CDFs with the instrument switched off and on.

4.45 Continuous Treatment IV and Average Derivative

Let p_i be a continuous treatment, such as price, and let $q_i(p)$ be a causal response function, such as quantity demanded at price p.

Let

$$p_{1i}$$

be the potential price when $z_i = 1$, and

$$p_{0i}$$

be the potential price when $z_i = 0$.

Assume monotonicity:

$$p_{1i} \geq p_{0i}.$$

By the fundamental theorem of calculus,

$$q_i(p_{1i}) - q_i(p_{0i}) = \int_{p_{0i}}^{p_{1i}} q_i'(t)\, \mathrm{d}t.$$

Taking expectations,

$$\mathbb{E}[q_i \mid z_i = 1] - \mathbb{E}[q_i \mid z_i = 0] = \mathbb{E}\left[\int_{p_{0i}}^{p_{1i}} q_i'(t)\, \mathrm{d}t\right].$$

This can be rewritten as

$$= \int \mathbb{E}[q_i'(t) \mid p_{1i} \geq t > p_{0i}]\mathbb{P}(p_{1i} \geq t > p_{0i})\, \mathrm{d}t.$$

The first stage is

$$\mathbb{E}[p_i \mid z_i = 1] - \mathbb{E}[p_i \mid z_i = 0] = \mathbb{E}[p_{1i} - p_{0i}] = \int \mathbb{P}(p_{1i} \geq t > p_{0i})\, \mathrm{d}t.$$

Therefore,

$$\boxed{\frac{\mathbb{E}[q_i \mid z_i = 1] - \mathbb{E}[q_i \mid z_i = 0]}{\mathbb{E}[p_i \mid z_i = 1] - \mathbb{E}[p_i \mid z_i = 0]} = \frac{\int \mathbb{E}[q_i'(t) \mid p_{1i} \geq t > p_{0i}]\mathbb{P}(p_{1i} \geq t > p_{0i})\, \mathrm{d}t}{\int \mathbb{P}(p_{1i} \geq t > p_{0i})\, \mathrm{d}t}.}$$

Thus, continuous-treatment IV identifies a weighted average derivative of the causal response function.

4.46 Linear Causal Response Special Case

Suppose

$$q_i(p) = \alpha_{0i} + \alpha_{1i} p.$$

Then

$$q_i'(p) = \alpha_{1i}.$$

The IV estimand becomes

$$\frac{\mathbb{E}[q_i \mid z_i = 1] - \mathbb{E}[q_i \mid z_i = 0]}{\mathbb{E}[p_i \mid z_i = 1] - \mathbb{E}[p_i \mid z_i = 0]} = \frac{\mathbb{E}[\alpha_{1i}(p_{1i} - p_{0i})]}{\mathbb{E}[p_{1i} - p_{0i}]}.$$

Therefore,

$$\boxed{IV = \frac{\mathbb{E}[\alpha_{1i}(p_{1i} - p_{0i})]}{\mathbb{E}[p_{1i} - p_{0i}]}.}$$

This is a first-stage-weighted average of heterogeneous slopes α_{1i}.

4.47 *Common Causal Response Special Case*

Suppose

$$q_i(p) = Q(p) + \eta_i.$$

Then

$$q_i'(p) = Q'(p).$$

The continuous IV estimand becomes

$$\boxed{\int Q'(t)\omega(t)\,\mathrm{d}t,}$$

where

$$\boxed{\omega(t) = \frac{\mathbb{P}(p_{1i} \geq t > p_{0i})}{\int \mathbb{P}(p_{1i} \geq r > p_{0i})\,\mathrm{d}r}.}$$

Thus, IV averages the derivative of the common causal response function over the region where the instrument shifts the treatment.

4.48 *Covariate Ambivalence Mistake*

Suppose the first stage is estimated as

$$s_i = X_{0i}'\pi_{10} + Z_i'\pi_{11} + \xi_{1i},$$

but the second stage includes additional covariates X_{1i}:

$$y_i = X_{0i}'\alpha_0 + X_{1i}'\alpha_1 + \rho\widehat{s}_i + u_i.$$

This is not valid 2SLS because the first-stage residual

$$s_i - \widehat{s}_i$$

is orthogonal to X_{0i} and Z_i, but not necessarily to X_{1i}. If

$$\mathrm{Cov}(X_{1i}, s_i - \widehat{s}_i) \neq 0,$$

then the second-stage error

$$\eta_i + \rho(s_i - \widehat{s}_i)$$

is correlated with X_{1i}. Therefore, the second-stage regression is inconsistent.

The rule is:

$$\boxed{\text{Use the same exogenous covariates in the first and second stages.}}$$

4.49 Forbidden Regression with Nonlinear First Stage

Suppose the structural equation is

$$y_i = X_i'\alpha + \rho d_i + \eta_i,$$

where d_i is binary. A researcher estimates a Probit first stage:

$$\mathbb{E}[d_i \mid X_i, Z_i] = \Phi(X_i'\pi_0 + Z_i'\pi_1),$$

and obtains fitted values

$$\widehat{d}_{pi} = \Phi(X_i'\widehat{\pi}_0 + Z_i'\widehat{\pi}_1).$$

The forbidden second stage is

$$y_i = X_i'\alpha + \rho\widehat{d}_{pi} + \left[\eta_i + \rho(d_i - \widehat{d}_{pi})\right].$$

The problem is that nonlinear fitted values do not guarantee

$$\mathbb{E}[\widehat{d}_{pi}(d_i - \widehat{d}_{pi})] = 0$$

unless the nonlinear model is correctly specified. OLS first-stage fitted values have this orthogonality by construction; Probit fitted values do not.

A safer procedure is:

$$\boxed{\text{Use nonlinear fitted values as instruments, not as plug-in regressors.}}$$

4.50 Forbidden Regression with Nonlinear Structural Terms

Suppose the causal model is quadratic:

$$y_i = X_i'\alpha + \rho_1 s_i + \rho_2 s_i^2 + \eta_i.$$

A mistaken manual procedure estimates one first stage

$$s_i = X_i' \pi + Z_i' \gamma + \xi_i$$

and then runs

$$y_i = X_i' \alpha + \rho_1 \widehat{s}_i + \rho_2 \widehat{s}_i^2 + \text{error}.$$

This is forbidden because

$$\widehat{s}_i^2$$

is not generally orthogonal to

$$s_i^2 - \widehat{s}_i^2.$$

The correct approach is to treat both s_i and s_i^2 as endogenous:

$$\boxed{\text{Instrument } s_i \text{ and } s_i^2 \text{ separately.}}$$

That requires at least two excluded instruments.

4.51 Peer Effects: Derivation of Equation (4.6.8)

Consider the cross-sectional peer-effects model

$$Y_{ij} = \mu + \pi_0 s_i + \pi_1 S_j + v_i,$$

where S_j is average schooling in group j. Define

$$\tau_i = s_i - S_j.$$

Rewrite s_i as

$$s_i = \tau_i + S_j.$$

Substitute into the model:

$$Y_{ij} = \mu + \pi_0 (\tau_i + S_j) + \pi_1 S_j + v_i.$$

Collect terms:

$$Y_{ij} = \mu + \pi_0 \tau_i + (\pi_0 + \pi_1) S_j + v_i.$$

Because τ_i and S_j are uncorrelated by construction, the coefficient from a bivariate regression of Y_{ij} on S_j is

$$\rho_1 = \pi_0 + \pi_1.$$

The coefficient on τ_i is
$$\pi_0 = \frac{\mathrm{Cov}(\tau_i, Y_{ij})}{\mathrm{Var}(\tau_i)}.$$
Since
$$\tau_i = s_i - S_j,$$
we have
$$\mathrm{Cov}(\tau_i, Y_{ij}) = \mathrm{Cov}(s_i, Y_{ij}) - \mathrm{Cov}(S_j, Y_{ij}).$$
Also,
$$\mathrm{Var}(\tau_i) = \mathrm{Var}(s_i) - \mathrm{Var}(S_j).$$
Let
$$\rho_0 = \frac{\mathrm{Cov}(s_i, Y_{ij})}{\mathrm{Var}(s_i)}$$
be the OLS coefficient from regressing Y_{ij} on individual schooling s_i, and let
$$\rho_1 = \frac{\mathrm{Cov}(S_j, Y_{ij})}{\mathrm{Var}(S_j)}$$
be the coefficient from regressing Y_{ij} on group average schooling S_j. Then
$$\pi_0 = \frac{\rho_0 \,\mathrm{Var}(s_i) - \rho_1 \,\mathrm{Var}(S_j)}{\mathrm{Var}(s_i) - \mathrm{Var}(S_j)}.$$
Define
$$\phi = \frac{\mathrm{Var}(s_i)}{\mathrm{Var}(s_i) - \mathrm{Var}(S_j)} = \frac{1}{1 - R^2},$$
where
$$R^2 = \frac{\mathrm{Var}(S_j)}{\mathrm{Var}(s_i)}.$$
Then
$$\pi_0 = \rho_0\phi + \rho_1(1 - \phi).$$
Rewrite:
$$\boxed{\pi_0 = \rho_1 + \phi(\rho_0 - \rho_1).}$$
Since
$$\rho_1 = \pi_0 + \pi_1,$$
we have
$$\pi_1 = \rho_1 - \pi_0.$$

Substitute:

$$\pi_1 = \rho_1 - [\rho_1 + \phi(\rho_0 - \rho_1)] \, .$$

Therefore,

$$\boxed{\pi_1 = \phi(\rho_1 - \rho_0).}$$

4.52 Why Regressing an Individual Variable on Its Group Mean Gives Coefficient One

Let s_{ij} be an individual variable and let

$$S_j = \frac{1}{n_j} \sum_{i \in j} s_{ij}$$

be the group mean. Consider the regression of s_{ij} on S_j. The slope is

$$\frac{\sum_j \sum_i s_{ij}(S_j - \bar{S})}{\sum_j \sum_i (S_j - \bar{S})^2}.$$

The numerator is

$$\sum_j (S_j - \bar{S}) \sum_i s_{ij}.$$

Since

$$\sum_i s_{ij} = n_j S_j,$$

the numerator becomes

$$\sum_j n_j S_j (S_j - \bar{S}).$$

This equals

$$\sum_j n_j (S_j - \bar{S})^2,$$

because

$$\sum_j n_j \bar{S}(S_j - \bar{S}) = 0.$$

The denominator is also

$$\sum_j n_j (S_j - \bar{S})^2.$$

Therefore, the slope is

$$\boxed{1.}$$

This shows why peer-effect regressions of an individual outcome on its own group mean can be mechanically uninformative.

4.53 Bivariate Probit Latent-Index Setup

Suppose treatment is determined by

$$d_i = \mathbf{1}[X_i'\gamma_0 + \gamma_1 z_i > v_i],$$

and the outcome is determined by

$$y_i = \mathbf{1}[X_i'\beta_0 + \beta_1 d_i > \varepsilon_i].$$

The source of endogeneity is correlation between v_i and ε_i. If

$$(v_i, \varepsilon_i)$$

are jointly normal, parameters can be estimated by maximum likelihood.

Potential outcomes are

$$y_{0i} = \mathbf{1}[X_i'\beta_0 > \varepsilon_i],$$

and

$$y_{1i} = \mathbf{1}[X_i'\beta_0 + \beta_1 > \varepsilon_i].$$

Potential treatments are

$$d_{0i} = \mathbf{1}[X_i'\gamma_0 > v_i],$$

and

$$d_{1i} = \mathbf{1}[X_i'\gamma_0 + \gamma_1 > v_i].$$

4.54 Bivariate Probit Average Treatment Effect

The average treatment effect is

$$ATE = \mathbb{E}[y_{1i} - y_{0i}].$$

Using the latent-index potential outcomes,

$$ATE = \mathbb{E}\left[\mathbf{1}[X_i'\beta_0 + \beta_1 > \varepsilon_i] - \mathbf{1}[X_i'\beta_0 > \varepsilon_i]\right].$$

Under normality,

$$\mathbb{P}(\varepsilon_i < X_i'\beta_0 + \beta_1 \mid X_i) = \Phi\left(\frac{X_i'\beta_0 + \beta_1}{\sigma_\varepsilon}\right),$$

and

$$\mathbb{P}(\varepsilon_i < X_i'\beta_0 \mid X_i) = \Phi\left(\frac{X_i'\beta_0}{\sigma_\varepsilon}\right).$$

Therefore,

$$\boxed{ATE = \mathbb{E}\left[\Phi\left(\frac{X_i'\beta_0 + \beta_1}{\sigma_\varepsilon}\right) - \Phi\left(\frac{X_i'\beta_0}{\sigma_\varepsilon}\right)\right].}$$

4.55 Bivariate Probit Treatment Effect on the Treated

The treatment effect on the treated is

$$TOT = \mathbb{E}[y_{1i} - y_{0i} \mid d_i = 1].$$

Since

$$d_i = 1 \iff X_i'\gamma_0 + \gamma_1 z_i > v_i,$$

we need the conditional distribution of ε_i given the treatment selection event. Under joint normality, this involves the bivariate normal CDF.

The corresponding expression is

$$\boxed{TOT = \mathbb{E}\left[\frac{\Phi_2\left(\frac{X_i'\beta_0+\beta_1}{\sigma_\varepsilon}, \frac{X_i'\gamma_0+\gamma_1 z_i}{\sigma_v}, \rho_{\varepsilon v}\right) - \Phi_2\left(\frac{X_i'\beta_0}{\sigma_\varepsilon}, \frac{X_i'\gamma_0+\gamma_1 z_i}{\sigma_v}, \rho_{\varepsilon v}\right)}{\Phi\left(\frac{X_i'\gamma_0+\gamma_1 z_i}{\sigma_v}\right)}\right].}$$

Here $\Phi_2(\cdot,\cdot,\rho)$ is the bivariate normal CDF with correlation ρ.

4.56 2SLS Bias Setup

Consider the structural equation

$$y = x\beta + \eta,$$

and the first stage

$$x = Z\pi + \xi.$$

Let

$$P_Z = Z(Z'Z)^{-1}Z'$$

be the projection matrix onto the instrument space. The 2SLS estimator is

$$\widehat{\beta}_{2SLS} = (x'P_Z x)^{-1}x'P_Z y.$$

Substitute

$$y = x\beta + \eta :$$

$$\widehat{\beta}_{2SLS} = (x'P_Z x)^{-1} x' P_Z (x\beta + \eta).$$

Therefore,

$$\widehat{\beta}_{2SLS} = \beta + (x'P_Z x)^{-1} x' P_Z \eta.$$

Thus,

$$\boxed{\widehat{\beta}_{2SLS} - \beta = (x'P_Z x)^{-1} x' P_Z \eta.}$$

4.57 Decomposing the Source of 2SLS Bias

Using the first stage

$$x = Z\pi + \xi,$$

we have

$$x' P_Z \eta = (Z\pi + \xi)' P_Z \eta.$$

Expand:

$$x' P_Z \eta = \pi' Z' P_Z \eta + \xi' P_Z \eta.$$

Since

$$P_Z Z = Z,$$

we have

$$\pi' Z' P_Z \eta = \pi' Z' \eta.$$

Therefore,

$$\widehat{\beta}_{2SLS} - \beta = (x'P_Z x)^{-1} \pi' Z' \eta + (x'P_Z x)^{-1} \xi' P_Z \eta.$$

Taking an approximate expectation by replacing the expectation of a ratio with the ratio of expectations,

$$\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \mathbb{E}[x'P_Z x]^{-1} \mathbb{E}[\pi' Z' \eta] + \mathbb{E}[x'P_Z x]^{-1} \mathbb{E}[\xi' P_Z \eta].$$

Since instruments are exogenous,

$$\mathbb{E}[\pi' Z' \eta] = 0.$$

Thus,

$$\boxed{\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \mathbb{E}[x'P_Z x]^{-1} \mathbb{E}[\xi' P_Z \eta].}$$

The bias arises because the first-stage error ξ is correlated with the structural error η.

4.58 Approximate Bias of 2SLS

Using

$$x = Z\pi + \xi,$$

we have

$$x'P_Z x = (Z\pi + \xi)'P_Z(Z\pi + \xi).$$

Expanding:

$$x'P_Z x = \pi'Z'Z\pi + 2\pi'Z'P_Z\xi + \xi'P_Z\xi.$$

Since

$$\mathbb{E}[\pi'Z'P_Z\xi] = 0,$$

we approximate

$$\mathbb{E}[x'P_Z x] = \mathbb{E}[\pi'Z'Z\pi] + \mathbb{E}[\xi'P_Z\xi].$$

Thus,

$$\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \left[\mathbb{E}[\pi'Z'Z\pi] + \mathbb{E}[\xi'P_Z\xi]\right]^{-1}\mathbb{E}[\xi'P_Z\eta].$$

4.59 Trace Derivation for 2SLS Bias

Assume

$$\mathbb{E}[\xi\xi'] = \sigma_\xi^2 I,$$

and

$$\mathbb{E}[\xi\eta'] = \sigma_{\eta\xi} I.$$

Because $\xi'P_Z\xi$ is a scalar,

$$\xi'P_Z\xi = (\xi'P_Z\xi).$$

Using cyclic invariance of trace:

$$(\xi'P_Z\xi) = (P_Z\xi\xi').$$

Taking expectations:

$$\mathbb{E}[\xi'P_Z\xi] = \mathbb{E}[(P_Z\xi\xi')] = (P_Z\mathbb{E}[\xi\xi']).$$

Substitute

$$\mathbb{E}[\xi\xi'] = \sigma_\xi^2 I:$$

$$= (P_Z\sigma_\xi^2 I) = \sigma_\xi^2(P_Z).$$

Since P_Z is idempotent, its trace equals its rank:

$$(P_Z) = q.$$

Therefore,

$$\boxed{\mathbb{E}[\xi' P_Z \xi] = \sigma_\xi^2 q.}$$

Similarly,

$$\mathbb{E}[\xi' P_Z \eta] = (P_Z \mathbb{E}[\eta \xi']) = \sigma_{\eta\xi}(P_Z) = \boxed{\sigma_{\eta\xi} q.}$$

4.60 2SLS Bias and the First-Stage F-Statistic

Substitute the trace results:

$$\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \left[\mathbb{E}[\pi' Z' Z \pi] + \sigma_\xi^2 q\right]^{-1} \sigma_{\eta\xi} q.$$

Divide numerator and denominator by $\sigma_\xi^2 q$:

$$\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\eta\xi}}{\sigma_\xi^2} \left[\frac{\mathbb{E}[\pi' Z' Z \pi]}{\sigma_\xi^2 q} + 1\right]^{-1}.$$

The term

$$F = \frac{\mathbb{E}[\pi' Z' Z \pi]}{\sigma_\xi^2 q}$$

is the population first-stage F-statistic. Therefore,

$$\boxed{\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\eta\xi}}{\sigma_\xi^2} \frac{1}{F+1}.}$$

As F becomes large, the approximate bias goes to zero. As F becomes small, the bias approaches the OLS-type endogeneity bias.

4.61 Why Many Weak Instruments Increase Bias

The population first-stage F-statistic is

$$F = \frac{\mathbb{E}[\pi' Z' Z \pi]}{\sigma_\xi^2 q}.$$

Adding irrelevant instruments increases q but does not increase the first-stage explanatory power

$$\mathbb{E}[\pi' Z' Z \pi].$$

Therefore, F falls.

Since approximate 2SLS bias is

$$\mathbb{E}[\widehat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\eta\xi}}{\sigma_{\xi}^2} \frac{1}{F+1},$$

a smaller F implies larger bias. Thus,

$$\boxed{\text{many weak instruments push 2SLS toward OLS.}}$$

4.62 Summary of Chapter 4

Chapter 4 explains instrumental variables as a research design for situations where the treatment or causal variable is endogenous. In the basic structural equation,

$$y_i = \alpha + \rho s_i + \eta_i,$$

OLS fails to identify ρ when

$$\text{Cov}(s_i, \eta_i) \neq 0.$$

An instrumental variable z_i solves this problem when it is correlated with the endogenous variable,

$$\text{Cov}(z_i, s_i) \neq 0,$$

but uncorrelated with the structural error,

$$\text{Cov}(z_i, \eta_i) = 0.$$

Under these two conditions, the causal parameter can be written as the IV covariance ratio:

$$\rho = \frac{\text{Cov}(y_i, z_i)}{\text{Cov}(s_i, z_i)}.$$

This ratio is the foundation of IV.

The chapter emphasizes the connection between IV and two regressions: the first stage and the reduced form. The first stage measures how the instrument changes the treatment:

$$s_i = X_i'\pi_{10} + \pi_{11}z_i + \xi_{1i},$$

while the reduced form measures how the instrument changes the outcome:

$$y_i = X_i'\pi_{20} + \pi_{21}z_i + \xi_{2i}.$$

The IV estimand is then

$$\rho = \frac{\pi_{21}}{\pi_{11}}.$$

This interpretation is important because IV does not directly compare treated and untreated units. Instead, it compares outcome changes induced by the instrument to treatment changes induced by the instrument.

The operational estimator is two-stage least squares. In the first stage, the endogenous variable is predicted using the instrument and covariates:

$$s_i = X_i'\pi_{10} + \pi_{11}z_i + \xi_{1i}.$$

In the second stage, the outcome is regressed on the fitted value from the first stage:

$$y_i = X_i'\alpha + \rho\widehat{s}_i + \text{error}.$$

With one endogenous variable and one instrument, 2SLS is algebraically equivalent to the ratio of reduced-form and first-stage coefficients:

$$\widehat{\rho}_{2SLS} = \frac{\widehat{\pi}_{21}}{\widehat{\pi}_{11}}.$$

A major conceptual contribution of the chapter is the LATE interpretation. When treatment effects are heterogeneous, IV generally does not estimate the average treatment effect for everyone. Instead, under independence, exclusion, first-stage relevance, and monotonicity, the Wald estimand identifies the local average treatment effect:

$$\frac{\mathbb{E}[y_i \mid z_i = 1] - \mathbb{E}[y_i \mid z_i = 0]}{\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]} = \mathbb{E}[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}].$$

The group

$$d_{1i} > d_{0i}$$

is the complier group: individuals whose treatment status is changed by the instrument. Thus, IV estimates the causal effect for the people moved by the instrument, not necessarily for always-takers, never-takers, or the whole population.

The chapter also shows how LATE generalizes. With multiple instruments, 2SLS combines multiple instrument-specific Wald estimands. In this sense, overidentified 2SLS can be understood as a weighted average of just-identified IV estimates. With covariates, IV can be justified by conditional independence:

$$\{y_{1i}, y_{0i}, d_{1i}, d_{0i}\} \perp z_i \mid X_i.$$

Covariates can make the IV assumptions more plausible and can improve precision. With multivalued or continuous treatments, IV estimates a weighted average of incremental causal effects or average derivatives along the margins shifted by the instrument.

The chapter further explains how to characterize compliers. Although individual compliers cannot be directly observed, their average characteristics can be recovered using IV-type comparisons. For a covariate x_i, the relative likelihood that compliers have characteristic $x_i = 1$ can be written as

$$\frac{\mathbb{P}(x_i = 1 \mid d_{1i} > d_{0i})}{\mathbb{P}(x_i = 1)} = \frac{\mathbb{E}[d_i \mid z_i = 1, x_i = 1] - \mathbb{E}[d_i \mid z_i = 0, x_i = 1]}{\mathbb{E}[d_i \mid z_i = 1] - \mathbb{E}[d_i \mid z_i = 0]}.$$

This helps interpret the external validity of IV estimates.

Finally, Chapter 4 warns that IV estimates can be fragile when instruments are weak or when there are many weak instruments. Weak instruments have small first-stage effects, so the denominator of the IV ratio is close to zero:

$$\text{Cov}(z_i, s_i) \approx 0.$$

This makes IV estimates imprecise and potentially biased toward OLS. With many instruments, 2SLS can overfit the endogenous treatment in the first stage, recreating some of the endogenous variation that IV was supposed to remove. The practical lesson is to examine first-stage strength, avoid excessive weak instruments, compare 2SLS with alternatives such as LIML when appropriate, and always interpret IV through the reduced form, first stage, exclusion restriction, and complier population.

Overall, Chapter 4's core logic is:

$$\begin{gathered}\text{endogeneity problem} \longrightarrow \text{instrument} \longrightarrow \text{first stage and reduced form} \\ \longrightarrow \text{IV/2SLS ratio} \longrightarrow \text{LATE interpretation under heterogeneous effects.}\end{gathered}$$

5 Chapter 5 Derivations: Parallel Worlds, Fixed Effects, Difference-in-Differences, and Panel Data

5.1 *Individual Fixed Effects Setup*

Let y_{it} denote the outcome for individual i in period t, and let d_{it} denote treatment status. In the union-wage example, y_{it} is log earnings and d_{it} indicates union membership.

Potential outcomes are

$$y_{0it} = \text{outcome without treatment}, \qquad y_{1it} = \text{outcome with treatment}.$$

Observed outcome is

$$y_{it} = y_{0it} + (y_{1it} - y_{0it})d_{it}.$$

The identifying assumption is that treatment is as good as randomly assigned conditional on unobserved individual ability A_i, observed covariates X_{it}, and time t:

$$\mathbb{E}[y_{0it} \mid A_i, X_{it}, t, d_{it}] = \mathbb{E}[y_{0it} \mid A_i, X_{it}, t].$$

This says that after controlling for A_i, X_{it}, and t, treatment assignment contains no additional information about untreated potential outcomes.

The key fixed-effects assumption is that A_i is time-invariant and enters additively:

$$\mathbb{E}[y_{0it} \mid A_i, X_{it}, t] = \alpha + \lambda_t + A_i'\gamma + X_{it}'\delta.$$

Assume a constant additive treatment effect:

$$\mathbb{E}[y_{1it} \mid A_i, X_{it}, t] = \mathbb{E}[y_{0it} \mid A_i, X_{it}, t] + \rho.$$

Then

$$\mathbb{E}[y_{it} \mid A_i, X_{it}, t, d_{it}] = \alpha + \lambda_t + \rho d_{it} + A_i'\gamma + X_{it}'\delta.$$

Define

$$\alpha_i \equiv \alpha + A_i'\gamma.$$

Then the fixed-effects model is

$$\boxed{y_{it} = \alpha_i + \lambda_t + \rho d_{it} + X_{it}'\delta + \varepsilon_{it}.}$$

Here ρ is the causal effect of interest.

5.2 *Why Fixed Effects Remove Time-Invariant Omitted Variables*

The fixed-effects model is

$$y_{it} = \alpha_i + \lambda_t + \rho d_{it} + X_{it}'\delta + \varepsilon_{it}.$$

The omitted variable α_i is allowed to be correlated with d_{it}. Therefore, a pooled OLS regression that omits α_i may suffer from omitted variables bias.

However, because α_i does not vary over time, it can be eliminated by subtracting individual means.

Define individual averages:

$$\bar{y}_i = \frac{1}{T_i}\sum_t y_{it}, \qquad \bar{d}_i = \frac{1}{T_i}\sum_t d_{it}, \qquad \bar{X}_i = \frac{1}{T_i}\sum_t X_{it},$$

$$\bar{\lambda}_i = \frac{1}{T_i}\sum_t \lambda_t, \qquad \bar{\varepsilon}_i = \frac{1}{T_i}\sum_t \varepsilon_{it}.$$

Taking the average of the fixed-effects equation over time gives

$$\bar{y}_i = \alpha_i + \bar{\lambda}_i + \rho\bar{d}_i + \bar{X}_i'\delta + \bar{\varepsilon}_i.$$

Subtract this individual mean equation from the original equation:

$$y_{it} - \bar{y}_i = (\alpha_i - \alpha_i) + (\lambda_t - \bar{\lambda}_i) + \rho(d_{it} - \bar{d}_i) + (X_{it} - \bar{X}_i)'\delta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

The individual fixed effect disappears:

$$\boxed{y_{it} - \bar{y}_i = \lambda_t - \bar{\lambda}_i + \rho(d_{it} - \bar{d}_i) + (X_{it} - \bar{X}_i)'\delta + (\varepsilon_{it} - \bar{\varepsilon}_i).}$$

Thus, fixed effects identify ρ using only within-individual variation over time.

5.3 *Within Estimator*

Define demeaned variables:

$$\ddot{y}_{it} = y_{it} - \bar{y}_i, \qquad \ddot{d}_{it} = d_{it} - \bar{d}_i, \qquad \ddot{X}_{it} = X_{it} - \bar{X}_i,$$

and

$$\ddot{\varepsilon}_{it} = \varepsilon_{it} - \bar{\varepsilon}_i.$$

The fixed-effects model after demeaning is

$$\ddot{y}_{it} = (\lambda_t - \bar{\lambda}_i) + \rho \ddot{d}_{it} + \ddot{X}_{it}' \delta + \ddot{\varepsilon}_{it}.$$

If time effects are included through time dummies, then the within estimator is obtained by OLS on the demeaned equation:

$$\boxed{\widehat{\rho}_{FE} = \text{coefficient on } \ddot{d}_{it} \text{ in a regression of } \ddot{y}_{it} \text{ on } \ddot{d}_{it}, \ddot{X}_{it}, \text{ and time effects.}}$$

In the simplest case with no X_{it} and no time effects, the within estimator is

$$\boxed{\widehat{\rho}_{FE} = \frac{\sum_i \sum_t (d_{it} - \bar{d}_i)(y_{it} - \bar{y}_i)}{\sum_i \sum_t (d_{it} - \bar{d}_i)^2}.}$$

The estimator uses only deviations from each individual's own average.

5.4 Fixed Effects as Regression with Individual Dummies

The fixed-effects model can also be written as

$$y_{it} = \sum_i \alpha_i \mathbf{1}[I = i] + \sum_t \lambda_t \mathbf{1}[T = t] + \rho d_{it} + X_{it}' \delta + \varepsilon_{it}.$$

This is an ordinary regression with one dummy variable for each individual and one dummy variable for each period.

By the Frisch-Waugh-Lovell theorem, the coefficient on d_{it} can be obtained by:

1. Regressing d_{it} on individual dummies, time dummies, and X_{it}, and saving the residual $\widetilde{d}_{it}$.
2. Regressing y_{it} on individual dummies, time dummies, and X_{it}, and saving the residual $\widetilde{y}_{it}$.
3. Regressing $\widetilde{y}_{it}$ on $\widetilde{d}_{it}$.

In the case with only individual effects, residualizing on individual dummies is exactly the same as subtracting individual means:

$$\widetilde{d}_{it} = d_{it} - \bar{d}_i, \qquad \widetilde{y}_{it} = y_{it} - \bar{y}_i.$$

Therefore,

$$\boxed{\text{OLS with individual dummies} = \text{OLS on deviations from individual means.}}$$

5.5 *First-Difference Estimator*

Starting from

$$y_{it} = \alpha_i + \lambda_t + \rho d_{it} + X_{it}'\delta + \varepsilon_{it},$$

subtract the previous-period equation:

$$y_{i,t-1} = \alpha_i + \lambda_{t-1} + \rho d_{i,t-1} + X_{i,t-1}'\delta + \varepsilon_{i,t-1}.$$

Then

$$y_{it} - y_{i,t-1} = (\alpha_i - \alpha_i) + (\lambda_t - \lambda_{t-1}) + \rho(d_{it} - d_{i,t-1}) + (X_{it} - X_{i,t-1})'\delta + (\varepsilon_{it} - \varepsilon_{i,t-1}).$$

Define

$$\Delta y_{it} = y_{it} - y_{i,t-1}, \qquad \Delta d_{it} = d_{it} - d_{i,t-1}, \qquad \Delta X_{it} = X_{it} - X_{i,t-1},$$

and

$$\Delta\lambda_t = \lambda_t - \lambda_{t-1}, \qquad \Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}.$$

Then

$$\boxed{\Delta y_{it} = \Delta\lambda_t + \rho\Delta d_{it} + \Delta X_{it}'\delta + \Delta\varepsilon_{it}.}$$

This is the first-difference estimator.

5.6 *Equivalence of First Differences and Fixed Effects with Two Periods*

Suppose there are only two periods, $t = 0, 1$. The fixed-effects within transformation gives

$$y_{i1} - \bar{y}_i = y_{i1} - \frac{y_{i1} + y_{i0}}{2} = \frac{y_{i1} - y_{i0}}{2} = \frac{\Delta y_i}{2},$$

and

$$y_{i0} - \bar{y}_i = y_{i0} - \frac{y_{i1} + y_{i0}}{2} = -\frac{y_{i1} - y_{i0}}{2} = -\frac{\Delta y_i}{2}.$$

Similarly,

$$d_{i1} - \bar{d}_i = \frac{\Delta d_i}{2}, \qquad d_{i0} - \bar{d}_i = -\frac{\Delta d_i}{2}.$$

The within estimator is

$$\widehat{\rho}_{FE} = \frac{\sum_i \sum_{t=0}^{1}(d_{it} - \bar{d}_i)(y_{it} - \bar{y}_i)}{\sum_i \sum_{t=0}^{1}(d_{it} - \bar{d}_i)^2}.$$

Substitute the expressions above:

$$\sum_{t=0}^{1}(d_{it} - \bar{d}_i)(y_{it} - \bar{y}_i) = \frac{\Delta d_i}{2}\frac{\Delta y_i}{2} + \left(-\frac{\Delta d_i}{2}\right)\left(-\frac{\Delta y_i}{2}\right) = \frac{\Delta d_i \Delta y_i}{2}.$$

Likewise,

$$\sum_{t=0}^{1}(d_{it}-\bar{d}_i)^2 = \left(\frac{\Delta d_i}{2}\right)^2 + \left(-\frac{\Delta d_i}{2}\right)^2 = \frac{(\Delta d_i)^2}{2}.$$

Therefore,

$$\widehat{\rho}_{FE} = \frac{\sum_i \Delta d_i \Delta y_i/2}{\sum_i(\Delta d_i)^2/2} = \frac{\sum_i \Delta d_i \Delta y_i}{\sum_i(\Delta d_i)^2}.$$

Thus,

$$\boxed{\widehat{\rho}_{FE} = \widehat{\rho}_{FD} \quad \text{when } T = 2.}$$

5.7 Why Fixed Effects Are Vulnerable to Measurement Error

Suppose the true treatment is d_{it}^*, but we observe

$$d_{it} = d_{it}^* + u_{it},$$

where u_{it} is classical measurement error:

$$\text{Cov}(d_{it}^*, u_{it}) = 0, \qquad \text{Cov}(\varepsilon_{it}, u_{it}) = 0.$$

In levels, the signal-to-noise ratio may be high if d_{it}^* is persistent:

$$\text{Var}(d_{it}^*) \gg \text{Var}(u_{it}).$$

But fixed effects use within variation:

$$d_{it} - \bar{d}_i = (d_{it}^* - \bar{d}_i^*) + (u_{it} - \bar{u}_i).$$

If true treatment is highly persistent, then

$$\text{Var}(d_{it}^* - \bar{d}_i^*)$$

may be small. However, measurement error often changes over time, so

$$\text{Var}(u_{it} - \bar{u}_i)$$

may remain large relative to the true within variation.

The attenuation factor in a simple within regression is approximately

$$\boxed{\frac{\text{Var}(d_{it}^* - \bar{d}_i^*)}{\text{Var}(d_{it}^* - \bar{d}_i^*) + \text{Var}(u_{it} - \bar{u}_i)}.}$$

When the true within variation is small, this factor can be much less than one. Thus, fixed-effects estimates may be strongly attenuated toward zero.

5.8 Difference-in-Differences Potential Outcomes Setup

Let s index states or groups and let t index time periods. In the Card and Krueger minimum-wage example, $s \in \{NJ, PA\}$ and $t \in \{Feb, Nov\}$.

Let

$$y_{1ist} = \text{potential employment if exposed to the high minimum wage,}$$

and

$$y_{0ist} = \text{potential employment if exposed to the low minimum wage.}$$

Observed outcome is

$$y_{ist} = y_{0ist} + (y_{1ist} - y_{0ist})d_{st},$$

where $d_{st} = 1$ for treated group-period cells and zero otherwise.

The key DD assumption is an additive structure for untreated potential outcomes:

$$\boxed{\mathbb{E}[y_{0ist} \mid s, t] = \gamma_s + \lambda_t.}$$

This is the common-trends or parallel-trends assumption written in levels of untreated potential outcomes.

Assume a constant treatment effect:

$$\mathbb{E}[y_{1ist} - y_{0ist} \mid s, t] = \beta.$$

Then

$$\mathbb{E}[y_{ist} \mid s, t] = \gamma_s + \lambda_t + \beta d_{st}.$$

5.9 Deriving the Two-by-Two Difference-in-Differences Estimand

Let T denote the treated group and C the control group. Let *Post* denote the post-treatment period and *Pre* the pre-treatment period. Treatment occurs only for the treated group in the post period:

$$d_{T,Post} = 1, \qquad d_{T,Pre} = d_{C,Post} = d_{C,Pre} = 0.$$

The model is

$$\mathbb{E}[y_{ist} \mid s, t] = \gamma_s + \lambda_t + \beta d_{st}.$$

For the control group:

$$\mathbb{E}[y \mid C, Post] = \gamma_C + \lambda_{Post},$$

and

$$\mathbb{E}[y \mid C, Pre] = \gamma_C + \lambda_{Pre}.$$

Therefore,

$$\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre] = \lambda_{Post} - \lambda_{Pre}.$$

For the treated group:

$$\mathbb{E}[y \mid T, Post] = \gamma_T + \lambda_{Post} + \beta,$$

and

$$\mathbb{E}[y \mid T, Pre] = \gamma_T + \lambda_{Pre}.$$

Therefore,

$$\mathbb{E}[y \mid T, Post] - \mathbb{E}[y \mid T, Pre] = \lambda_{Post} - \lambda_{Pre} + \beta.$$

Subtract the control change from the treated change:

$$[\mathbb{E}[y \mid T, Post] - \mathbb{E}[y \mid T, Pre]] - [\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre]] = \beta.$$

Thus,

$$\boxed{\beta_{DD} = [\mathbb{E}[y \mid T, Post] - \mathbb{E}[y \mid T, Pre]] - [\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre]]\,.}$$

5.10 DD as a Causal Estimand Under Parallel Trends

The causal effect for the treated group in the post period is

$$\mathbb{E}[y_{1i,T,Post} - y_{0i,T,Post} \mid T, Post].$$

The problem is that

$$\mathbb{E}[y_{0i,T,Post} \mid T, Post]$$

is unobserved.

The observed treated-post outcome is

$$\mathbb{E}[y \mid T, Post] = \mathbb{E}[y_1 \mid T, Post].$$

The observed treated-pre outcome is

$$\mathbb{E}[y \mid T, Pre] = \mathbb{E}[y_0 \mid T, Pre].$$

The observed control change is

$$\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre] = \mathbb{E}[y_0 \mid C, Post] - \mathbb{E}[y_0 \mid C, Pre].$$

Parallel trends says

$$\mathbb{E}[y_0 \mid T, Post] - \mathbb{E}[y_0 \mid T, Pre] = \mathbb{E}[y_0 \mid C, Post] - \mathbb{E}[y_0 \mid C, Pre].$$

Therefore,

$$\mathbb{E}[y_0 \mid T, Post] = \mathbb{E}[y_0 \mid T, Pre] + \left[\mathbb{E}[y_0 \mid C, Post] - \mathbb{E}[y_0 \mid C, Pre]\right].$$

Using observed outcomes,

$$\mathbb{E}[y_0 \mid T, Post] = \mathbb{E}[y \mid T, Pre] + \left[\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre]\right].$$

Thus the causal effect is

$$\mathbb{E}[y \mid T, Post] - \left\{\mathbb{E}[y \mid T, Pre] + \left[\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre]\right]\right\}.$$

Rearranging gives

$$\boxed{ATT_{DD} = \left[\mathbb{E}[y \mid T, Post] - \mathbb{E}[y \mid T, Pre]\right] - \left[\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre]\right].}$$

5.11 Regression Difference-in-Differences

Let NJ_s be a dummy for New Jersey, and let $Post_t$ be a dummy for the post-treatment period. The regression-DD model is

$$\boxed{y_{ist} = \alpha + \gamma NJ_s + \lambda Post_t + \beta(NJ_s \cdot Post_t) + \varepsilon_{ist}.}$$

This model is saturated because there are four group-time cells and four parameters.

The conditional means are:

$$\mathbb{E}[y \mid PA, Pre] = \alpha,$$

$$\mathbb{E}[y \mid NJ, Pre] = \alpha + \gamma,$$

$$\mathbb{E}[y \mid PA, Post] = \alpha + \lambda,$$

and

$$\mathbb{E}[y \mid NJ, Post] = \alpha + \gamma + \lambda + \beta.$$

Therefore,

$$\gamma = \mathbb{E}[y \mid NJ, Pre] - \mathbb{E}[y \mid PA, Pre],$$

$$\lambda = \mathbb{E}[y \mid PA, Post] - \mathbb{E}[y \mid PA, Pre],$$

and

$$\beta = \mathbb{E}[y \mid NJ, Post] - \mathbb{E}[y \mid NJ, Pre] - [\mathbb{E}[y \mid PA, Post] - \mathbb{E}[y \mid PA, Pre]] .$$

Hence,

$$\boxed{\text{The coefficient on the interaction term equals the DD estimand.}}$$

5.12 General Regression-DD with Group and Time Fixed Effects

With many groups s and many periods t, the regression-DD model is

$$\boxed{y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + X_{ist}'\delta + \varepsilon_{ist}.}$$

Here:

$$\gamma_s = \text{group fixed effect},$$

$$\lambda_t = \text{time fixed effect},$$

$$d_{st} = \text{policy or treatment variable},$$

and X_{ist} contains additional controls.

The coefficient β is identified from deviations of treatment from group and time averages. By regression anatomy,

$$\boxed{\beta = \frac{\mathrm{Cov}(y_{ist}, \widetilde{d}_{st})}{\mathrm{Var}(\widetilde{d}_{st})},}$$

where $\widetilde{d}_{st}$ is the residual from regressing d_{st} on group fixed effects, time fixed effects, and controls.

Thus, regression-DD compares treated and untreated observations after removing permanent group differences and common time shocks.

5.13 *Treatment Intensity Difference-in-Differences*

Treatment does not always turn on as a zero-one variable. Let f_s measure baseline treatment intensity or exposure. In Card's minimum-wage application, f_s is the fraction of teenagers in state s affected by a federal minimum-wage increase.

The model is

$$\boxed{y_{ist} = \gamma_s + \lambda_t + \beta(f_s \cdot Post_t) + \varepsilon_{ist}.}$$

The treatment variable is

$$d_{st} = f_s \cdot Post_t.$$

The coefficient β measures the effect of a one-unit increase in exposure after the policy change.

With two periods, taking first differences gives

$$\Delta y_{is} = \lambda + \beta f_s + \Delta\varepsilon_{is}.$$

At the group level, let

$$\Delta\bar{y}_s = \bar{y}_{s,Post} - \bar{y}_{s,Pre}.$$

Then

$$\boxed{\Delta\bar{y}_s = \lambda + \beta f_s + \Delta\bar{\varepsilon}_s.}$$

Thus, with two periods, treatment-intensity DD is a cross-sectional regression of outcome changes on baseline exposure.

5.14 *Weighted Group-Level Regression-DD*

Suppose individual data are aggregated to group-time means:

$$\bar{y}_{st} = \frac{1}{n_{st}} \sum_{i \in (s,t)} y_{ist}.$$

The group-level DD model is

$$\bar{y}_{st} = \gamma_s + \lambda_t + \beta d_{st} + \bar{\varepsilon}_{st}.$$

If the estimand is the micro-data regression coefficient and cells have different sizes n_{st}, then group-level regression should be weighted by n_{st}.

The weighted least squares problem is

$$\boxed{\min_{\gamma,\lambda,\beta} \sum_{s,t} n_{st} \left(\bar{y}_{st} - \gamma_s - \lambda_t - \beta d_{st}\right)^2.}$$

This reproduces the corresponding micro-data regression when the only regressors vary at the group-time level.

5.15 Adding Covariates to DD

Suppose the untreated potential outcome satisfies

$$\mathbb{E}[y_{0ist} \mid s, t, X_{st}] = \gamma_s + \lambda_t + X_{st}'\delta.$$

Then the regression-DD equation is

$$\boxed{y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + X_{st}'\delta + \varepsilon_{ist}.}$$

The inclusion of X_{st} is useful when X_{st} captures group-specific time-varying confounders.

However, if X_{st} is itself affected by treatment, then it is a bad control. In that case, controlling for X_{st} can remove part of the treatment effect or introduce selection bias. Thus, valid controls should be predetermined or otherwise unaffected by the treatment.

5.16 DD with Individual-Level Covariates

Suppose we estimate

$$\boxed{y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + X_{ist}'\delta + \varepsilon_{ist}.}$$

Here d_{st} varies only at the group-time level, while X_{ist} varies at the individual level.

Individual-level covariates can improve precision by explaining residual outcome variation:

$$\mathrm{Var}(\varepsilon_{ist} \mid X_{ist}) < \mathrm{Var}(y_{ist}).$$

However, individual covariates do not usually address the central DD omitted-variable problem, which is typically about group-specific time-varying shocks. The key identifying assumption still concerns the evolution of untreated potential outcomes across groups:

$$\mathbb{E}[y_{0ist} \mid s, t] = \gamma_s + \lambda_t$$

or its covariate-adjusted version.

5.17 *Event-Study and Leads-Lags DD*

Suppose treatment changes at different times across groups. Let $d_{s,t-\tau}$ denote treatment status τ periods before t, and let $d_{s,t+\tau}$ denote future treatment status.

A dynamic DD or event-study equation is

$$\boxed{y_{ist} = \gamma_s + \lambda_t + \sum_{\tau=0}^{m} \beta_\tau d_{s,t-\tau} + \sum_{\tau=1}^{q} \beta_{+\tau} d_{s,t+\tau} + X'_{ist}\delta + \varepsilon_{ist}.}$$

The coefficients

$$\beta_0, \beta_1, \ldots, \beta_m$$

describe contemporaneous and lagged treatment effects.

The coefficients

$$\beta_{+1}, \ldots, \beta_{+q}$$

are lead coefficients. Under a causal interpretation with no anticipation and valid parallel trends,

$$\boxed{\beta_{+\tau} = 0 \quad \text{for all } \tau \geq 1.}$$

Nonzero lead coefficients suggest that treated groups were already on different trends before treatment.

5.18 *State-Specific Time Trends*

A robustness check for DD allows each group to have its own linear time trend:

$$\boxed{y_{ist} = \gamma_{0s} + \gamma_{1s}t + \lambda_t + \beta d_{st} + X'_{ist}\delta + \varepsilon_{ist}.}$$

Here

$$\gamma_{0s}$$

is the group-specific intercept, and

$$\gamma_{1s}$$

is the group-specific trend.

This model assumes that untreated potential outcomes follow

$$\mathbb{E}[y_{0ist} \mid s, t] = \gamma_{0s} + \gamma_{1s}t + \lambda_t.$$

Then identification comes from deviations from group-specific linear trends. The treatment effect is identified if, absent treatment, treated and control groups would have continued along their own pre-existing trends.

5.19 Why At Least Three Periods Are Needed for Group-Specific Trends

Consider a model with group fixed effects, period fixed effects, treatment, and group-specific linear trends:

$$y_{st} = \gamma_s + \lambda_t + \theta_s t + \beta d_{st} + \varepsilon_{st}.$$

If there are only two periods, then for each group the linear trend $\theta_s t$ can perfectly fit that group's change over time. With two periods, there is no remaining within-group time variation after estimating group-specific trends.

Formally, for group s, the two observations are

$$y_{s0} = \gamma_s + \lambda_0 + \theta_s \cdot 0 + \beta d_{s0} + \varepsilon_{s0},$$

$$y_{s1} = \gamma_s + \lambda_1 + \theta_s \cdot 1 + \beta d_{s1} + \varepsilon_{s1}.$$

The group-specific trend θ_s absorbs the group-specific change

$$y_{s1} - y_{s0}.$$

Therefore, treatment changes that also occur at the group-time level cannot be separately distinguished from θ_s. Hence,

$$\boxed{\text{Group-specific linear trends require at least three periods, and usually more for credible estimation.}}$$

5.20 Triple Differences Setup

Triple differences use three dimensions: group s, time t, and subgroup a. In the Medicaid example, a may index the age of the youngest child.

The model is

$$\boxed{y_{iast} = \gamma_{st} + \lambda_{at} + \theta_{as} + \beta d_{ast} + X'_{iast}\delta + \varepsilon_{iast}.}$$

Here:

$$\gamma_{st} = \text{state-by-time effects},$$

$$\lambda_{at} = \text{age-by-time effects},$$

$$\theta_{as} = \text{age-by-state effects},$$

and

$$d_{ast} = \text{treatment indicator for affected age groups in treated states and periods.}$$

The coefficient β is identified by comparing changes across affected and unaffected subgroups, across treated and untreated states, and over time.

5.21 *Deriving the Triple-Differences Estimand*

Let there be:

$$s \in \{T, C\} \quad \text{treated and control states,}$$

$$t \in \{Post, Pre\},$$

and

$$a \in \{A, U\} \quad \text{affected and unaffected subgroups.}$$

Define the mean outcome:

$$\mu_{sat} = \mathbb{E}[y \mid s, a, t].$$

The triple-differences estimand is

$$\boxed{\begin{aligned} DDD = {} & [(\mu_{T,A,Post} - \mu_{T,A,Pre}) - (\mu_{T,U,Post} - \mu_{T,U,Pre})] \\ & - [(\mu_{C,A,Post} - \mu_{C,A,Pre}) - (\mu_{C,U,Post} - \mu_{C,U,Pre})]. \end{aligned}}$$

This can also be written as

$$DDD = DD_A - DD_U,$$

where DD_A is the DD for the affected subgroup and DD_U is the DD for the unaffected subgroup.

The identifying assumption is that, absent treatment, the treated-control difference-in-differences for the affected subgroup would have been the same as that for the unaffected subgroup.

5.22 *Fixed Effects Versus Lagged Dependent Variables*

Fixed effects are motivated by

$$\boxed{\mathbb{E}[y_{0it} \mid \alpha_i, X_{it}, d_{it}] = \mathbb{E}[y_{0it} \mid \alpha_i, X_{it}].}$$

This says selection into treatment is driven by time-invariant unobserved heterogeneity α_i, plus observed covariates.

The lagged-dependent-variable model is motivated by

$$\boxed{\mathbb{E}[y_{0it} \mid y_{i,t-h}, X_{it}, d_{it}] = \mathbb{E}[y_{0it} \mid y_{i,t-h}, X_{it}].}$$

This says selection into treatment is driven by past outcomes, such as past earnings, rather than by a time-invariant fixed effect.

The LDV regression is

$$\boxed{y_{it} = \alpha + \theta y_{i,t-h} + \lambda_t + \beta d_{it} + X_{it}'\delta + \varepsilon_{it}.}$$

Here β is interpreted as causal if conditioning on lagged outcomes and covariates removes selection bias.

5.23 *Fixed Effects and Lagged Outcomes Together*

A more general model includes both fixed effects and lagged dependent variables:

$$\boxed{y_{it} = \alpha_i + \theta y_{i,t-h} + \lambda_t + \beta d_{it} + X_{it}'\delta + \nu_{it}.}$$

The identifying assumption is

$$\boxed{\mathbb{E}[y_{0it} \mid \alpha_i, y_{i,t-h}, X_{it}, d_{it}] = \mathbb{E}[y_{0it} \mid \alpha_i, y_{i,t-h}, X_{it}].}$$

This is weaker conceptually because it allows both fixed unobserved heterogeneity and lagged outcomes to matter. However, it is much harder to estimate consistently.

5.24 *Nickell Bias in the Fixed Effects Model with Lagged Outcomes*

Consider the model

$$y_{it} = \alpha_i + \theta y_{i,t-1} + \lambda_t + \beta d_{it} + X_{it}'\delta + \nu_{it}.$$

First-difference to remove α_i:

$$\Delta y_{it} = \theta \Delta y_{i,t-1} + \Delta\lambda_t + \beta \Delta d_{it} + \Delta X_{it}'\delta + \Delta\nu_{it}.$$

The problem is that

$$\Delta y_{i,t-1} = y_{i,t-1} - y_{i,t-2}$$

contains $y_{i,t-1}$, and

$$y_{i,t-1} = \alpha_i + \theta y_{i,t-2} + \lambda_{t-1} + \beta d_{i,t-1} + X'_{i,t-1}\delta + \nu_{i,t-1}.$$

Therefore, $\Delta y_{i,t-1}$ contains $\nu_{i,t-1}$.

The differenced error is

$$\Delta \nu_{it} = \nu_{it} - \nu_{i,t-1}.$$

This also contains $-\nu_{i,t-1}$. Therefore,

$$\mathrm{Cov}(\Delta y_{i,t-1}, \Delta \nu_{it}) \neq 0.$$

Hence,

$$\boxed{\text{OLS on the differenced dynamic panel equation is inconsistent.}}$$

This is the Nickell bias problem.

5.25 Instrumenting the Lagged Difference

The differenced dynamic panel equation is

$$\Delta y_{it} = \theta \Delta y_{i,t-1} + \Delta \lambda_t + \beta \Delta d_{it} + \Delta X'_{it}\delta + \Delta \nu_{it}.$$

Because

$$\mathrm{Cov}(\Delta y_{i,t-1}, \Delta \nu_{it}) \neq 0,$$

one possible instrument for $\Delta y_{i,t-1}$ is $y_{i,t-2}$.

For $y_{i,t-2}$ to be valid, we need

$$\mathrm{Cov}(y_{i,t-2}, \Delta \nu_{it}) = 0.$$

Since

$$\Delta \nu_{it} = \nu_{it} - \nu_{i,t-1},$$

this requires

$$\mathrm{Cov}(y_{i,t-2}, \nu_{it}) = 0$$

and

$$\mathrm{Cov}(y_{i,t-2}, \nu_{i,t-1}) = 0.$$

This is plausible only if the idiosyncratic errors are serially uncorrelated. If ν_{it} is serially correlated, then $y_{i,t-2}$ may be correlated with $\nu_{i,t-1}$, making the instrument invalid.

Thus,

$$\boxed{\text{Dynamic panel IV requires strong assumptions on serial correlation.}}$$

5.26 Why Fixed Effects and Lagged Dependent Variable Models Are Not Nested

The LDV model is

$$y_{it} = \alpha + \theta y_{i,t-1} + \lambda_t + \beta d_{it} + X'_{it}\delta + \varepsilon_{it}.$$

One might think the fixed-effects model appears when $\theta = 1$. But if $\theta = 1$, then

$$y_{it} = \alpha + y_{i,t-1} + \lambda_t + \beta d_{it} + X'_{it}\delta + \varepsilon_{it}.$$

Subtract $y_{i,t-1}$ from both sides:

$$\Delta y_{it} = \alpha + \lambda_t + \beta d_{it} + X'_{it}\delta + \varepsilon_{it}.$$

This is not the fixed-effects first-difference equation:

$$\Delta y_{it} = \Delta\lambda_t + \beta\Delta d_{it} + \Delta X'_{it}\delta + \Delta\varepsilon_{it}.$$

In the $\theta = 1$ LDV model, the right-hand-side variables remain in levels, while in the fixed-effects first-difference model, the right-hand-side variables are differenced.

Therefore,

$$\boxed{\text{The fixed-effects model is not a special case of the LDV model.}}$$

5.27 Bracketing Logic: If Fixed Effects Is Correct but LDV Is Estimated

To simplify, ignore covariates and time effects. Suppose the correct model is fixed effects:

$$y_{it} = a_i + \beta d_{it} + \varepsilon_{it},$$

where $\beta > 0$, and ε_{it} is serially uncorrelated and uncorrelated with a_i and d_{it}.

Assume treatment is zero in the previous period:

$$d_{i,t-1} = 0.$$

Then

$$y_{i,t-1} = a_i + \varepsilon_{i,t-1}.$$

Suppose we mistakenly estimate an LDV model:

$$y_{it} = \alpha + \theta y_{i,t-1} + \beta_{LDV} d_{it} + u_{it}.$$

By regression anatomy, the probability limit of $\widehat{\beta}_{LDV}$ is

$$\operatorname{plim} \widehat{\beta}_{LDV} = \frac{\operatorname{Cov}(y_{it}, \widetilde{d}_{it})}{\operatorname{Var}(\widetilde{d}_{it})},$$

where

$$\widetilde{d}_{it} = d_{it} - \psi y_{i,t-1}$$

is the residual from regressing d_{it} on $y_{i,t-1}$.

Now substitute

$$a_i = y_{i,t-1} - \varepsilon_{i,t-1}$$

into the correct model:

$$y_{it} = y_{i,t-1} + \beta d_{it} + \varepsilon_{it} - \varepsilon_{i,t-1}.$$

Taking covariance with $\widetilde{d}_{it}$,

$$\operatorname{Cov}(y_{it}, \widetilde{d}_{it}) = \operatorname{Cov}(y_{i,t-1}, \widetilde{d}_{it}) + \beta \operatorname{Cov}(d_{it}, \widetilde{d}_{it}) + \operatorname{Cov}(\varepsilon_{it}, \widetilde{d}_{it}) - \operatorname{Cov}(\varepsilon_{i,t-1}, \widetilde{d}_{it}).$$

Because $\widetilde{d}_{it}$ is residualized on $y_{i,t-1}$,

$$\operatorname{Cov}(y_{i,t-1}, \widetilde{d}_{it}) = 0.$$

Also,

$$\operatorname{Cov}(\varepsilon_{it}, \widetilde{d}_{it}) = 0.$$

Therefore,

$$\operatorname{Cov}(y_{it}, \widetilde{d}_{it}) = \beta \operatorname{Var}(\widetilde{d}_{it}) - \operatorname{Cov}(\varepsilon_{i,t-1}, \widetilde{d}_{it}).$$

Thus,

$$\operatorname{plim} \widehat{\beta}_{LDV} = \beta - \frac{\operatorname{Cov}(\varepsilon_{i,t-1}, \widetilde{d}_{it})}{\operatorname{Var}(\widetilde{d}_{it})}.$$

Since

$$\widetilde{d}_{it} = d_{it} - \psi y_{i,t-1} = d_{it} - \psi(a_i + \varepsilon_{i,t-1}),$$

and $\varepsilon_{i,t-1}$ is uncorrelated with d_{it} and a_i,

$$\operatorname{Cov}(\varepsilon_{i,t-1}, \widetilde{d}_{it}) = -\psi \operatorname{Var}(\varepsilon_{i,t-1}) = -\psi\sigma_\varepsilon^2.$$

Therefore,

$$\operatorname{plim} \widehat{\beta}_{LDV} = \beta + \frac{\psi\sigma_\varepsilon^2}{\operatorname{Var}(\widetilde{d}_{it})}.$$

If treatment is negatively related to lagged earnings, as in training-program applications, then

$$\psi < 0.$$

Hence,

$$\boxed{\operatorname{plim} \widehat{\beta}_{LDV} < \beta.}$$

So if the fixed-effects model is correct but we estimate an LDV model, a positive treatment effect tends to be underestimated.

5.28 Bracketing Logic: If LDV Is Correct but Fixed Effects Is Estimated

Now suppose the correct model is an LDV model:

$$y_{it} = \alpha + \theta y_{i,t-1} + \beta d_{it} + \varepsilon_{it},$$

where $0 < \theta < 1$, and $\beta > 0$.

Assume treatment is zero in the previous period:

$$d_{i,t-1} = 0.$$

If we mistakenly estimate a first-differenced fixed-effects model, the estimand is

$$\frac{\operatorname{Cov}(y_{it} - y_{i,t-1}, d_{it} - d_{i,t-1})}{\operatorname{Var}(d_{it} - d_{i,t-1})} = \frac{\operatorname{Cov}(y_{it} - y_{i,t-1}, d_{it})}{\operatorname{Var}(d_{it})}.$$

Subtract $y_{i,t-1}$ from both sides of the correct LDV model:

$$y_{it} - y_{i,t-1} = \alpha + (\theta - 1)y_{i,t-1} + \beta d_{it} + \varepsilon_{it}.$$

Take covariance with d_{it}:

$$\operatorname{Cov}(y_{it} - y_{i,t-1}, d_{it}) = (\theta - 1)\operatorname{Cov}(y_{i,t-1}, d_{it}) + \beta \operatorname{Var}(d_{it}) + \operatorname{Cov}(\varepsilon_{it}, d_{it}).$$

Assume
$$\mathrm{Cov}(\varepsilon_{it}, d_{it}) = 0.$$
Then
$$\operatorname{plim} \widehat{\beta}_{FE} = \beta + (\theta - 1)\frac{\mathrm{Cov}(y_{i,t-1}, d_{it})}{\mathrm{Var}(d_{it})}.$$
In training-program applications, trainees tend to have low lagged earnings:
$$\mathrm{Cov}(y_{i,t-1}, d_{it}) < 0.$$
Because
$$0 < \theta < 1,$$
we have
$$\theta - 1 < 0.$$
Therefore,
$$(\theta - 1)\,\mathrm{Cov}(y_{i,t-1}, d_{it}) > 0.$$
Hence,
$$\boxed{\operatorname{plim} \widehat{\beta}_{FE} > \beta.}$$
So if the LDV model is correct but we estimate fixed effects, a positive treatment effect tends to be overestimated.

5.29 Fixed Effects and LDV as Bounds

The previous two results imply a useful bracketing property.

If the fixed-effects model is correct but the LDV model is estimated, then for a positive treatment effect,
$$\widehat{\beta}_{LDV}$$
tends to be too small.

If the LDV model is correct but the fixed-effects model is estimated, then
$$\widehat{\beta}_{FE}$$
tends to be too large.

Therefore, in applications where treated units are selected because of low lagged outcomes, and where $0 < \theta < 1$, fixed-effects and LDV estimates can be interpreted as bounds:
$$\boxed{\widehat{\beta}_{LDV} \lesssim \beta \lesssim \widehat{\beta}_{FE}.}$$

This bracketing logic is not a theorem for every setting, but it is a useful diagnostic for panel-data causal inference.

5.30 Summary of Chapter 5

Chapter 5 studies research designs that use time, cohort, or group variation to control for unobserved confounding. The main idea is that when important omitted variables are unobserved but fixed over time, panel data can remove them by comparing units to themselves over time rather than comparing units in levels. The fixed-effects model begins from

$$y_{it} = \alpha_i + \lambda_t + \rho d_{it} + X_{it}'\delta + \varepsilon_{it},$$

where α_i captures time-invariant individual heterogeneity, λ_t captures common time shocks, and ρ is the causal effect of interest. By subtracting individual means, the fixed effect disappears:

$$y_{it} - \bar{y}_i = (\lambda_t - \bar{\lambda}_i) + \rho(d_{it} - \bar{d}_i) + (X_{it} - \bar{X}_i)'\delta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

Thus, fixed effects identify treatment effects from within-unit changes over time.

The chapter then connects fixed effects to first differences. Starting from

$$y_{it} = \alpha_i + \lambda_t + \rho d_{it} + X_{it}'\delta + \varepsilon_{it},$$

subtracting the previous-period equation gives

$$\Delta y_{it} = \Delta\lambda_t + \rho\Delta d_{it} + \Delta X_{it}'\delta + \Delta\varepsilon_{it}.$$

This removes the same time-invariant α_i. With exactly two periods, the fixed-effects estimator and the first-difference estimator are algebraically equivalent:

$$\widehat{\rho}_{FE} = \widehat{\rho}_{FD}.$$

Both approaches rely on the idea that the omitted confounder is fixed over time.

Difference-in-differences extends this logic to group-level treatment changes. In a two-group, two-period design, the DD estimand is

$$\beta_{DD} = [\mathbb{E}[y \mid T, Post] - \mathbb{E}[y \mid T, Pre]] - [\mathbb{E}[y \mid C, Post] - \mathbb{E}[y \mid C, Pre]]\,.$$

This identifies the treatment effect if, absent treatment, the treated and control groups would have followed parallel trends:

$$\mathbb{E}[y_0 \mid T, Post] - \mathbb{E}[y_0 \mid T, Pre] = \mathbb{E}[y_0 \mid C, Post] - \mathbb{E}[y_0 \mid C, Pre].$$

The regression version is

$$y_{ist} = \alpha + \gamma T_s + \lambda Post_t + \beta(T_s \cdot Post_t) + \varepsilon_{ist},$$

where the coefficient on the interaction term is the DD estimate.

With many groups and time periods, the generalized DD model is

$$y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + X'_{ist}\delta + \varepsilon_{ist}.$$

Here γ_s absorbs permanent group differences, and λ_t absorbs common aggregate shocks. The identifying assumption is that, after these controls, treated and untreated groups would have had comparable counterfactual trends. In practice, this assumption is often assessed using event-study specifications with leads and lags:

$$y_{ist} = \gamma_s + \lambda_t + \sum_{\tau} \beta_\tau d_{s,t-\tau} + \sum_{\ell} \beta_{+\ell} d_{s,t+\ell} + X'_{ist}\delta + \varepsilon_{ist}.$$

The lead coefficients should be close to zero if there are no pre-existing differential trends:

$$\beta_{+\ell} = 0.$$

The chapter also introduces triple differences. Triple differences add a third comparison dimension, such as affected versus unaffected subgroups. A typical model is

$$y_{iast} = \gamma_{st} + \lambda_{at} + \theta_{as} + \beta d_{ast} + X'_{iast}\delta + \varepsilon_{iast}.$$

The idea is to compare changes across treated and control groups, before and after treatment, and across affected and unaffected subgroups. This can make the identifying assumption more plausible when ordinary DD is not convincing.

A major part of the chapter compares fixed-effects models with lagged dependent-variable models. Fixed effects are motivated by the assumption

$$\mathbb{E}[y_{0it} \mid \alpha_i, X_{it}, d_{it}] = \mathbb{E}[y_{0it} \mid \alpha_i, X_{it}],$$

which says that selection is driven by time-invariant unobserved heterogeneity. Lagged dependent-variable models instead rely on

$$\mathbb{E}[y_{0it} \mid y_{i,t-h}, X_{it}, d_{it}] = \mathbb{E}[y_{0it} \mid y_{i,t-h}, X_{it}],$$

which says that selection is driven by past outcomes. The corresponding LDV regression is

$$y_{it} = \alpha + \theta y_{i,t-h} + \lambda_t + \beta d_{it} + X_{it}'\delta + \varepsilon_{it}.$$

This model is attractive when treated units are selected because of recent shocks, such as a pre-program earnings dip.

The chapter warns that fixed effects and lagged dependent-variable models are not nested. Including both gives

$$y_{it} = \alpha_i + \theta y_{i,t-h} + \lambda_t + \beta d_{it} + X_{it}'\delta + \nu_{it},$$

but estimating this model is difficult because differencing creates correlation between the differenced lagged dependent variable and the differenced error:

$$\Delta y_{it} = \theta \Delta y_{i,t-1} + \Delta\lambda_t + \beta \Delta d_{it} + \Delta X_{it}'\delta + \Delta\nu_{it}.$$

Since both $\Delta y_{i,t-1}$ and $\Delta \nu_{it}$ contain $\nu_{i,t-1}$, OLS is inconsistent. This is the Nickell-bias problem.

A useful practical message is the bracketing logic. If the LDV model is correct but fixed effects are estimated, positive treatment effects tend to be overestimated. If the fixed-effects model is correct but an LDV model is estimated, positive treatment effects tend to be underestimated. Thus, under certain conditions, the two estimates can be viewed as approximate bounds:

$$\widehat{\beta}_{LDV} \lesssim \beta \lesssim \widehat{\beta}_{FE}.$$

This is not a universal theorem, but it provides a useful diagnostic when the two identifying stories are both plausible.

Overall, Chapter 5's core logic is:

$$\text{unobserved confounding} \longrightarrow \text{time or group variation} \longrightarrow \text{fixed effects / differences}$$
$$\longrightarrow \text{parallel trends} \longrightarrow \text{DD, DDD, FE, or LDV identification.}$$

The chapter teaches that panel-data methods do not solve selection automatically. They replace selection-on-observables with assumptions about the timing and persistence of unobserved confounders. Fixed effects work when omitted variables are stable; DD works when counterfactual trends are parallel; lagged dependent-variable models work when past outcomes summarize the relevant selection process.

6 Chapter 6 Derivations: Regression Discontinuity Designs

6.1 Sharp Regression Discontinuity Assignment Rule

Sharp regression discontinuity is used when treatment status is a deterministic and discontinuous function of a running variable x_i. Let x_0 denote the cutoff. The sharp RD assignment rule is

$$d_i = \begin{cases} 1, & \text{if } x_i \geq x_0, \\ 0, & \text{if } x_i < x_0. \end{cases}$$

Equivalently,

$$\boxed{d_i = \mathbf{1}[x_i \geq x_0].}$$

This assignment rule is deterministic because once x_i is known, d_i is also known. It is discontinuous because treatment switches sharply at x_0.

6.2 Sharp RD with Linear Untreated Potential Outcomes

Suppose untreated potential outcomes satisfy

$$\mathbb{E}[y_{0i} \mid x_i] = \alpha + \beta x_i.$$

Suppose the treatment effect is constant:

$$y_{1i} = y_{0i} + \rho.$$

Observed outcome is

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i.$$

Since

$$y_{1i} - y_{0i} = \rho,$$

we have

$$y_i = y_{0i} + \rho d_i.$$

Taking conditional expectation given x_i,

$$\mathbb{E}[y_i \mid x_i] = \mathbb{E}[y_{0i} \mid x_i] + \rho d_i.$$

Using the linear specification,

$$\mathbb{E}[y_i \mid x_i] = \alpha + \beta x_i + \rho d_i.$$

Thus the regression model is

$$\boxed{y_i = \alpha + \beta x_i + \rho d_i + \eta_i,}$$

where

$$\mathbb{E}[\eta_i \mid x_i] = 0.$$

The coefficient ρ captures the discontinuous jump in the conditional expectation of y_i at x_0.

6.3 Why the Sharp RD Coefficient Is the Jump at the Cutoff

The sharp RD model is

$$\mathbb{E}[y_i \mid x_i] = \alpha + \beta x_i + \rho \mathbf{1}[x_i \geq x_0].$$

Consider the right-hand limit at the cutoff:

$$\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] = \alpha + \beta x_0 + \rho.$$

Consider the left-hand limit:

$$\lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x] = \alpha + \beta x_0.$$

Subtracting,

$$\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x] = \rho.$$

Therefore,

$$\boxed{\rho = \lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x].}$$

Sharp RD identifies the treatment effect as the discontinuous jump in the conditional mean at the cutoff.

6.4 *Smooth Counterfactual CEF and Sharp RD Identification*

Suppose untreated potential outcomes satisfy

$$\mathbb{E}[y_{0i} \mid x_i] = f(x_i),$$

where $f(\cdot)$ is smooth and continuous at x_0. Assume a constant treatment effect:

$$y_{1i} = y_{0i} + \rho.$$

Observed outcome is

$$y_i = y_{0i} + \rho d_i.$$

Taking conditional expectation:

$$\mathbb{E}[y_i \mid x_i] = f(x_i) + \rho \mathbf{1}[x_i \geq x_0].$$

The right-hand limit is

$$\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] = \lim_{x \downarrow x_0} f(x) + \rho.$$

The left-hand limit is

$$\lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x] = \lim_{x \uparrow x_0} f(x).$$

Since $f(\cdot)$ is continuous at x_0,

$$\lim_{x \downarrow x_0} f(x) = \lim_{x \uparrow x_0} f(x) = f(x_0).$$

Therefore,

$$\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x] = \rho.$$

Thus,

$$\boxed{\text{Continuity of } \mathbb{E}[y_{0i} \mid x_i] \text{ at } x_0 \text{ identifies the sharp RD effect.}}$$

6.5 *Sharp RD with a Polynomial Control Function*

If $f(x_i)$ is unknown but smooth, we may approximate it with a polynomial:

$$f(x_i) = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p.$$

Then the sharp RD regression is

$$\boxed{y_i = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + \rho d_i + \eta_i.}$$

Here

$$d_i = \mathbf{1}[x_i \geq x_0].$$

The coefficient ρ is identified by the discontinuity in d_i, while the polynomial terms absorb the smooth relationship between x_i and untreated outcomes.

6.6 *Why Polynomial Sharp RD Can Mistake Nonlinearity for a Jump*

Suppose the true untreated CEF is nonlinear:

$$\mathbb{E}[y_{0i} \mid x_i] = f(x_i),$$

but the estimated model uses a misspecified approximation:

$$f(x_i) \approx \alpha + \beta_1 x_i + \cdots + \beta_p x_i^p.$$

If the approximation error changes sharply near x_0, then the estimated jump may capture misspecified curvature rather than a causal treatment effect.

Formally, write

$$f(x_i) = \alpha + \beta_1 x_i + \cdots + \beta_p x_i^p + r(x_i),$$

where $r(x_i)$ is approximation error. The estimated regression becomes

$$y_i = \alpha + \beta_1 x_i + \cdots + \beta_p x_i^p + \rho d_i + \{r(x_i) + \eta_i\}.$$

If

$$\mathbb{E}[r(x_i) \mid x_i \downarrow x_0] \neq \mathbb{E}[r(x_i) \mid x_i \uparrow x_0],$$

then the coefficient on d_i can absorb this difference. Therefore,

$$\boxed{\text{RD requires careful modeling of the smooth counterfactual relationship near the cutoff.}}$$

6.7 *Centered Running Variable*

Define the centered running variable

$$\widetilde{x}_i = x_i - x_0.$$

Then the cutoff is normalized to zero:

$$\widetilde{x}_i = 0 \iff x_i = x_0.$$

The treatment indicator becomes

$$d_i = \mathbf{1}[\widetilde{x}_i \geq 0].$$

Centering is useful because the intercepts on the left and right of the cutoff directly describe conditional means at the cutoff.

6.8 Sharp RD with Different Polynomial Trends on Each Side

Let the untreated and treated CEFs be

$$\mathbb{E}[y_{0i} \mid x_i] = f_0(x_i),$$

and

$$\mathbb{E}[y_{1i} \mid x_i] = f_1(x_i).$$

Using the centered running variable $\widetilde{x}_i = x_i - x_0$, approximate both functions by p-th order polynomials:

$$f_0(x_i) = \alpha + \beta_{01}\widetilde{x}_i + \beta_{02}\widetilde{x}_i^2 + \cdots + \beta_{0p}\widetilde{x}_i^p,$$

and

$$f_1(x_i) = \alpha + \rho + \beta_{11}\widetilde{x}_i + \beta_{12}\widetilde{x}_i^2 + \cdots + \beta_{1p}\widetilde{x}_i^p.$$

Because

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i,$$

we have

$$\mathbb{E}[y_i \mid x_i] = \mathbb{E}[y_{0i} \mid x_i] + \left(\mathbb{E}[y_{1i} \mid x_i] - \mathbb{E}[y_{0i} \mid x_i]\right) d_i.$$

Substitute the two polynomial functions:

$$\mathbb{E}[y_i \mid x_i] = \alpha + \beta_{01}\widetilde{x}_i + \beta_{02}\widetilde{x}_i^2 + \cdots + \beta_{0p}\widetilde{x}_i^p$$

$$+ \left[\rho + (\beta_{11} - \beta_{01})\widetilde{x}_i + (\beta_{12} - \beta_{02})\widetilde{x}_i^2 + \cdots + (\beta_{1p} - \beta_{0p})\widetilde{x}_i^p\right] d_i.$$

Define

$$\beta_j^* = \beta_{1j} - \beta_{0j}.$$

Then the regression equation is

$$\boxed{y_i = \alpha + \beta_{01}\widetilde{x}_i + \beta_{02}\widetilde{x}_i^2 + \cdots + \beta_{0p}\widetilde{x}_i^p + \rho d_i + \beta_1^* d_i\widetilde{x}_i + \beta_2^* d_i\widetilde{x}_i^2 + \cdots + \beta_p^* d_i\widetilde{x}_i^p + \eta_i.}$$

This model allows the regression function to have different slopes and curvatures on the two sides of the cutoff.

6.9 *Treatment Effect at the Cutoff in the Interacted Polynomial Model*

The interacted polynomial model is

$$y_i = \alpha + \sum_{j=1}^{p} \beta_{0j}\widetilde{x}_i^j + \rho d_i + \sum_{j=1}^{p} \beta_j^* d_i\widetilde{x}_i^j + \eta_i.$$

For $x_i = x_0$, we have

$$\widetilde{x}_i = 0.$$

The left-hand conditional mean at the cutoff is

$$\lim_{\widetilde{x}\uparrow 0} \mathbb{E}[y_i \mid \widetilde{x}_i = \widetilde{x}] = \alpha.$$

The right-hand conditional mean at the cutoff is

$$\lim_{\widetilde{x}\downarrow 0} \mathbb{E}[y_i \mid \widetilde{x}_i = \widetilde{x}] = \alpha + \rho.$$

All interaction terms disappear at $\widetilde{x}_i = 0$ because

$$d_i\widetilde{x}_i^j = 0 \quad \text{at } \widetilde{x}_i = 0.$$

Therefore,

$$\boxed{\rho = \lim_{\widetilde{x}\downarrow 0} \mathbb{E}[y_i \mid \widetilde{x}_i = \widetilde{x}] - \lim_{\widetilde{x}\uparrow 0} \mathbb{E}[y_i \mid \widetilde{x}_i = \widetilde{x}].}$$

Centering the running variable ensures that the coefficient on d_i is the treatment effect exactly at the cutoff.

6.10 *Treatment Effect Away from the Cutoff*

In the interacted polynomial model,

$$\mathbb{E}[y_i \mid \widetilde{x}_i] = \alpha + \sum_{j=1}^{p} \beta_{0j}\widetilde{x}_i^j + \rho d_i + \sum_{j=1}^{p} \beta_j^* d_i\widetilde{x}_i^j.$$

For a point $c > 0$ to the right of the cutoff, the treatment effect relative to the left-side counterfactual is

$$\tau(c) = \rho + \beta_1^* c + \beta_2^* c^2 + \cdots + \beta_p^* c^p.$$

Therefore,

$$\boxed{\tau(c) = \rho + \sum_{j=1}^{p} \beta_j^* c^j.}$$

The coefficient ρ is the effect at the cutoff, while the interaction terms allow the effect to vary with the distance from the cutoff.

6.11 Sharp RD as a Limit of Local Comparisons

Consider a small window around the cutoff:

$$[x_0 - \delta, x_0 + \delta].$$

For observations just below the cutoff,

$$x_0 - \delta < x_i < x_0,$$

treatment is zero, so

$$y_i = y_{0i}.$$

If the window is very small, then

$$\mathbb{E}[y_i \mid x_0 - \delta < x_i < x_0] \approx \mathbb{E}[y_{0i} \mid x_i = x_0].$$

For observations just above the cutoff,

$$x_0 < x_i < x_0 + \delta,$$

treatment is one, so

$$y_i = y_{1i}.$$

Thus

$$\mathbb{E}[y_i \mid x_0 < x_i < x_0 + \delta] \approx \mathbb{E}[y_{1i} \mid x_i = x_0].$$

Taking the difference and letting $\delta \to 0$,

$$\boxed{\lim_{\delta \to 0} \{\mathbb{E}[y_i \mid x_0 < x_i < x_0 + \delta] - \mathbb{E}[y_i \mid x_0 - \delta < x_i < x_0]\} = \mathbb{E}[y_{1i} - y_{0i} \mid x_i = x_0].}$$

This is the nonparametric sharp RD estimand.

6.12 *Why Sharp RD Is Local*

The sharp RD estimand is

$$\mathbb{E}[y_{1i} - y_{0i} \mid x_i = x_0].$$

It is local because it identifies the treatment effect only for units at the cutoff. Units far from the cutoff are not directly comparable because their running variable values differ substantially.

Thus,

$$\boxed{\text{Sharp RD identifies the treatment effect at } x_0, \text{ not necessarily the population average effect.}}$$

6.13 *Local Constant RD Estimator*

A simple local estimator uses observations in a bandwidth δ around the cutoff. Define the right-side sample:

$$\mathcal{R}_\delta = \{i : x_0 < x_i < x_0 + \delta\},$$

and the left-side sample:

$$\mathcal{L}_\delta = \{i : x_0 - \delta < x_i < x_0\}.$$

The local constant sharp RD estimator is

$$\boxed{\widehat{\tau}_{LC} = \frac{1}{N_R} \sum_{i \in \mathcal{R}_\delta} y_i - \frac{1}{N_L} \sum_{i \in \mathcal{L}_\delta} y_i.}$$

As $\delta \to 0$, this estimates the discontinuity in the conditional mean at the cutoff. In finite samples, however, a very small δ gives few observations and therefore imprecise estimates.

6.14 *Boundary Bias in Local Constant RD*

The local constant estimator estimates

$$\mathbb{E}[y_i \mid x_0 < x_i < x_0 + \delta] - \mathbb{E}[y_i \mid x_0 - \delta < x_i < x_0].$$

Suppose the untreated CEF is smooth but sloped near the cutoff:

$$\mathbb{E}[y_{0i} \mid x_i] = f(x_i), \qquad f'(x_0) \neq 0.$$

Then the average of $f(x_i)$ on the right side is generally not exactly $f(x_0)$, and the average on the left side is also not exactly $f(x_0)$. The difference between these averages creates boundary bias.

For small δ, if the running variable is approximately uniformly distributed near x_0, then

$$\mathbb{E}[x_i - x_0 \mid x_0 < x_i < x_0 + \delta] \approx \frac{\delta}{2},$$

and

$$\mathbb{E}[x_i - x_0 \mid x_0 - \delta < x_i < x_0] \approx -\frac{\delta}{2}.$$

Using a first-order expansion,

$$f(x_i) \approx f(x_0) + f'(x_0)(x_i - x_0).$$

Therefore, the right-left difference in untreated potential outcomes is approximately

$$f'(x_0)\left(\frac{\delta}{2} - \left(-\frac{\delta}{2}\right)\right) = f'(x_0)\delta.$$

Thus,

$$\boxed{\text{Local constant RD can be biased when the underlying CEF has slope near the cutoff.}}$$

This motivates local linear regression.

6.15 Local Linear Sharp RD

Local linear regression fits separate linear functions on the two sides of the cutoff, giving more weight to observations close to x_0. Let

$$u_i = x_i - x_0.$$

The local linear sharp RD model can be written as

$$y_i = \alpha + \tau d_i + \beta_0 u_i + \beta_1 d_i u_i + \eta_i,$$

estimated using weights

$$K\left(\frac{u_i}{h}\right),$$

where $K(\cdot)$ is a kernel function and h is the bandwidth.

The weighted least squares problem is

$$\boxed{(\widehat{\alpha}, \widehat{\tau}, \widehat{\beta}_0, \widehat{\beta}_1) = \arg\min_{\alpha,\tau,\beta_0,\beta_1} \sum_i K\left(\frac{x_i - x_0}{h}\right)[y_i - \alpha - \tau d_i - \beta_0(x_i - x_0) - \beta_1 d_i(x_i - x_0)]^2.}$$

The RD estimate is

$$\boxed{\widehat{\tau}_{LL} = \widehat{\tau}.}$$

This estimator estimates the jump at the cutoff after fitting local linear trends on each side.

6.16 *Bandwidth Tradeoff*

The bandwidth h determines which observations receive meaningful weight. A small bandwidth uses observations very close to the cutoff. This reduces bias because the smooth counterfactual function is easier to approximate locally:

$$h \downarrow \quad \Rightarrow \quad \text{bias decreases.}$$

But a small bandwidth also reduces the effective sample size:

$$h \downarrow \quad \Rightarrow \quad \text{variance increases.}$$

A larger bandwidth improves precision but requires stronger functional-form assumptions:

$$h \uparrow \quad \Rightarrow \quad \text{variance decreases, but bias may increase.}$$

Thus,

$$\boxed{\text{RD bandwidth choice is a bias–variance tradeoff.}}$$

6.17 *Discontinuity-Sample Robustness Check*

A practical RD robustness check is to estimate the model in smaller and smaller neighborhoods around the cutoff:

$$[x_0 - \delta, x_0 + \delta].$$

As δ shrinks, fewer polynomial terms should be needed to model the smooth relationship between x_i and y_i. For example, one may compare

$$y_i = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \rho d_i + \eta_i$$

in the full sample with

$$y_i = \alpha + \beta_1 x_i + \rho d_i + \eta_i$$

or even

$$y_i = \alpha + \rho d_i + \eta_i$$

in a narrow discontinuity sample.

A credible RD design should show that

$$\boxed{\widehat{\rho} \text{ remains broadly stable as the bandwidth shrinks and polynomial order falls.}}$$

6.18 *Pre-Treatment Covariate Balance Check*

Let w_i be a predetermined covariate that cannot be affected by treatment. In a valid RD design, w_i should not jump at the cutoff. The corresponding test regression is

$$w_i = \alpha + \beta_1 x_i + \cdots + \beta_p x_i^p + \theta d_i + v_i.$$

The null hypothesis is

$$\boxed{H_0 : \theta = 0.}$$

Equivalently,

$$\boxed{\lim_{x \downarrow x_0} \mathbb{E}[w_i \mid x_i = x] = \lim_{x \uparrow x_0} \mathbb{E}[w_i \mid x_i = x].}$$

A significant discontinuity in predetermined variables suggests that units just above and below the cutoff may not be comparable.

6.19 *Density Manipulation Check*

A key RD concern is manipulation of the running variable near the cutoff. If agents can precisely manipulate x_i, then observations just above and just below x_0 may differ systematically.

Let $f_X(x)$ denote the density of the running variable. A basic implication of no precise manipulation is

$$\boxed{\lim_{x \downarrow x_0} f_X(x) = \lim_{x \uparrow x_0} f_X(x).}$$

A jump in the density at x_0 suggests bunching or sorting around the cutoff.

This is the intuition behind the McCrary density test:

$$\boxed{H_0 : \text{the density of } x_i \text{ is continuous at } x_0.}$$

6.20 Lee Election RD Example

In Lee's election RD design, the running variable is the Democratic margin of victory in election t:

$$x_i = \text{Democratic vote share} - \text{Republican vote share}.$$

Treatment is Democratic incumbency:

$$d_i = \mathbf{1}[x_i \geq 0].$$

The outcome is whether the Democrat wins in election $t+1$:

$$y_i = \mathbf{1}[\text{Democrat wins election } t+1].$$

The sharp RD estimand is

$$\boxed{\tau_{Lee} = \lim_{x \downarrow 0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow 0} \mathbb{E}[y_i \mid x_i = x].}$$

Because treatment switches mechanically at a zero vote margin, close elections provide quasi-random variation in incumbency.

6.21 Placebo Outcome Check in the Lee Design

A placebo outcome in the Lee design is the number of past Democratic victories before election t. Denote this variable by w_i. Because w_i was determined before election t, it should not be affected by crossing the cutoff in election t.

The placebo RD estimand is

$$\theta_w = \lim_{x \downarrow 0} \mathbb{E}[w_i \mid x_i = x] - \lim_{x \uparrow 0} \mathbb{E}[w_i \mid x_i = x].$$

A valid RD design predicts

$$\boxed{\theta_w = 0.}$$

A discontinuity in past victories would suggest that candidates barely winning and barely losing are not comparable.

6.22 Fuzzy RD Treatment Probability

In fuzzy RD, treatment is not a deterministic function of the running variable. Instead, the probability or expected value of treatment jumps at the cutoff:

$$\mathbb{P}(d_i = 1 \mid x_i) = \begin{cases} g_1(x_i), & \text{if } x_i \geq x_0, \\ g_0(x_i), & \text{if } x_i < x_0, \end{cases}$$

with

$$g_1(x_0) \neq g_0(x_0).$$

Let

$$t_i = \mathbf{1}[x_i \geq x_0].$$

Then the treatment probability can be written as

$$\mathbb{E}[d_i \mid x_i] = g_0(x_i) + \{g_1(x_i) - g_0(x_i)\}t_i.$$

Thus,

$$\boxed{t_i \text{ creates a discontinuity in the conditional probability of treatment.}}$$

6.23 Fuzzy RD First Stage with Polynomial Controls

Approximate the left-side treatment probability by a polynomial and allow a different polynomial on the right side:

$$g_0(x_i) = \gamma_{00} + \gamma_{01}x_i + \gamma_{02}x_i^2 + \cdots + \gamma_{0p}x_i^p.$$

Let the right-left difference be

$$g_1(x_i) - g_0(x_i) = \gamma_0^* + \gamma_1^* x_i + \gamma_2^* x_i^2 + \cdots + \gamma_p^* x_i^p.$$

Then

$$\begin{aligned} \mathbb{E}[d_i \mid x_i] = \gamma_{00} + \gamma_{01}x_i + \gamma_{02}x_i^2 + \cdots + \gamma_{0p}x_i^p \\ + \gamma_0^* t_i + \gamma_1^* x_i t_i + \gamma_2^* x_i^2 t_i + \cdots + \gamma_p^* x_i^p t_i. \end{aligned}$$

The corresponding first-stage equation is

$$\boxed{d_i = \gamma_{00} + \gamma_{01}x_i + \gamma_{02}x_i^2 + \cdots + \gamma_{0p}x_i^p + \gamma_0^* t_i + \gamma_1^* x_i t_i + \cdots + \gamma_p^* x_i^p t_i + \xi_{1i}.}$$

The excluded instruments are

$$t_i, \quad x_i t_i, \quad x_i^2 t_i, \quad \dots, \quad x_i^p t_i.$$

6.24 Simple Fuzzy RD First Stage

A simpler fuzzy RD uses only the cutoff indicator t_i as the excluded instrument:

$$\boxed{d_i = \gamma_0 + \gamma_1 x_i + \gamma_2 x_i^2 + \cdots + \gamma_p x_i^p + \pi t_i + \xi_{1i}.}$$

Here π is the first-stage jump in treatment at the cutoff, after controlling for smooth functions of x_i. The relevance condition is

$$\boxed{\pi \neq 0.}$$

6.25 Fuzzy RD Reduced Form

The structural equation is

$$y_i = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + \rho d_i + \eta_i.$$

The simple first stage is

$$d_i = \gamma_0 + \gamma_1 x_i + \gamma_2 x_i^2 + \cdots + \gamma_p x_i^p + \pi t_i + \xi_{1i}.$$

Substitute the first stage into the structural equation:

$$y_i = \alpha + \beta_1 x_i + \cdots + \beta_p x_i^p + \rho \left(\gamma_0 + \gamma_1 x_i + \cdots + \gamma_p x_i^p + \pi t_i + \xi_{1i} \right) + \eta_i.$$

Collect terms:

$$y_i = \mu + \kappa_1 x_i + \kappa_2 x_i^2 + \cdots + \kappa_p x_i^p + \rho \pi t_i + \xi_{2i},$$

where

$$\mu = \alpha + \rho \gamma_0,$$

$$\kappa_j = \beta_j + \rho \gamma_j, \qquad j = 1, \dots, p,$$

and

$$\xi_{2i} = \rho \xi_{1i} + \eta_i.$$

Therefore, the reduced form is

$$\boxed{y_i = \mu + \kappa_1 x_i + \kappa_2 x_i^2 + \cdots + \kappa_p x_i^p + \rho\pi t_i + \xi_{2i}.}$$

6.26 Fuzzy RD as an IV Ratio

In the simple fuzzy RD model, the first-stage coefficient on t_i is

$$\pi.$$

The reduced-form coefficient on t_i is

$$\rho\pi.$$

Therefore,

$$\boxed{\rho = \frac{\text{reduced-form jump in } y_i}{\text{first-stage jump in } d_i} = \frac{\rho\pi}{\pi}.}$$

In sample implementation, this is a 2SLS estimator using t_i as an instrument for d_i, controlling for a smooth polynomial in x_i.

6.27 Nonparametric Fuzzy RD Wald Estimand

Consider a small neighborhood around the cutoff. The reduced-form jump is

$$\lim_{\delta \to 0} \left\{ \mathbb{E}[y_i \mid x_0 < x_i < x_0 + \delta] - \mathbb{E}[y_i \mid x_0 - \delta < x_i < x_0] \right\}.$$

The first-stage jump is

$$\lim_{\delta \to 0} \left\{ \mathbb{E}[d_i \mid x_0 < x_i < x_0 + \delta] - \mathbb{E}[d_i \mid x_0 - \delta < x_i < x_0] \right\}.$$

The fuzzy RD estimand is their ratio:

$$\boxed{\rho_{FRD} = \lim_{\delta \to 0} \frac{\mathbb{E}[y_i \mid x_0 < x_i < x_0 + \delta] - \mathbb{E}[y_i \mid x_0 - \delta < x_i < x_0]}{\mathbb{E}[d_i \mid x_0 < x_i < x_0 + \delta] - \mathbb{E}[d_i \mid x_0 - \delta < x_i < x_0]}.}$$

This is a Wald estimand localized at the cutoff.

6.28 Fuzzy RD as Local Average Treatment Effect

Let

$$d_i(1)$$

be potential treatment status just to the right of the cutoff, and let

$$d_i(0)$$

be potential treatment status just to the left of the cutoff. Let potential outcomes be y_{1i} and y_{0i}.

Assume local independence, exclusion, relevance, and monotonicity:

$$d_i(1) \geq d_i(0).$$

The reduced-form jump is

$$\mathbb{E}[y_i \mid x_i = x_0^+] - \mathbb{E}[y_i \mid x_i = x_0^-] = \mathbb{E}[(y_{1i} - y_{0i})(d_i(1) - d_i(0)) \mid x_i = x_0].$$

Under monotonicity,

$$d_i(1) - d_i(0) = 1$$

for compliers and zero otherwise. Therefore,

$$\mathbb{E}[(y_{1i} - y_{0i})(d_i(1) - d_i(0)) \mid x_i = x_0]$$

$$= \mathbb{E}[y_{1i} - y_{0i} \mid d_i(1) > d_i(0), x_i = x_0]\mathbb{P}(d_i(1) > d_i(0) \mid x_i = x_0).$$

The first-stage jump is

$$\mathbb{E}[d_i \mid x_i = x_0^+] - \mathbb{E}[d_i \mid x_i = x_0^-] = \mathbb{P}(d_i(1) > d_i(0) \mid x_i = x_0).$$

Taking the ratio gives

$$\boxed{\rho_{FRD} = \mathbb{E}[y_{1i} - y_{0i} \mid d_i(1) > d_i(0), x_i = x_0].}$$

Thus fuzzy RD identifies a LATE for compliers at the cutoff.

6.29 Fuzzy RD with Treatment-Running-Variable Interactions

With heterogeneous treatment effects by x_i, use the second-stage model

$$y_i = \alpha + \sum_{j=1}^{p} \beta_{0j}\tilde{x}_i^j + \rho d_i + \sum_{j=1}^{p} \beta_j^* d_i \tilde{x}_i^j + \eta_i.$$

The endogenous variables are

$$d_i, \quad d_i\widetilde{x}_i, \quad d_i\widetilde{x}_i^2, \quad \ldots, \quad d_i\widetilde{x}_i^p.$$

The excluded instruments are

$$t_i, \quad t_i\widetilde{x}_i, \quad t_i\widetilde{x}_i^2, \quad \ldots, \quad t_i\widetilde{x}_i^p.$$

The first stage for d_i is

$$\boxed{d_i = \gamma_{00} + \gamma_{01}\widetilde{x}_i + \gamma_{02}\widetilde{x}_i^2 + \cdots + \gamma_{0p}\widetilde{x}_i^p + \gamma_0^* t_i + \gamma_1^* \widetilde{x}_i t_i + \cdots + \gamma_p^* \widetilde{x}_i^p t_i + \xi_{1i}.}$$

Analogous first stages are constructed for each endogenous interaction term.

6.30 *Why Centering Matters in Fuzzy RD with Interactions*

If the running variable is centered,

$$\widetilde{x}_i = x_i - x_0,$$

then at the cutoff,

$$\widetilde{x}_i = 0.$$

In the second-stage model,

$$y_i = \alpha + \sum_{j=1}^{p} \beta_{0j}\widetilde{x}_i^j + \rho d_i + \sum_{j=1}^{p} \beta_j^* d_i \widetilde{x}_i^j + \eta_i,$$

all interaction terms vanish at the cutoff. Therefore, ρ is the causal effect at $x_i = x_0$.

If the running variable is not centered, the coefficient on d_i corresponds to the treatment effect at $x_i = 0$, not at $x_i = x_0$. Hence,

$$\boxed{\text{Centering at the cutoff is necessary for } \rho \text{ to measure the RD effect at } x_0.}$$

6.31 *Multiple Discontinuities*

Suppose there are multiple cutoffs:

$$x_1^0, x_2^0, \ldots, x_K^0.$$

A general RD design can use multiple discontinuity indicators:

$$t_{ki} = \mathbf{1}[x_i \geq x_k^0], \qquad k = 1, \dots, K.$$

In a fuzzy RD design, each t_{ki} may generate a jump in treatment:

$$\mathbb{E}[d_i \mid x_i] = f(x_i) + \sum_{k=1}^{K} \pi_k t_{ki}.$$

If each cutoff is valid, the instruments

$$t_{1i}, \dots, t_{Ki}$$

can be used jointly in 2SLS. The resulting IV estimate combines the local treatment effects generated by each discontinuity.

6.32 Maimonides' Rule Predicted Class Size

In the Angrist and Lavy class-size application, let e_s be enrollment in school s. If classes are capped at 40 students and grades are split into equal-sized classes, the predicted number of classes is

$$\text{int}\left(\frac{e_s - 1}{40}\right) + 1,$$

where $\text{int}(\cdot)$ denotes the integer part.

Predicted class size is enrollment divided by the predicted number of classes:

$$\boxed{m_{sc} = \frac{e_s}{\text{int}\left(\frac{e_s - 1}{40}\right) + 1}.}$$

This function has sharp drops at

$$e_s = 41, 81, 121, \dots$$

because the predicted number of classes increases by one at those points.

6.33 Maimonides' Rule as a Fuzzy RD Instrument

Let n_{sc} denote actual class size, and let m_{sc} denote predicted class size from Maimonides' Rule. The first stage is

$$n_{sc} = \delta_0 + \delta_1 m_{sc} + f(e_s) + u_{sc}.$$

The instrument is relevant if

$$\boxed{\delta_1 \neq 0.}$$

Because actual class size is not perfectly determined by the rule, the design is fuzzy rather than sharp. The rule creates discontinuities in predicted class size, which shift actual class size but do not directly affect test scores except through class size.

6.34 Class Size Structural Equation

Let y_{isc} be the test score of student i in school s, class c. Let n_{sc} be actual class size. The structural equation is

$$\boxed{y_{isc} = \alpha_0 + \alpha_1 pds_s + \beta_1 e_s + \beta_2 e_s^2 + \cdots + \beta_p e_s^p + \rho n_{sc} + \eta_{isc}.}$$

Here pds_s is the proportion of disadvantaged students in the school, and e_s is enrollment.

Actual class size n_{sc} may be endogenous because schools with different enrollments and backgrounds may differ in unobserved ways. Maimonides' Rule provides an instrument:

$$m_{sc} = \frac{e_s}{\operatorname{int}\left(\frac{e_s - 1}{40}\right) + 1}.$$

6.35 2SLS for the Class Size RD Design

The first stage is

$$n_{sc} = \gamma_0 + \gamma_1 pds_s + \lambda_1 e_s + \lambda_2 e_s^2 + \cdots + \lambda_p e_s^p + \pi m_{sc} + \xi_{sc}.$$

The second stage is

$$y_{isc} = \alpha_0 + \alpha_1 pds_s + \beta_1 e_s + \beta_2 e_s^2 + \cdots + \beta_p e_s^p + \rho \widehat{n}_{sc} + \text{second-stage error}.$$

Equivalently, in proper IV notation:

$$\boxed{n_{sc} \text{ is instrumented by } m_{sc}, \text{ controlling for } pds_s \text{ and smooth functions of } e_s.}$$

The coefficient ρ is interpreted as the causal effect of class size for the enrollment margins where Maimonides' Rule changes class size.

6.36 *Why Enrollment Must Be Controlled in Maimonides' Rule Designs*

Predicted class size m_{sc} is a deterministic function of enrollment e_s. Enrollment may itself be related to test scores because larger schools may differ from smaller schools:

$$\mathbb{E}[\eta_{isc} \mid e_s] \neq 0.$$

Therefore, the RD design must compare schools near the discontinuities while controlling for smooth functions of enrollment:

$$e_s, \quad e_s^2, \quad \ldots, \quad e_s^p.$$

The identifying idea is that while test scores may vary smoothly with enrollment, class size changes discontinuously at the Maimonides cutoffs. Thus,

$$\boxed{\begin{array}{c}\text{Identification comes from discontinuous changes in predicted class size,}\\ \text{not from smooth variation in enrollment.}\end{array}}$$

6.37 *Effect Size Calculation in the Class Size Example*

Suppose the estimated class-size effect is approximately

$$\widehat{\rho} = -0.25.$$

This means one additional student lowers the test score by about 0.25 points.

A 7-student reduction in class size changes scores by

$$-7\widehat{\rho}.$$

Substituting $\widehat{\rho} = -0.25$,

$$-7(-0.25) = 1.75.$$

If the standard deviation of class-average scores is approximately

$$\sigma = 9.6,$$

then the effect size is

$$\frac{1.75}{9.6} \approx 0.18.$$

Thus,

$$\boxed{\text{A 7-student class-size reduction raises scores by about } 0.18\sigma.}$$

6.38 Precision Cost of Narrow Discontinuity Samples

Let $N(h)$ denote the number of observations within bandwidth h:

$$N(h) = \#\{i : |x_i - x_0| \le h\}.$$

As h decreases,

$$N(h) \downarrow .$$

The standard error of a local RD estimate is approximately proportional to

$$\frac{1}{\sqrt{N(h)}}.$$

Thus,

$$h \downarrow \quad \Rightarrow \quad N(h) \downarrow \quad \Rightarrow \quad SE(\widehat{\tau}) \uparrow .$$

Therefore,

$$\boxed{\text{Narrower RD samples reduce bias but increase standard errors.}}$$

6.39 Sharp RD, Fuzzy RD, and the IV Analogy

Sharp RD identifies

$$\boxed{\tau_{SRD} = \lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x].}$$

This is a treatment-control comparison at the cutoff because treatment changes from 0 to 1 exactly at x_0.

Fuzzy RD identifies

$$\boxed{\tau_{FRD} = \frac{\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x]}{\lim_{x \downarrow x_0} \mathbb{E}[d_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[d_i \mid x_i = x]}.}$$

This is an IV estimand because the discontinuity indicator shifts treatment probability but does not deterministically assign treatment.

Thus,

$$\boxed{\text{Sharp RD is like selection on a smooth running variable; fuzzy RD is local IV.}}$$

6.40 *Summary of Chapter 6*

Chapter 6 studies regression discontinuity designs. The core idea is that many real-world policies assign treatment using explicit rules. When treatment changes discontinuously at a known cutoff, observations just above and just below the cutoff can be compared as if they come from a local experiment. Let x_i be the running variable and x_0 be the cutoff. In a sharp RD design, treatment is deterministically assigned by

$$d_i = \mathbf{1}[x_i \geq x_0].$$

The key identifying idea is that untreated potential outcomes should evolve smoothly through the cutoff:

$$\lim_{x \downarrow x_0} \mathbb{E}[y_{0i} \mid x_i = x] = \lim_{x \uparrow x_0} \mathbb{E}[y_{0i} \mid x_i = x].$$

If this continuity condition holds, then any jump in the observed outcome at the cutoff can be interpreted as the causal effect of treatment:

$$\tau_{SRD} = \lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x].$$

Sharp RD can be written as a regression model. If untreated outcomes are locally linear,

$$\mathbb{E}[y_{0i} \mid x_i] = \alpha + \beta x_i,$$

and treatment effects are constant, then

$$y_i = \alpha + \beta x_i + \rho d_i + \eta_i.$$

Here ρ captures the discontinuous jump in the conditional mean of y_i at the cutoff. More generally, the smooth part of the relationship between y_i and x_i can be modeled using polynomials:

$$y_i = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + \rho d_i + \eta_i.$$

The chapter emphasizes that functional form is especially important in RD because there is no exact overlap at the same value of x_i: below the cutoff all observations are untreated, and above the cutoff all observations are treated.

A practical way to write RD is to center the running variable at the cutoff:

$$\widetilde{x}_i = x_i - x_0.$$

Then the treatment indicator is

$$d_i = \mathbf{1}[\widetilde{x}_i \geq 0].$$

Centering makes the coefficient on d_i directly interpretable as the treatment effect at the cutoff. A flexible sharp RD specification allows different trends on the two sides:

$$y_i = \alpha + \sum_{j=1}^{p} \beta_{0j}\widetilde{x}_i^j + \rho d_i + \sum_{j=1}^{p} \beta_j^* d_i \widetilde{x}_i^j + \eta_i.$$

At the cutoff, $\widetilde{x}_i = 0$, so all interaction terms vanish and

$$\rho = \lim_{\widetilde{x} \downarrow 0} \mathbb{E}[y_i \mid \widetilde{x}_i = \widetilde{x}] - \lim_{\widetilde{x} \uparrow 0} \mathbb{E}[y_i \mid \widetilde{x}_i = \widetilde{x}].$$

The chapter also explains local RD estimation. Instead of relying on high-order global polynomials, one can restrict attention to a narrow bandwidth around the cutoff:

$$[x_0 - h, x_0 + h].$$

The nonparametric sharp RD estimand is

$$\lim_{h \to 0} \left\{ \mathbb{E}[y_i \mid x_0 < x_i < x_0 + h] - \mathbb{E}[y_i \mid x_0 - h < x_i < x_0] \right\}.$$

The bandwidth creates a bias–variance tradeoff. A smaller bandwidth improves local comparability but increases standard errors:

$$h \downarrow \quad \Rightarrow \quad \text{bias decreases, variance increases.}$$

This motivates local linear regression, which estimates separate local slopes on each side of the cutoff:

$$y_i = \alpha + \tau d_i + \beta_0(x_i - x_0) + \beta_1 d_i (x_i - x_0) + \eta_i,$$

using kernel weights that give more importance to observations close to x_0.

Fuzzy RD arises when crossing the cutoff does not perfectly determine treatment, but instead changes the probability or intensity of treatment. Let

$$t_i = \mathbf{1}[x_i \geq x_0].$$

In a fuzzy RD design, t_i is an instrument for actual treatment d_i. The first-stage jump is

$$\lim_{x \downarrow x_0} \mathbb{E}[d_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[d_i \mid x_i = x],$$

and the reduced-form jump is

$$\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x].$$

The fuzzy RD estimand is their ratio:

$$\tau_{FRD} = \frac{\lim_{x \downarrow x_0} \mathbb{E}[y_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[y_i \mid x_i = x]}{\lim_{x \downarrow x_0} \mathbb{E}[d_i \mid x_i = x] - \lim_{x \uparrow x_0} \mathbb{E}[d_i \mid x_i = x]}.$$

Thus, fuzzy RD is a local IV design.

The fuzzy RD estimand has a LATE interpretation. Under local independence, exclusion, relevance, and monotonicity, fuzzy RD identifies the average treatment effect for compliers at the cutoff:

$$\tau_{FRD} = \mathbb{E}[y_{1i} - y_{0i} \mid d_i(1) > d_i(0), x_i = x_0].$$

These are individuals whose treatment status changes when the running variable crosses the cutoff. The effect is therefore local in two senses: it is local to the cutoff x_0, and local to the complier group.

The chapter's main empirical example is Angrist and Lavy's class-size study. Israeli schools are subject to Maimonides' Rule, under which classes are capped at 40 students. Predicted class size is

$$m_{sc} = \frac{e_s}{\text{int}\left(\frac{e_s - 1}{40}\right) + 1},$$

where e_s is grade enrollment. This rule creates discontinuous drops in predicted class size at enrollments $41, 81, 121, \ldots$. Because actual class size does not perfectly follow the rule, the design is fuzzy. Predicted class size from Maimonides' Rule is used as an instrument for actual class size:

$$y_{isc} = \alpha_0 + \alpha_1 pds_s + \beta_1 e_s + \beta_2 e_s^2 + \cdots + \beta_p e_s^p + \rho n_{sc} + \eta_{isc},$$

where n_{sc} is actual class size and m_{sc} instruments for it.

The key lesson is that RD designs convert institutional rules into research designs. Identification comes from discontinuities generated by rules, while smooth variation in the running variable must be controlled flexibly. The practical checklist is:

$$\text{known cutoff} \longrightarrow \text{jump in treatment or treatment probability} \longrightarrow \text{smooth potential outcomes}$$
$$\longrightarrow \text{local comparison around cutoff} \longrightarrow \text{sharp RD or fuzzy RD estimate.}$$

Overall, Chapter 6's core logic is:

$$\text{rule-based assignment} \longrightarrow \text{discontinuity at cutoff}$$
$$\longrightarrow \text{smooth counterfactual outcome} \longrightarrow \text{local causal effect.}$$

Sharp RD estimates the jump in outcomes directly. Fuzzy RD estimates the ratio of the outcome jump to the treatment jump and is interpreted as a local IV estimate for compliers at the cutoff.

7 Chapter 7 Derivations: Quantile Regression

7.1 *Conditional Quantile Function*

Let y_i be a continuously distributed outcome, and let X_i be a vector of covariates. The conditional distribution function of y_i given X_i is

$$F_Y(y \mid X_i) = \mathbb{P}(y_i \leq y \mid X_i).$$

The conditional quantile function at quantile index $\tau \in (0, 1)$ is

$$\boxed{Q_\tau(y_i \mid X_i) = F_Y^{-1}(\tau \mid X_i).}$$

Equivalently, for a general distribution,

$$\boxed{Q_\tau(y_i \mid X_i) = \inf\{y : F_Y(y \mid X_i) \geq \tau\}.}$$

For example,

$$Q_{0.10}(y_i \mid X_i)$$

is the conditional lower decile, and

$$Q_{0.50}(y_i \mid X_i)$$

is the conditional median.

The conditional quantile function is the quantile analogue of the conditional expectation function:

$$\mathbb{E}[y_i \mid X_i].$$

The CEF describes the conditional mean, while the CQF describes the conditional distribution at different points.

7.2 The Check Function

Quantile regression is based on the check function

$$\boxed{\rho_\tau(u) = (\tau - \mathbf{1}[u \leq 0])u.}$$

Equivalently, the check function can be written piecewise as

$$\rho_\tau(u) = \begin{cases} \tau u, & u > 0, \\ (\tau - 1)u, & u \leq 0. \end{cases}$$

Since $u \leq 0$ implies $(\tau - 1)u = (1 - \tau)|u|$, we can also write

$$\boxed{\rho_\tau(u) = \tau u \mathbf{1}[u > 0] + (1 - \tau)(-u)\mathbf{1}[u \leq 0].}$$

Thus, positive residuals and negative residuals are weighted asymmetrically.

When $\tau = 1/2$,

$$\rho_{1/2}(u) = \frac{1}{2}|u|.$$

Therefore, median regression is equivalent to least absolute deviations.

7.3 Why the Check Function Picks Out a Quantile

Let q be a scalar predictor of y_i. The population problem is

$$q_\tau = \arg\min_q \mathbb{E}[\rho_\tau(y_i - q)].$$

Define

$$L(q) = \mathbb{E}[\rho_\tau(y_i - q)].$$

For observations with $y_i > q$, the residual $u_i = y_i - q > 0$, so

$$\rho_\tau(y_i - q) = \tau(y_i - q).$$

For observations with $y_i \leq q$, the residual $u_i = y_i - q \leq 0$, so

$$\rho_\tau(y_i - q) = (\tau - 1)(y_i - q).$$

Differentiate with respect to q. When $y_i > q$,

$$\frac{\partial}{\partial q}\tau(y_i - q) = -\tau.$$

When $y_i \leq q$,

$$\frac{\partial}{\partial q}(\tau - 1)(y_i - q) = 1 - \tau.$$

Therefore,

$$\frac{\partial L(q)}{\partial q} = -\tau \mathbb{P}(y_i > q) + (1 - \tau)\mathbb{P}(y_i \leq q).$$

Set the derivative equal to zero:

$$-\tau[1 - F_Y(q)] + (1 - \tau)F_Y(q) = 0.$$

Expand:

$$-\tau + \tau F_Y(q) + F_Y(q) - \tau F_Y(q) = 0.$$

Thus,

$$F_Y(q) = \tau.$$

Therefore,

$$\boxed{q_\tau = Q_\tau(y_i).}$$

This shows that minimizing expected check loss gives the τ-quantile.

7.4 Conditional Quantile Function as a Minimization Problem

The conditional quantile function solves

$$\boxed{Q_\tau(y_i \mid X_i) = \arg\min_{q(X_i)} \mathbb{E}[\rho_\tau(y_i - q(X_i))].}$$

To see why, fix $X_i = x$. Conditional on $X_i = x$, the same scalar derivation gives

$$q_\tau(x) = \arg\min_q \mathbb{E}[\rho_\tau(y_i - q) \mid X_i = x].$$

The first-order condition is

$$F_Y(q \mid X_i = x) = \tau.$$

Thus,

$$q_\tau(x) = Q_\tau(y_i \mid X_i = x).$$

Since this holds for every x,

$$\boxed{Q_\tau(y_i \mid X_i) = \arg\min_{q(X_i)} \mathbb{E}[\rho_\tau(y_i - q(X_i))].}$$

7.5 Linear Quantile Regression Model

Because the full CQF may be difficult to estimate nonparametrically, quantile regression approximates it with a linear function:

$$q(X_i) = X_i'b.$$

The population quantile regression coefficient is

$$\boxed{\beta_\tau = \arg\min_{b\in\mathbb{R}^d} \mathbb{E}[\rho_\tau(y_i - X_i'b)].}$$

The sample analog is

$$\boxed{\widehat{\beta}_\tau = \arg\min_{b\in\mathbb{R}^d} \sum_{i=1}^{n} \rho_\tau(y_i - X_i'b).}$$

This is the quantile-regression analogue of OLS:

$$\widehat{\beta}_{OLS} = \arg\min_b \sum_{i=1}^{n} (y_i - X_i'b)^2.$$

OLS uses squared loss and targets conditional means. Quantile regression uses asymmetric absolute loss and targets conditional quantiles.

7.6 First-Order Condition for Linear Quantile Regression

The population objective is

$$Q(b) = \mathbb{E}[\rho_\tau(y_i - X_i'b)].$$

Let

$$u_i(b) = y_i - X_i'b.$$

For $u_i(b) \neq 0$, the derivative of $\rho_\tau(u_i(b))$ with respect to b is

$$\frac{\partial \rho_\tau(u_i(b))}{\partial b} = -\left(\tau - \mathbf{1}[u_i(b) \leq 0]\right) X_i.$$

Therefore, a population first-order condition is

$$\mathbb{E}\left[X_i\left(\tau - \mathbf{1}[y_i - X_i'\beta_\tau \leq 0]\right)\right] = 0.$$

Equivalently,

$$\boxed{\mathbb{E}\left[X_i\left(\mathbf{1}[y_i \leq X_i'\beta_\tau] - \tau\right)\right] = 0.}$$

When the conditional quantile is exactly linear,

$$Q_\tau(y_i \mid X_i) = X_i'\beta_\tau,$$

we have

$$\mathbb{P}(y_i \leq X_i'\beta_\tau \mid X_i) = \tau.$$

Thus,

$$\mathbb{E}[\mathbf{1}[y_i \leq X_i'\beta_\tau] - \tau \mid X_i] = 0,$$

which implies

$$\mathbb{E}\left[X_i(\mathbf{1}[y_i \leq X_i'\beta_\tau] - \tau)\right] = 0.$$

7.7 Median Regression as Least Absolute Deviations

Set

$$\tau = \frac{1}{2}.$$

The check function becomes

$$\rho_{1/2}(u) = \left(\frac{1}{2} - \mathbf{1}[u \leq 0]\right) u.$$

If $u > 0$,

$$\rho_{1/2}(u) = \frac{1}{2}u.$$

If $u \leq 0$,

$$\rho_{1/2}(u) = \left(\frac{1}{2} - 1\right) u = -\frac{1}{2}u = \frac{1}{2}|u|.$$

Therefore,

$$\rho_{1/2}(u) = \frac{1}{2}|u|.$$

The median-regression estimator solves

$$\widehat{\beta}_{1/2} = \arg\min_b \sum_i \frac{1}{2}|y_i - X_i' b|.$$

The factor $1/2$ does not affect the minimizer, so

$$\boxed{\widehat{\beta}_{1/2} = \arg\min_b \sum_i |y_i - X_i' b|.}$$

Thus, median regression is least absolute deviations regression.

7.8 Linear CQF Theorem for Quantile Regression

Suppose the conditional quantile function is exactly linear:

$$Q_\tau(y_i \mid X_i) = X_i' \beta_\tau^0.$$

Then β_τ^0 solves the population quantile regression problem

$$\beta_\tau = \arg\min_b \mathbb{E}[\rho_\tau(y_i - X_i' b)].$$

Derivation

The first-order condition for population quantile regression is

$$\mathbb{E}\left[X_i(\mathbf{1}[y_i \leq X_i' b] - \tau)\right] = 0.$$

Evaluate this at $b = \beta_\tau^0$:

$$\mathbb{E}\left[X_i(\mathbf{1}[y_i \leq X_i' \beta_\tau^0] - \tau)\right].$$

Since

$$X_i' \beta_\tau^0 = Q_\tau(y_i \mid X_i),$$

we have

$$\mathbb{P}(y_i \le X_i'\beta_\tau^0 \mid X_i) = \tau.$$

Thus,

$$\mathbb{E}[\mathbf{1}[y_i \le X_i'\beta_\tau^0] - \tau \mid X_i] = 0.$$

Multiplying by X_i and taking expectations,

$$\mathbb{E}\left[X_i(\mathbf{1}[y_i \le X_i'\beta_\tau^0] - \tau)\right] = 0.$$

Therefore,

$$\boxed{\beta_\tau = \beta_\tau^0.}$$

If the CQF is linear, quantile regression recovers it.

7.9 Location-Shift Model and Constant Quantile Coefficients

Suppose log wages satisfy the classical homoskedastic Normal model

$$y_i \sim \mathcal{N}(X_i'\beta, \sigma_\varepsilon^2),$$

so that

$$y_i = X_i'\beta + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

with ε_i independent of X_i.

Then

$$\mathbb{P}(y_i \le q \mid X_i) = \mathbb{P}(X_i'\beta + \varepsilon_i \le q \mid X_i) = \mathbb{P}(\varepsilon_i \le q - X_i'\beta).$$

Standardizing,

$$= \Phi\left(\frac{q - X_i'\beta}{\sigma_\varepsilon}\right).$$

Set this equal to τ:

$$\Phi\left(\frac{q - X_i'\beta}{\sigma_\varepsilon}\right) = \tau.$$

Apply Φ^{-1}:

$$\frac{q - X_i'\beta}{\sigma_\varepsilon} = \Phi^{-1}(\tau).$$

Thus,

$$q = X_i'\beta + \sigma_\varepsilon\Phi^{-1}(\tau).$$

Therefore,

$$\boxed{Q_\tau(y_i \mid X_i) = X_i'\beta + \sigma_\varepsilon\Phi^{-1}(\tau).}$$

If X_i includes an intercept, then changing τ changes only the intercept. The slope coefficients are the same across all quantiles.

Thus, in a pure location-shift model,

$$\boxed{\text{quantile regression coefficients are constant across quantiles, except for the intercept.}}$$

7.10 Heteroskedastic Location-Scale Model

Now suppose

$$y_i \sim \mathcal{N}(X_i'\beta, \sigma^2(X_i)),$$

where

$$\sigma(X_i) = \lambda' X_i$$

and $\lambda' X_i > 0$. Then

$$y_i = X_i'\beta + (\lambda' X_i)\varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0,1).$$

The conditional quantile is determined by

$$\mathbb{P}(y_i \leq q \mid X_i) = \tau.$$

Substitute the model:

$$\mathbb{P}(X_i'\beta + (\lambda' X_i)\varepsilon_i \leq q \mid X_i) = \tau.$$

This is

$$\mathbb{P}\left(\varepsilon_i \leq \frac{q - X_i'\beta}{\lambda' X_i}\right) = \tau.$$

Thus,

$$\frac{q - X_i'\beta}{\lambda' X_i} = \Phi^{-1}(\tau).$$

Solving for q,

$$q = X_i'\beta + (\lambda' X_i)\Phi^{-1}(\tau).$$

Therefore,

$$\boxed{Q_\tau(y_i \mid X_i) = X_i'[\beta + \lambda\Phi^{-1}(\tau)].}$$

Hence, the quantile coefficient vector is

$$\boxed{\beta_\tau = \beta + \lambda\Phi^{-1}(\tau).}$$

If $\lambda \neq 0$, then quantile regression coefficients vary across quantiles. This captures changes in within-group inequality.

7.11 *Interpreting Fanning-Out Quantile Coefficients*

Suppose X_i includes schooling s_i, and let $\beta_{\tau,s}$ be the schooling coefficient at quantile τ. If

$$\beta_{0.90,s} > \beta_{0.50,s} > \beta_{0.10,s},$$

then schooling raises the upper tail of the wage distribution more than the lower tail.

This does not mean that the same person receives a larger return at the upper tail. Instead, it means that the conditional wage distribution is more spread out at higher schooling levels.

Thus,

$$\boxed{\begin{array}{c}\text{Quantile regression coefficients describe changes in distributions,}\\ \text{not necessarily individual-level effects.}\end{array}}$$

7.12 *Conventional Quantile Regression Asymptotic Variance*

Let

$$u_{i\tau} = y_i - X_i'\beta_\tau.$$

Suppose the linear CQF is correctly specified. The first-order condition is

$$\mathbb{E}[X_i(\mathbf{1}[u_{i\tau} \leq 0] - \tau)] = 0.$$

The asymptotic variance of $\widehat{\beta}_\tau$ is

$$\boxed{\tau(1-\tau)\left\{\mathbb{E}[f_{u_\tau}(0 \mid X_i)X_iX_i']\right\}^{-1}\mathbb{E}[X_iX_i']\left\{\mathbb{E}[f_{u_\tau}(0 \mid X_i)X_iX_i']\right\}^{-1}.}$$

Here

$$f_{u_\tau}(0 \mid X_i)$$

is the conditional density of the quantile-regression residual at zero.

7.13 *Homoskedastic Simplification of Quantile Regression Variance*

If

$$f_{u_\tau}(0 \mid X_i) = f_{u_\tau}(0)$$

does not depend on X_i, then

$$\mathbb{E}[f_{u_\tau}(0 \mid X_i)X_iX_i'] = f_{u_\tau}(0)\mathbb{E}[X_iX_i'].$$

Substitute into the variance formula:

$$\tau(1-\tau)\left[f_{u_\tau}(0)\mathbb{E}[X_iX_i']\right]^{-1}\mathbb{E}[X_iX_i']\left[f_{u_\tau}(0)\mathbb{E}[X_iX_i']\right]^{-1}.$$

Since

$$\left[f_{u_\tau}(0)\mathbb{E}[X_iX_i']\right]^{-1}=\frac{1}{f_{u_\tau}(0)}\mathbb{E}[X_iX_i']^{-1},$$

the variance becomes

$$\tau(1-\tau)\frac{1}{f_{u_\tau}(0)^2}\mathbb{E}[X_iX_i']^{-1}\mathbb{E}[X_iX_i']\mathbb{E}[X_iX_i']^{-1}.$$

Canceling,

$$\boxed{V(\widehat{\beta}_\tau)=\frac{\tau(1-\tau)}{f_{u_\tau}(0)^2}\mathbb{E}[X_iX_i']^{-1}.}$$

7.14 Censored Quantile Regression Setup

Suppose the latent outcome of interest is y_i, but we observe a censored outcome $y_{i,obs}$. For topcoding at c, the observed outcome is

$$\boxed{y_{i,obs}=\min(y_i,c).}$$

Equivalently, observations above c are recorded as c.

If the conditional quantile of interest is below the censoring point,

$$Q_\tau(y_i\mid X_i)<c,$$

then censoring does not affect the τ-quantile. For example, topcoding very high earnings does not affect the conditional median if the median is below the topcode.

7.15 Why Censoring Does Not Affect Quantiles Below the Censoring Point

Let

$$y_{i,obs}=\min(y_i,c).$$

For any $q<c$,

$$\mathbb{P}(y_{i,obs}\le q\mid X_i)=\mathbb{P}(\min(y_i,c)\le q\mid X_i).$$

Since $q<c$, the event $\min(y_i,c)\le q$ is equivalent to

$$y_i\le q.$$

Therefore,

$$\mathbb{P}(y_{i,obs} \le q \mid X_i) = \mathbb{P}(y_i \le q \mid X_i).$$

Thus, if

$$Q_\tau(y_i \mid X_i) < c,$$

then

$$\boxed{Q_\tau(y_{i,obs} \mid X_i) = Q_\tau(y_i \mid X_i).}$$

So censored quantile regression can recover quantiles below the censoring threshold.

7.16 *Powell Censored Quantile Regression*

Suppose the conditional quantile function is censored from above:

$$Q_\tau(y_i \mid X_i) = \min(c, X_i'\beta_\tau^c).$$

The idea is to estimate the linear latent quantile $X_i'\beta_\tau^c$ using observations whose fitted quantile is below the censoring point:

$$X_i'\beta_\tau^c < c.$$

Powell's censored quantile regression parameter solves

$$\boxed{\beta_\tau^c = \arg\min_{b\in\mathbb{R}^d} \mathbb{E}\left[\mathbf{1}[X_i'\beta_\tau^c < c]\rho_\tau(y_i - X_i'b)\right].}$$

In words, the quantile regression objective is minimized only over observations for which the relevant conditional quantile is predicted to be uncensored.

7.17 *Iterated Algorithm for Censored Quantile Regression*

The censored quantile regression objective depends on the unknown parameter β_τ^c through the selection indicator

$$\mathbf{1}[X_i'\beta_\tau^c < c].$$

A practical algorithm proceeds as follows:

1. Estimate ordinary quantile regression ignoring censoring:

$$\widehat{\beta}_\tau^{(0)} = \arg\min_b \sum_i \rho_\tau(y_i - X_i'b).$$

2. Keep observations satisfying
$$X_i'\widehat{\beta}_\tau^{(0)} < c.$$
3. Re-estimate quantile regression using only these observations:
$$\widehat{\beta}_\tau^{(1)} = \arg\min_b \sum_{i: X_i'\widehat{\beta}_\tau^{(0)} < c} \rho_\tau(y_i - X_i'b).$$
4. Repeat:
$$\widehat{\beta}_\tau^{(k+1)} = \arg\min_b \sum_{i: X_i'\widehat{\beta}_\tau^{(k)} < c} \rho_\tau(y_i - X_i'b).$$

The procedure stops when
$$\widehat{\beta}_\tau^{(k+1)} \approx \widehat{\beta}_\tau^{(k)}.$$

The estimator is useful when enough uncensored observations remain.

7.18 Quantile Regression Specification Error

Let the true conditional quantile function be
$$Q_\tau(y_i \mid X_i).$$

For a linear approximation $X_i'b$, define the quantile-regression specification error
$$\boxed{\Delta_\tau(X_i, b) = X_i'b - Q_\tau(y_i \mid X_i).}$$

If the CQF is exactly linear, then for the true coefficient vector β_τ,
$$\Delta_\tau(X_i, \beta_\tau) = 0$$
for all X_i.

If the CQF is nonlinear, quantile regression chooses the best linear approximation to it under a particular weighting scheme.

7.19 Quantile Regression Approximation Property

The population quantile regression vector
$$\beta_\tau = \arg\min_b \mathbb{E}[\rho_\tau(y_i - X_i'b)]$$

can be understood as solving

$$\boxed{\beta_\tau = \arg\min_b \mathbb{E}\left[w_\tau(X_i, b)\Delta_\tau(X_i, b)^2\right].}$$

Here the weight is

$$\boxed{w_\tau(X_i, b) = \int_0^1 (1-u) f_{\varepsilon(\tau)}\left(u\Delta_\tau(X_i, b) \mid X_i\right) \mathrm{d}u,}$$

where

$$\varepsilon_i(\tau) = y_i - Q_\tau(y_i \mid X_i).$$

Equivalently,

$$w_\tau(X_i, b) = \int_0^1 (1-u) f_Y\left(uX_i'b + (1-u)Q_\tau(y_i \mid X_i) \mid X_i\right) \mathrm{d}u.$$

Thus, quantile regression approximates the CQF with weights that depend on the density of y_i around the conditional quantile.

7.20 Local Density Approximation to Quantile Weights

When the conditional density is smooth and b is close to β_τ, the specification error

$$\Delta_\tau(X_i, b)$$

is small. Then

$$uX_i'b + (1-u)Q_\tau(y_i \mid X_i)$$

is close to

$$Q_\tau(y_i \mid X_i).$$

Therefore,

$$f_Y\left(uX_i'b + (1-u)Q_\tau(y_i \mid X_i) \mid X_i\right) \approx f_Y(Q_\tau(y_i \mid X_i) \mid X_i).$$

Thus,

$$w_\tau(X_i, b) \approx f_Y(Q_\tau(y_i \mid X_i) \mid X_i) \int_0^1 (1-u)\,\mathrm{d}u.$$

Since

$$\int_0^1 (1-u)\,\mathrm{d}u = \left[u - \frac{u^2}{2}\right]_0^1 = \frac{1}{2},$$

we have

$$\boxed{w_\tau(X_i, b) \approx \frac{1}{2} f_Y(Q_\tau(y_i \mid X_i) \mid X_i).}$$

Thus, quantile regression gives more weight to covariate values where the outcome density is high near the target quantile.

7.21 Minimum Distance Approximation to the CQF

An alternative way to approximate the CQF is to regress the true conditional quantile on X_i:

$$\widetilde{\beta}_\tau = \arg\min_b \mathbb{E}\left[(Q_\tau(y_i \mid X_i) - X_i'b)^2\right].$$

Using the specification error,

$$\Delta_\tau(X_i, b) = X_i'b - Q_\tau(y_i \mid X_i),$$

this is

$$\boxed{\widetilde{\beta}_\tau = \arg\min_b \mathbb{E}[\Delta_\tau(X_i, b)^2].}$$

This minimum-distance estimator weights covariate values according to the distribution of X_i. Quantile regression instead weights by both the distribution of X_i and the density of y_i around the conditional quantile.

7.22 Conditional-to-Marginal Quantile Problem

Even if the conditional quantile function is linear,

$$Q_\tau(y_i \mid X_i) = X_i'\beta_\tau,$$

it does not follow that

$$Q_\tau(y_i) = Q_\tau(X_i)'\beta_\tau.$$

This contrasts with means. If

$$\mathbb{E}[y_i \mid X_i] = X_i'\beta,$$

then by iterated expectations,

$$\mathbb{E}[y_i] = \mathbb{E}[X_i]'\beta.$$

But quantiles do not aggregate linearly.

Therefore,

$$\boxed{\text{Marginal quantiles require the entire family of conditional quantiles, not just one conditional quantile.}}$$

7.23 Extracting the Marginal Distribution from Conditional Quantiles

Let

$$F_Y(y \mid X_i) = \mathbb{P}(y_i < y \mid X_i).$$

By definition of the conditional quantile function,

$$\boxed{F_Y(y \mid X_i) = \int_0^1 \mathbf{1}[F_Y^{-1}(\tau \mid X_i) < y] \, \mathrm{d}\tau.}$$

If the CQF is linear,

$$F_Y^{-1}(\tau \mid X_i) = Q_\tau(y_i \mid X_i) = X_i'\beta_\tau.$$

Substitute:

$$F_Y(y \mid X_i) = \int_0^1 \mathbf{1}[X_i'\beta_\tau < y] \, \mathrm{d}\tau.$$

To get the marginal distribution, integrate over the distribution of X_i:

$$\boxed{F_Y(y) = \int \int_0^1 \mathbf{1}[X_i'\beta_\tau < y] \, \mathrm{d}\tau \, \mathrm{d}F_X(x).}$$

The marginal quantile is then obtained by inversion:

$$\boxed{Q_\tau(y_i) = \inf\{y : F_Y(y) \geq \tau\}.}$$

7.24 Sample Estimator of the Marginal Distribution

Suppose quantile regressions are estimated on a grid

$$\tau_1, \tau_2, \ldots, \tau_M.$$

Let

$$\widehat{\beta}_{\tau_m}$$

be the estimated quantile regression coefficient at τ_m. A sample analog of the marginal distribution is

$$\boxed{\widehat{F}_Y(y) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{M}\sum_{m=1}^{M}\mathbf{1}[X_i'\widehat{\beta}_{\tau_m} < y].}$$

Then the marginal quantile estimator is

$$\boxed{\widehat{Q}_\tau(y_i) = \inf\{y : \widehat{F}_Y(y) \geq \tau\}.}$$

This procedure requires estimating many quantile regressions and then integrating over the empirical distribution of X_i.

7.25 Quantile Treatment Effects: Basic Setup

Let $d_i \in \{0, 1\}$ be treatment status. Potential outcomes are

$$y_{1i} = \text{outcome if treated}, \qquad y_{0i} = \text{outcome if untreated}.$$

Observed outcome is

$$y_i = y_{0i} + (y_{1i} - y_{0i})d_i.$$

Quantile treatment analysis compares the marginal or conditional distributions of y_{1i} and y_{0i}.

For example, the conditional quantile treatment effect at quantile τ is

$$\boxed{Q_\tau(y_{1i} \mid X_i) - Q_\tau(y_{0i} \mid X_i).}$$

This is a difference between two quantiles, not the quantile of the individual treatment effect $y_{1i} - y_{0i}$.

7.26 Quantile Difference Is Not the Quantile of Treatment Effects

In general,

$$\boxed{Q_\tau(y_{1i} - y_{0i}) \neq Q_\tau(y_{1i}) - Q_\tau(y_{0i}).}$$

The reason is that we never observe both y_{1i} and y_{0i} for the same individual, and quantiles are nonlinear operators.

For means, we have

$$\mathbb{E}[y_{1i} - y_{0i}] = \mathbb{E}[y_{1i}] - \mathbb{E}[y_{0i}].$$

But for quantiles,

$$Q_\tau(A - B)$$

is generally not equal to

$$Q_\tau(A) - Q_\tau(B).$$

Thus, quantile treatment effects describe differences in distributions, not the distribution of individual treatment effects.

7.27 Rank Preservation

Suppose treatment preserves each individual's rank in the outcome distribution. Then an individual at the lower decile without treatment remains at the lower decile with treatment.

Under rank preservation, the difference

$$Q_\tau(y_{1i}) - Q_\tau(y_{0i})$$

can be interpreted as the treatment effect for individuals at rank τ.

Without rank preservation, the same expression means only that the τ-quantile of the treated-potential-outcome distribution differs from the τ-quantile of the untreated-potential-outcome distribution.

Thus,

$$\boxed{\text{Without rank preservation, QTEs are distributional effects, not individual-rank effects.}}$$

7.28 Complier Quantile Regression Model

Let $z_i \in \{0, 1\}$ be a binary instrument, and define potential treatment statuses:

$$d_{1i} = \text{treatment status if } z_i = 1,$$

$$d_{0i} = \text{treatment status if } z_i = 0.$$

Compliers are individuals satisfying

$$d_{1i} > d_{0i}.$$

The quantile treatment effect model for compliers assumes

$$\boxed{Q_\tau(y_i \mid X_i, d_i, d_{1i} > d_{0i}) = \alpha_\tau d_i + X_i'\beta_\tau.}$$

Here α_τ is the coefficient of interest.

7.29 Why the Complier Quantile Coefficient Is Causal

Under the LATE assumptions, treatment d_i is independent of potential outcomes conditional on X_i within the complier population:

$$d_i \perp (y_{1i}, y_{0i}) \mid X_i, d_{1i} > d_{0i}.$$

For compliers with $d_i = 1$, the observed outcome is

$$y_i = y_{1i}.$$

For compliers with $d_i = 0$, the observed outcome is

$$y_i = y_{0i}.$$

Therefore,

$$Q_\tau(y_i \mid X_i, d_i = 1, d_{1i} > d_{0i}) = Q_\tau(y_{1i} \mid X_i, d_{1i} > d_{0i}),$$

and

$$Q_\tau(y_i \mid X_i, d_i = 0, d_{1i} > d_{0i}) = Q_\tau(y_{0i} \mid X_i, d_{1i} > d_{0i}).$$

Using the linear complier quantile model,

$$Q_\tau(y_i \mid X_i, d_i = 1, d_{1i} > d_{0i}) = \alpha_\tau + X_i'\beta_\tau,$$

and

$$Q_\tau(y_i \mid X_i, d_i = 0, d_{1i} > d_{0i}) = X_i'\beta_\tau.$$

Subtracting,

$$\boxed{Q_\tau(y_{1i} \mid X_i, d_{1i} > d_{0i}) - Q_\tau(y_{0i} \mid X_i, d_{1i} > d_{0i}) = \alpha_\tau.}$$

Thus, α_τ is the conditional quantile treatment effect for compliers.

7.30 Abadie Kappa Weights

Define the instrument propensity score

$$p(X_i) = \mathbb{P}(z_i = 1 \mid X_i).$$

The Abadie kappa weight is

$$\boxed{\kappa_i = 1 - \frac{d_i(1 - z_i)}{1 - p(X_i)} - \frac{(1 - d_i)z_i}{p(X_i)}.}$$

The key result is that for any suitable function $g(y_i, d_i, X_i)$,

$$\boxed{\mathbb{E}[g(y_i, d_i, X_i) \mid d_{1i} > d_{0i}] = \frac{\mathbb{E}[\kappa_i g(y_i, d_i, X_i)]}{\mathbb{E}[\kappa_i]}.}$$

Thus, kappa weighting allows us to recover moments and objective functions for the unobserved complier population.

7.31 QTE Minimization Problem

If we could observe compliers, we would estimate

$$(\alpha_\tau, \beta_\tau) = \arg\min_{a,b} \mathbb{E}[\rho_\tau(y_i - ad_i - X_i'b) \mid d_{1i} > d_{0i}].$$

Using Abadie's kappa theorem, this is equivalent to

$$\boxed{(\alpha_\tau, \beta_\tau) = \arg\min_{a,b} \mathbb{E}[\kappa_i \rho_\tau(y_i - ad_i - X_i'b)].}$$

The sample QTE estimator is the sample analog:

$$\boxed{(\widehat{\alpha}_\tau, \widehat{\beta}_\tau) = \arg\min_{a,b} \sum_{i=1}^{n} \widehat{\kappa}_i \rho_\tau(y_i - ad_i - X_i'b).}$$

Here $\widehat{\kappa}_i$ is constructed from an estimate of $p(X_i)$.

7.32 Why Negative Kappa Weights Are a Problem

The kappa weight is

$$\kappa_i = 1 - \frac{d_i(1 - z_i)}{1 - p(X_i)} - \frac{(1 - d_i)z_i}{p(X_i)}.$$

When $d_i \neq z_i$, the subtractive terms can make κ_i negative. Quantile regression with negative weights can make the objective function non-convex:

$$\sum_i \kappa_i \rho_\tau(y_i - ad_i - X_i'b).$$

A non-convex objective is harder to minimize and may not have the usual linear-programming representation.

The solution is to use conditional kappa weights:

$$\mathbb{E}[\kappa_i \mid y_i, d_i, X_i].$$

These are probabilities and therefore lie between zero and one.

7.33 Conditional Kappa Weight

By iterating expectations,

$$\mathbb{E}[\kappa_i \rho_\tau(y_i - ad_i - X_i' b)] = \mathbb{E}[\mathbb{E}[\kappa_i \mid y_i, d_i, X_i] \rho_\tau(y_i - ad_i - X_i' b)].$$

Therefore, the same population minimizer solves

$$\boxed{\min_{a,b} \mathbb{E}[\mathbb{E}[\kappa_i \mid y_i, d_i, X_i] \rho_\tau(y_i - ad_i - X_i' b)].}$$

Moreover,

$$\boxed{\mathbb{E}[\kappa_i \mid y_i, d_i, X_i] = \mathbb{P}(d_{1i} > d_{0i} \mid y_i, d_i, X_i).}$$

Thus, the conditional kappa weight is the probability that an observation is a complier given its observed outcome, treatment status, and covariates.

7.34 Formula for Conditional Kappa Weight

The conditional kappa weight can be written as

$$\boxed{\mathbb{E}[\kappa_i \mid y_i, d_i, X_i] = 1 - \frac{d_i \left(1 - \mathbb{E}[z_i \mid y_i, d_i = 1, X_i]\right)}{1 - p(X_i)} - \frac{(1 - d_i)\mathbb{E}[z_i \mid y_i, d_i = 0, X_i]}{p(X_i)}.}$$

This formula follows from taking conditional expectations of each component of κ_i.

For treated observations, $d_i = 1$, so

$$\mathbb{E}[\kappa_i \mid y_i, d_i = 1, X_i] = 1 - \frac{1 - \mathbb{E}[z_i \mid y_i, d_i = 1, X_i]}{1 - p(X_i)}.$$

For untreated observations, $d_i = 0$, so

$$\mathbb{E}[\kappa_i \mid y_i, d_i = 0, X_i] = 1 - \frac{\mathbb{E}[z_i \mid y_i, d_i = 0, X_i]}{p(X_i)}.$$

7.35 Practical QTE Algorithm

A practical QTE algorithm proceeds as follows:

1. Estimate
$$p(X_i) = \mathbb{P}(z_i = 1 \mid X_i)$$
in the full sample, for example by Probit or Logit.

2. Estimate
$$\mathbb{E}[z_i \mid y_i, d_i = 1, X_i]$$
in the treated subsample.

3. Estimate
$$\mathbb{E}[z_i \mid y_i, d_i = 0, X_i]$$
in the untreated subsample.

4. Construct
$$\widehat{\omega}_i = \widehat{\mathbb{E}}[\kappa_i \mid y_i, d_i, X_i]$$
using the conditional-kappa formula.

5. Trim weights outside the unit interval:
$$\widehat{\omega}_i < 0 \Rightarrow \widehat{\omega}_i = 0, \qquad \widehat{\omega}_i > 1 \Rightarrow \widehat{\omega}_i = 1.$$

6. Estimate weighted quantile regression:
$$(\widehat{\alpha}_\tau, \widehat{\beta}_\tau) = \arg\min_{a,b} \sum_i \widehat{\omega}_i \rho_\tau(y_i - ad_i - X_i'b).$$

7. Bootstrap the full procedure to obtain standard errors.

7.36 QTE and Ordinary Quantile Regression

If no instrument is needed and treatment is conditionally independent of potential outcomes given X_i, then the relevant weights are all equal to one:

$$\omega_i = 1.$$

The QTE objective becomes

$$\min_{a,b} \mathbb{E}[\rho_\tau(y_i - ad_i - X_i'b)],$$

which is ordinary linear quantile regression with d_i and X_i as regressors.

Thus,

$$\boxed{\text{QTE is to quantile regression what IV/2SLS is to OLS.}}$$

7.37 JTPA Experiment: Instrumental-Variables Setup

In the JTPA application, define

$$y_i = \text{30-month earnings},$$

$$d_i = \text{actual enrollment in JTPA services},$$

and

$$z_i = \text{randomized offer of JTPA services}.$$

Because some individuals offered training do not enroll, treatment received is not equal to treatment assigned:

$$d_i \neq z_i$$

for some individuals.

The randomized offer z_i is used as an instrument for actual training d_i. The first stage is

$$d_i = \pi_0 + \pi_1 z_i + X_i' \pi_2 + v_i,$$

and the reduced form is

$$y_i = \gamma_0 + \gamma_1 z_i + X_i' \gamma_2 + u_i.$$

The conventional IV effect on average earnings is

$$\boxed{\rho_{IV} = \frac{\gamma_1}{\pi_1}.}$$

The QTE estimator extends this logic from average effects to quantile effects for compliers.

7.38 Why QTE Can Differ from Descriptive Quantile Regression

Ordinary quantile regression comparing trainees and non-trainees estimates differences in the observed conditional earnings distributions:

$$Q_\tau(y_i \mid d_i = 1, X_i) - Q_\tau(y_i \mid d_i = 0, X_i).$$

This comparison may reflect selection bias because actual trainees are self-selected.

QTE instead estimates

$$Q_\tau(y_{1i} \mid X_i, d_{1i} > d_{0i}) - Q_\tau(y_{0i} \mid X_i, d_{1i} > d_{0i}),$$

which is the causal quantile effect for compliers.

Thus,

$$\boxed{\begin{array}{c}\text{Quantile regression is descriptive unless treatment is conditionally exogenous;}\\ \text{QTE uses an instrument to recover causal quantile effects.}\end{array}}$$

7.39 Summary of Chapter 7

Quantile regression replaces the conditional mean function with the conditional quantile function:

$$\mathbb{E}[y_i \mid X_i] \quad \longrightarrow \quad Q_\tau(y_i \mid X_i).$$

The key population object is

$$\beta_\tau = \arg\min_b \mathbb{E}[\rho_\tau(y_i - X_i'b)].$$

When the CQF is linear, this recovers the true conditional quantile function. When the CQF is nonlinear, quantile regression gives a weighted linear approximation to it.

Censored quantile regression is useful when the quantile of interest lies below or above the censoring point. Quantile treatment effects extend quantile regression to causal inference, using Abadie kappa weights and IV logic to estimate causal quantile effects for compliers.

8 Chapter 8 Derivations: Nonstandard Standard Error Issues

8.1 OLS in Matrix Form

Let the regression model be

$$y_i = X_i'\beta + e_i,$$

where X_i is a $k \times 1$ vector of regressors. Stack the observations:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \qquad X = \begin{bmatrix} X_1' \\ X_2' \\ \vdots \\ X_N' \end{bmatrix}.$$

The OLS estimator is

$$\boxed{\widehat{\beta} = (X'X)^{-1}X'y.}$$

Equivalently,

$$\widehat{\beta} = \left(\sum_{i=1}^{N} X_i X_i'\right)^{-1} \sum_{i=1}^{N} X_i y_i.$$

8.2 Asymptotic Variance of OLS

The population regression residual is

$$e_i = y_i - X_i'\beta.$$

Under independent sampling, the asymptotic distribution is

$$\sqrt{N}(\widehat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega).$$

The heteroskedasticity-robust asymptotic covariance matrix is

$$\boxed{\Omega_r = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i X_i' e_i^2]\mathbb{E}[X_i X_i']^{-1}.}$$

When residuals are homoskedastic,

$$\mathbb{E}[e_i^2 \mid X_i] = \sigma^2,$$

so

$$\mathbb{E}[X_i X_i' e_i^2] = \mathbb{E}[X_i X_i' \mathbb{E}[e_i^2 \mid X_i]] = \sigma^2 \mathbb{E}[X_i X_i'].$$

Therefore,

$$\Omega_r = \mathbb{E}[X_i X_i']^{-1}\sigma^2\mathbb{E}[X_i X_i']\mathbb{E}[X_i X_i']^{-1} = \sigma^2\mathbb{E}[X_i X_i']^{-1}.$$

Thus,

$$\boxed{\Omega_c = \sigma^2\mathbb{E}[X_i X_i']^{-1}.}$$

8.3 Fixed-Regressor Robust Variance Formula

To study finite-sample bias, treat X as fixed in repeated samples. Let

$$e = \begin{bmatrix} e_1 \\ \vdots \\ e_N \end{bmatrix}, \qquad \Psi = \mathbb{E}[ee'] = (\sigma_1^2, \ldots, \sigma_N^2).$$

Then

$$\boxed{\Omega_r = \left(\frac{X'X}{N}\right)^{-1}\left(\frac{X'\Psi X}{N}\right)\left(\frac{X'X}{N}\right)^{-1}.}$$

Under homoskedasticity,

$$\Psi = \sigma^2 I_N,$$

so

$$\Omega_c = \left(\frac{X'X}{N}\right)^{-1}\left(\frac{X'\sigma^2 I_N X}{N}\right)\left(\frac{X'X}{N}\right)^{-1}.$$

Since

$$X' I_N X = X'X,$$

we get

$$\Omega_c = \sigma^2\left(\frac{X'X}{N}\right)^{-1}.$$

8.4 Conventional Variance Estimator

The conventional OLS variance estimator is

$$\boxed{\widehat{V}_c(\widehat{\beta}) = (X'X)^{-1}\widehat{\sigma}^2,}$$

where

$$\widehat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}\widehat{e}_i^2.$$

The estimated residual vector is

$$\widehat{e} = y - X\widehat{\beta}.$$

Using

$$\widehat{\beta} = (X'X)^{-1}X'y,$$

we have

$$\widehat{e} = y - X(X'X)^{-1}X'y.$$

Define the projection matrix

$$H = X(X'X)^{-1}X',$$

and the residual-maker matrix

$$M = I_N - H.$$

Then

$$\boxed{\widehat{e} = My.}$$

Since

$$y = X\beta + e,$$

we get

$$\widehat{e} = M(X\beta + e) = MX\beta + Me.$$

Because

$$MX = (I_N - H)X = X - X = 0,$$

we have

$$\boxed{\widehat{e} = Me.}$$

8.5 Expected Squared Residuals

Let m_i be the i-th column of M. Then

$$\widehat{e_i} = m_i' e.$$

Therefore,

$$\mathbb{E}[\widehat{e}_i^2] = \mathbb{E}[m_i' e e' m_i] = m_i' \mathbb{E}[ee'] m_i.$$

Since

$$\mathbb{E}[ee'] = \Psi,$$

we get

$$\boxed{\mathbb{E}[\widehat{e}_i^2] = m_i' \Psi m_i.}$$

8.6 Leverage and the Residual-Maker Matrix

Let ℓ_i be the i-th column of I_N, and let h_i be the i-th column of the projection matrix H. Since

$$M = I_N - H,$$

the i-th column of M is

$$m_i = \ell_i - h_i.$$

Therefore,

$$\mathbb{E}[\widehat{e}_i^2] = (\ell_i - h_i)' \Psi (\ell_i - h_i).$$

Expand:

$$\mathbb{E}[\widehat{e}_i^2] = \ell_i' \Psi \ell_i - 2\ell_i' \Psi h_i + h_i' \Psi h_i.$$

Since Ψ is diagonal with i-th diagonal element σ_i^2,

$$\ell_i' \Psi \ell_i = \sigma_i^2.$$

Also,

$$\ell_i' \Psi h_i = \sigma_i^2 h_{ii},$$

where

$$h_{ii} = X_i'(X'X)^{-1} X_i$$

is the i-th diagonal element of H, called leverage. Thus,

$$\boxed{\mathbb{E}[\hat{e}_i^2] = \sigma_i^2 - 2\sigma_i^2 h_{ii} + h_i' \Psi h_i.}$$

8.7 Expected Squared Residuals Under Homoskedasticity

Under homoskedasticity,

$$\Psi = \sigma^2 I_N.$$

Then

$$h_i' \Psi h_i = \sigma^2 h_i' h_i.$$

Because H is symmetric and idempotent,

$$h_i' h_i = h_{ii}.$$

Therefore,

$$\mathbb{E}[\hat{e}_i^2] = \sigma^2 - 2\sigma^2 h_{ii} + \sigma^2 h_{ii}.$$

Hence,

$$\boxed{\mathbb{E}[\hat{e}_i^2] = \sigma^2(1 - h_{ii}).}$$

Since

$$0 \leq h_{ii} \leq 1,$$

we have

$$\mathbb{E}[\hat{e}_i^2] \leq \sigma^2.$$

Thus, squared OLS residuals are biased downward as estimates of the true residual variance.

8.8 Degrees-of-Freedom Bias in the Conventional Estimator

The conventional residual variance estimator is

$$\widehat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}\widehat{e}_i^2.$$

Using

$$\mathbb{E}[\widehat{e}_i^2] = \sigma^2(1-h_{ii}),$$

we get

$$\mathbb{E}[\widehat{\sigma}^2] = \frac{1}{N}\sum_{i=1}^{N}\sigma^2(1-h_{ii}).$$

Thus,

$$\mathbb{E}[\widehat{\sigma}^2] = \sigma^2\left(1-\frac{1}{N}\sum_{i=1}^{N}h_{ii}\right).$$

Because

$$\sum_{i=1}^{N}h_{ii} = (H) = k,$$

we get

$$\mathbb{E}[\widehat{\sigma}^2] = \sigma^2\left(1-\frac{k}{N}\right) = \sigma^2\frac{N-k}{N}.$$

Therefore,

$$\boxed{\mathbb{E}\left[\frac{1}{N}\sum_i \widehat{e}_i^2\right] = \sigma^2\frac{N-k}{N}.}$$

The unbiased homoskedastic residual variance estimator is

$$\boxed{\widehat{\sigma}_{df}^2 = \frac{1}{N-k}\sum_{i=1}^{N}\widehat{e}_i^2.}$$

8.9 Leverage in a Bivariate Regression

Consider a bivariate regression with a constant and one regressor x_i. The leverage is

$$h_{ii} = X_i'(X'X)^{-1}X_i.$$

For the model with intercept and x_i, this simplifies to

$$\boxed{h_{ii} = \frac{1}{N} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{N}(x_j - \bar{x})^2}.}$$

This formula shows that leverage is high when x_i is far from the sample mean. High-leverage observations exert more influence on fitted values and tend to produce smaller OLS residuals.

8.10 Bias of Robust Standard Errors

The HC0 robust variance estimator is

$$\boxed{\widehat{V}_{HC0}(\widehat{\beta}) = (X'X)^{-1}\left(\sum_{i=1}^{N} X_i X_i' \widehat{e}_i^2\right)(X'X)^{-1}.}$$

The problem is that

$$\mathbb{E}[\widehat{e}_i^2] < \sigma_i^2$$

in finite samples. Thus, the middle term

$$\sum_i X_i X_i' \widehat{e}_i^2$$

tends to underestimate

$$\sum_i X_i X_i' \sigma_i^2.$$

Therefore,

$$\boxed{\widehat{V}_{HC0}(\widehat{\beta}) \text{ is generally biased downward in finite samples.}}$$

This bias is especially severe when leverage values h_{ii} are large.

8.11 HC0, HC1, HC2, and HC3

All HC estimators have the form

$$\widehat{V}_{HCj}(\widehat{\beta}) = (X'X)^{-1}\left(\sum_{i=1}^{N} X_i X_i' \widehat{\sigma}_{i,j}^2\right)(X'X)^{-1}.$$

The different estimators use different residual-variance estimates:

$$\boxed{HC0: \quad \widehat{\sigma}_{i,0}^2 = \widehat{e}_i^2.}$$

$$\boxed{HC1: \quad \widehat{\sigma}^2_{i,1} = \frac{N}{N-k}\widehat{e}^2_i.}$$

$$\boxed{HC2: \quad \widehat{\sigma}^2_{i,2} = \frac{\widehat{e}^2_i}{1-h_{ii}}.}$$

$$\boxed{HC3: \quad \widehat{\sigma}^2_{i,3} = \frac{\widehat{e}^2_i}{(1-h_{ii})^2}.}$$

HC1 applies a global degrees-of-freedom correction. HC2 corrects each squared residual by its own leverage. HC3 is more conservative and approximates a jackknife correction.

8.12 Why HC2 Is Unbiased Under Homoskedasticity

Under homoskedasticity,

$$\mathbb{E}[\widehat{e}^2_i] = \sigma^2(1-h_{ii}).$$

HC2 uses

$$\widehat{\sigma}^2_{i,2} = \frac{\widehat{e}^2_i}{1-h_{ii}}.$$

Therefore,

$$\mathbb{E}[\widehat{\sigma}^2_{i,2}] = \frac{\mathbb{E}[\widehat{e}^2_i]}{1-h_{ii}} = \frac{\sigma^2(1-h_{ii})}{1-h_{ii}} = \sigma^2.$$

Thus,

$$\boxed{HC2 \text{ gives an unbiased estimate of each residual variance under homoskedasticity.}}$$

8.13 Bivariate Example: Robust Bias Can Be Worse Than Conventional Bias

Consider a bivariate regression after partialling out the constant. Let $\widetilde{x}_i = x_i - \bar{x}$. The slope is

$$\widehat{\beta}_1 = \frac{\sum_i \widetilde{x}_i y_i}{\sum_i \widetilde{x}^2_i}.$$

Let

$$s^2_x = \frac{1}{N}\sum_i \widetilde{x}^2_i.$$

Then

$$\sum_i \widetilde{x}^2_i = Ns^2_x.$$

The leverage, after partialling out the constant, is

$$h_{ii} = \frac{\widetilde{x}_i^2}{\sum_j \widetilde{x}_j^2} = \frac{\widetilde{x}_i^2}{N s_x^2}.$$

The conventional variance estimator has expectation

$$\mathbb{E}[\widehat{V}_c] = \frac{\sigma^2}{N s_x^2} \left[\frac{\sum_i (1 - h_{ii})}{N} \right].$$

Since

$$\sum_i h_{ii} = 1$$

in the no-intercept residualized regression,

$$\mathbb{E}[\widehat{V}_c] = \frac{\sigma^2}{N s_x^2} \left(1 - \frac{1}{N} \right).$$

The robust estimator has expectation

$$\mathbb{E}[\widehat{V}_r] = \frac{\sigma^2}{N s_x^2} \sum_i (1 - h_{ii}) h_{ii}.$$

Since

$$\sum_i (1 - h_{ii}) h_{ii} = \sum_i h_{ii} - \sum_i h_{ii}^2 = 1 - \sum_i h_{ii}^2,$$

we have

$$\boxed{\mathbb{E}[\widehat{V}_r] = \frac{\sigma^2}{N s_x^2} \left(1 - \sum_i h_{ii}^2 \right).}$$

The conventional expectation is

$$\boxed{\mathbb{E}[\widehat{V}_c] = \frac{\sigma^2}{N s_x^2} \left(1 - \frac{1}{N} \right).}$$

Since Jensen's inequality implies

$$\sum_i h_{ii}^2 > \frac{1}{N}$$

unless leverage is constant, the robust estimator is more biased downward:

$$1 - \sum_i h_{ii}^2 < 1 - \frac{1}{N}.$$

Therefore,

$$\boxed{\text{With unequal leverage, HC0 can be more downward biased than the conventional estimator.}}$$

8.14 Pairs Bootstrap

The pairs bootstrap treats the empirical distribution of the observed data as the population. Each bootstrap sample is generated by drawing N pairs

$$(y_i, X_i)$$

with replacement from the original sample.

For bootstrap draw $b = 1, \ldots, B$, compute

$$\widehat{\beta}^{*(b)}.$$

The bootstrap standard error is

$$\boxed{SE_{boot}(\widehat{\beta}) = \left[\frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\beta}^{*(b)} - \bar{\beta}^*\right)^2\right]^{1/2},}$$

where

$$\bar{\beta}^* = \frac{1}{B}\sum_{b=1}^{B}\widehat{\beta}^{*(b)}.$$

The pairs bootstrap allows the joint distribution of (y_i, X_i) to be resampled directly.

8.15 Residual Bootstrap

A residual bootstrap keeps X_i fixed. Estimate the original regression:

$$y_i = X_i'\widehat{\beta} + \widehat{e}_i.$$

Draw residuals $\widehat{e}_i^*$ with replacement from

$$\{\widehat{e}_1, \ldots, \widehat{e}_N\}.$$

Construct bootstrap outcomes:

$$y_i^* = X_i'\widehat{\beta} + \widehat{e}_i^*.$$

Then estimate

$$\widehat{\beta}^* = (X'X)^{-1}X'y^*.$$

This procedure mimics sampling with fixed regressors and independent homoskedastic residuals. It is less suitable when heteroskedasticity is central because it breaks the relationship between X_i and the residual variance.

8.16 Wild Bootstrap

The wild bootstrap preserves the relationship between X_i and the magnitude of residuals. A simple version draws

$$v_i^* = \begin{cases} 1, & \text{with probability } 1/2, \\ -1, & \text{with probability } 1/2. \end{cases}$$

Then construct

$$y_i^* = X_i'\widehat{\beta} + \widehat{e}_i v_i^*.$$

Because

$$\mathbb{E}[v_i^*] = 0, \qquad \mathbb{E}[(v_i^*)^2] = 1,$$

we have

$$\mathbb{E}[y_i^* \mid X_i] = X_i'\widehat{\beta},$$

and

$$\mathrm{Var}(y_i^* \mid X_i) = \widehat{e}_i^2.$$

Thus,

$$\boxed{\text{The wild bootstrap preserves heteroskedasticity patterns in the original sample.}}$$

8.17 Bootstrap t-Statistics

Bootstrap refinements are most reliable for pivotal statistics. A t-statistic is approximately pivotal because its asymptotic distribution is standard normal:

$$t = \frac{\widehat{\beta}_j - \beta_{j,0}}{SE(\widehat{\beta}_j)}.$$

In each bootstrap sample, compute

$$t^{*(b)} = \frac{\widehat{\beta}_j^{*(b)} - \widehat{\beta}_j}{SE^{*(b)}(\widehat{\beta}_j^{*(b)})}.$$

The bootstrap critical value is obtained from the empirical distribution of

$$|t^{*(b)}|.$$

Reject the null if

$$\boxed{|t| > c_{1-\alpha}^{boot},}$$

where $c_{1-\alpha}^{boot}$ is the $(1-\alpha)$-quantile of the bootstrap distribution of $|t^{*(b)}|$.

8.18 Dummy Treatment Example

Consider

$$y_i = \beta_0 + \beta_1 d_i + \varepsilon_i,$$

where $d_i \in \{0, 1\}$. The OLS estimator is the difference in sample means:

$$\boxed{\widehat{\beta}_1 = \bar{y}_1 - \bar{y}_0.}$$

Let

$$N_1 = \sum_i d_i, \qquad N_0 = \sum_i (1 - d_i), \qquad N = N_0 + N_1.$$

Let

$$S_j^2 = \sum_{d_i = j} (y_i - \bar{y}_j)^2, \qquad j = 0, 1.$$

The leverage values are

$$h_{ii} = \begin{cases} 1/N_0, & d_i = 0, \\ 1/N_1, & d_i = 1. \end{cases}$$

8.19 Conventional Standard Error in the Dummy Treatment Example

The conventional pooled residual variance estimator is

$$\widehat{\sigma}_{pooled}^2 = \frac{S_0^2 + S_1^2}{N - 2}.$$

The variance of the difference in means under equal variances is

$$\widehat{V}_{conv}(\widehat{\beta}_1) = \widehat{\sigma}^2_{pooled}\left(\frac{1}{N_0} + \frac{1}{N_1}\right).$$

Since

$$\frac{1}{N_0} + \frac{1}{N_1} = \frac{N}{N_0 N_1},$$

we get

$$\boxed{\widehat{V}_{conv}(\widehat{\beta}_1) = \frac{N}{N_0 N_1}\left(\frac{S_0^2 + S_1^2}{N-2}\right).}$$

If

$$r = \frac{N_1}{N},$$

then

$$N_1 = Nr, \qquad N_0 = N(1-r),$$

so

$$\frac{N}{N_0 N_1} = \frac{1}{Nr(1-r)}.$$

Thus,

$$\boxed{\widehat{V}_{conv}(\widehat{\beta}_1) = \frac{1}{Nr(1-r)}\left(\frac{S_0^2 + S_1^2}{N-2}\right).}$$

8.20 HC0–HC3 in the Dummy Treatment Example

For the difference in means, HC0 is

$$\boxed{\widehat{V}_{HC0} = \frac{S_0^2}{N_0^2} + \frac{S_1^2}{N_1^2}.}$$

HC1 applies the degrees-of-freedom correction:

$$\boxed{\widehat{V}_{HC1} = \frac{N}{N-2}\left(\frac{S_0^2}{N_0^2} + \frac{S_1^2}{N_1^2}\right).}$$

HC2 divides each squared residual by $1 - h_{ii}$. Since

$$1 - h_{ii} = 1 - \frac{1}{N_j} = \frac{N_j - 1}{N_j}$$

within group j, HC2 is

$$\boxed{\widehat{V}_{HC2} = \frac{S_0^2}{N_0(N_0-1)} + \frac{S_1^2}{N_1(N_1-1)}.}$$

HC3 divides by $(1-h_{ii})^2$, giving

$$\boxed{\widehat{V}_{HC3} = \frac{S_0^2}{(N_0-1)^2} + \frac{S_1^2}{(N_1-1)^2}.}$$

When the design is balanced, $N_0 = N_1$, these estimators are closer to each other. When the design is highly unbalanced, the differences can be substantial.

8.21 Rule of Thumb: Maximum of Conventional and Robust

Robust standard errors may be too small because of finite-sample bias and because the robust standard error itself is noisy. A practical rule of thumb is

$$\boxed{SE_{reported} = \max\{SE_{conv}, SE_{HCj}\}.}$$

This rule truncates unusually small robust estimates and reduces the risk of overstating precision. The cost is that when homoskedasticity truly holds, the reported standard error may be larger than necessary.

8.22 Clustered Data Setup

Suppose observations are indexed by individual i and group g. Consider

$$\boxed{y_{ig} = \beta_0 + \beta_1 x_g + e_{ig}.}$$

The regressor x_g varies only at the group level. A standard error problem arises when residuals are correlated within groups:

$$\boxed{\mathbb{E}[e_{ig}e_{jg}] = \rho\sigma_e^2 \qquad \text{for } i \neq j.}$$

Here ρ is the intra-class correlation coefficient. If $\rho > 0$, then observations in the same group contain less independent information than independent sampling would imply.

8.23 Random Effects Error Structure

A common model for within-cluster correlation is

$$\boxed{e_{ig} = v_g + \eta_{ig},}$$

where v_g is a group-level shock and η_{ig} is an individual-level shock. Assume

$$\text{Var}(v_g) = \sigma_v^2, \qquad \text{Var}(\eta_{ig}) = \sigma_\eta^2,$$

and

$$\text{Cov}(v_g, \eta_{ig}) = 0.$$

Then

$$\text{Var}(e_{ig}) = \sigma_v^2 + \sigma_\eta^2.$$

For two different individuals $i \neq j$ in the same group,

$$\text{Cov}(e_{ig}, e_{jg}) = \text{Cov}(v_g + \eta_{ig}, v_g + \eta_{jg}) = \text{Var}(v_g) = \sigma_v^2.$$

Thus, the intra-class correlation is

$$\boxed{\rho = \frac{\text{Cov}(e_{ig}, e_{jg})}{\text{Var}(e_{ig})} = \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\eta^2}.}$$

8.24 Simple Moulton Factor

Let $V(\widehat{\beta}_1)$ be the true sampling variance accounting for clustering, and let $V_c(\widehat{\beta}_1)$ be the conventional OLS variance formula that ignores clustering. If the regressor varies only at the group level and all groups have equal size n, then

$$\boxed{\frac{V(\widehat{\beta}_1)}{V_c(\widehat{\beta}_1)} = 1 + (n-1)\rho.}$$

The standard error inflation factor is the square root:

$$\boxed{\text{Moulton factor} = \sqrt{1 + (n-1)\rho}.}$$

If $\rho = 0$, there is no clustering problem. If $\rho = 1$, all observations within a group share the same residual component, and the variance inflation factor is n.

8.25 General Moulton Factor

When group sizes vary and the regressor may vary within groups, the variance inflation factor is approximately

$$\boxed{\frac{V(\widehat{\beta}_1)}{V_c(\widehat{\beta}_1)} = 1 + \left[\frac{\mathrm{Var}(n_g)}{\bar{n}} + \bar{n} - 1\right]\rho_x\rho.}$$

Here

$$\bar{n} = \text{average group size},$$

$$\rho = \text{intra-class correlation of residuals},$$

and

$$\rho_x = \text{intra-class correlation of the regressor}.$$

The regressor intra-class correlation is

$$\boxed{\rho_x = \frac{\sum_g \sum_{i\neq k}(x_{ig} - \bar{x})(x_{kg} - \bar{x})}{\mathrm{Var}(x_{ig})\sum_g n_g(n_g - 1)}.}$$

If x_{ig} is constant within groups, then

$$\rho_x = 1.$$

If x_{ig} is uncorrelated within groups, then

$$\rho_x = 0,$$

and the clustering problem is much less severe for that regressor.

8.26 Parametric Moulton Correction

Let SE_{conv} be the conventional standard error. The Moulton-corrected standard error is

$$\boxed{SE_{Moulton} = SE_{conv}\sqrt{1 + \left[\frac{\mathrm{Var}(n_g)}{\bar{n}} + \bar{n} - 1\right]\rho_x\rho}.}$$

In the equal-size group case with group-level regressors,

$$\boxed{SE_{Moulton} = SE_{conv}\sqrt{1 + (n-1)\rho}.}$$

This correction is simple and transparent but relies on the parametric intra-class correlation structure.

8.27 Cluster-Robust Variance Estimator

Let X_g be the regressor matrix for group g, and let $\widehat{e}_g$ be the vector of residuals for group g. The cluster-robust variance estimator is

$$\boxed{\widehat{V}_{cluster}(\widehat{\beta}) = (X'X)^{-1}\left(\sum_{g=1}^{G} X_g'\widehat{e}_g\widehat{e}_g'X_g\right)(X'X)^{-1}.}$$

With a degrees-of-freedom adjustment a, this becomes

$$\boxed{\widehat{V}_{cluster}(\widehat{\beta}) = (X'X)^{-1}\left(\sum_{g=1}^{G} X_g'\left(a\widehat{e}_g\widehat{e}_g'\right)X_g\right)(X'X)^{-1}.}$$

This estimator allows arbitrary correlation within clusters and heteroskedasticity across clusters. It is consistent as

$$G \to \infty.$$

It is not justified by large numbers of observations within a fixed number of clusters.

8.28 Why Cluster-Robust Inference Needs Many Clusters

The cluster-robust variance estimator sums over clusters:

$$\sum_{g=1}^{G} X_g'\widehat{e}_g\widehat{e}_g'X_g.$$

Consistency relies on a law of large numbers over g, not over individuals i. Therefore, the relevant sample size for inference is the number of clusters:

$$\boxed{\text{effective sample size for clustered inference } \approx G.}$$

If G is small, then the cluster-robust variance estimator may be biased downward and hypothesis tests may over-reject.

8.29 Group-Means Estimation

If the regressor of interest varies only at the group level, one can aggregate to group means:

$$\bar{y}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} y_{ig}.$$

The group-level regression is

$$\boxed{\bar{y}_g = \beta_0 + \beta_1 x_g + \bar{e}_g.}$$

If the target is the micro-data coefficient and the only regressor of interest is group-level, this regression can reproduce the relevant coefficient when weighted by group size.

The weighted least squares problem is

$$\boxed{\min_{\beta_0,\beta_1} \sum_g n_g (\bar{y}_g - \beta_0 - \beta_1 x_g)^2.}$$

Inference then uses the number of groups G, not the number of individuals.

8.30 Group-Means Estimation with Micro Covariates

Suppose the micro equation is

$$\boxed{y_{ig} = \beta_0 + \beta_1 x_g + w_{ig}'\delta + e_{ig},}$$

where w_{ig} varies within groups. A two-step procedure is useful.

First estimate

$$\boxed{y_{ig} = \mu_g + w_{ig}'\delta + \eta_{ig}.}$$

The estimated group effects $\widehat{\mu}_g$ are covariate-adjusted group means.

Under the structural model,

$$\mu_g = \beta_0 + \beta_1 x_g + v_g.$$

Second, regress estimated group effects on group-level regressors:

$$\boxed{\widehat{\mu}_g = \beta_0 + \beta_1 x_g + \left[v_g + (\widehat{\mu}_g - \mu_g)\right].}$$

Inference is then based on group-level variation. A conservative approach is to use a t-distribution with approximately $G - k$ degrees of freedom.

8.31 Block Bootstrap for Clustered Data

The block bootstrap preserves within-cluster dependence by resampling whole clusters. For each bootstrap draw:

1. Draw G clusters with replacement from the original set of clusters.
2. Include all observations within each selected cluster.
3. Re-estimate the model.
4. Store the bootstrap estimate.

The block-bootstrap standard error is

$$\boxed{SE_{block} = \left[\frac{1}{B-1}\sum_{b=1}^{B}(\widehat{\beta}^{*(b)} - \bar{\beta}^{*})^2\right]^{1/2}.}$$

Simple individual-level bootstrapping is invalid in clustered data because it breaks the dependence structure within clusters.

8.32 Difference-in-Differences with State-Year Shocks

Consider the DD model

$$\boxed{y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + \varepsilon_{ist}.}$$

A more realistic error structure is

$$\boxed{\varepsilon_{ist} = v_{st} + \eta_{ist},}$$

so that

$$y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + v_{st} + \eta_{ist}.$$

Here v_{st} is a state-year shock common to individuals in state s at time t.

8.33 Why Two-State Two-Period DD Can Be Inconsistent

In the two-state, two-period case, the DD estimator is

$$\widehat{\beta}_{DD} = (\bar{y}_{NJ,Nov} - \bar{y}_{NJ,Feb}) - (\bar{y}_{PA,Nov} - \bar{y}_{PA,Feb}).$$

Using the model

$$y_{st} = \gamma_s + \lambda_t + \beta d_{st} + v_{st},$$

the probability limit is

$$\text{plim}\, \widehat{\beta}_{DD} = \beta + (v_{NJ,Nov} - v_{NJ,Feb}) - (v_{PA,Nov} - v_{PA,Feb}).$$

Thus,

$$\boxed{\text{plim}\, \widehat{\beta}_{DD} \neq \beta}$$

unless

$$(v_{NJ,Nov} - v_{NJ,Feb}) - (v_{PA,Nov} - v_{PA,Feb}) = 0.$$

Increasing the number of individuals within each state-period cell does not eliminate the state-year shocks. The problem is not individual-level sampling error, but a failure of the common-trends assumption due to aggregate shocks.

8.34 Serial Correlation in Difference-in-Differences

With many states and periods, DD can average out state-year shocks. However, state-year shocks are often serially correlated:

$$\text{Cov}(v_{st}, v_{s,t-1}) \neq 0.$$

If shocks are persistent, then residuals remain correlated within a state over time, even after controlling for state fixed effects:

$$v_{st} - \bar{v}_s.$$

Therefore, standard errors should allow arbitrary serial correlation within states:

$$\boxed{\text{cluster standard errors at the state level, not merely at the state-year level.}}$$

This accounts for correlation across time within the same state.

8.35 Clustering One Level Higher

In panel DD settings, the model is often

$$y_{ist} = \gamma_s + \lambda_t + \beta d_{st} + \varepsilon_{ist}.$$

Because residuals may be serially correlated within states, cluster by state:

$$\boxed{\widehat{V}_{state} = (X'X)^{-1} \left(\sum_s X_s' \widehat{e}_s \widehat{e}_s' X_s \right) (X'X)^{-1}.}$$

Here X_s stacks all observations from state s over time, and $\widehat{e}_s$ stacks the corresponding residuals.

This allows arbitrary correlation among residuals within a state over time. But the number of clusters is now the number of states, which may be small.

8.36 Few Clusters Problem

Cluster-robust standard errors can be biased downward when the number of clusters G is small. The issue is analogous to finite-sample bias in HC robust standard errors:

$$\mathbb{E}[\widehat{e}_g \widehat{e}_g'] \neq \mathbb{E}[e_g e_g'].$$

Typically,

$$\mathbb{E}[\widehat{e}_g \widehat{e}_g']$$

is too small, leading to underestimated standard errors.

A practical concern is:

$$\boxed{\text{cluster-robust asymptotics require many clusters, not many observations per cluster.}}$$

8.37 Bias-Reduced Linearization

Bell and McCaffrey's bias-reduced linearization adjusts cluster residuals:

$$\widehat{\Psi}_g = a \widetilde{e}_g \widetilde{e}_g',$$

where

$$\widetilde{e}_g = A_g \widehat{e}_g.$$

The adjustment matrix A_g satisfies

$$\boxed{A_g' A_g = (I - H_g)^{-1},}$$

where

$$H_g = X_g (X'X)^{-1} X_g'.$$

This is a cluster-level analogue of HC2 because it corrects residuals for cluster leverage.

The resulting variance estimator is

$$\boxed{\widehat{V}_{BRL} = (X'X)^{-1} \left(\sum_g X_g' \widehat{\Psi}_g X_g \right) (X'X)^{-1}.}$$

This can improve finite-sample inference when there are few clusters, though it may not be feasible in all designs.

8.38 Using a t-Distribution with Cluster Degrees of Freedom

With few clusters, inference should often use a t-distribution rather than a standard normal approximation. The test statistic is

$$t = \frac{\widehat{\beta}_j - \beta_{j,0}}{SE_{cluster}(\widehat{\beta}_j)}.$$

Instead of comparing to $N(0,1)$, compare to

$$\boxed{t_{G-k},}$$

where G is the number of clusters and k is the number of estimated parameters relevant to the group-level regression.

This makes confidence intervals wider:

$$\boxed{\widehat{\beta}_j \pm t_{1-\alpha/2,G-k} SE_{cluster}(\widehat{\beta}_j).}$$

8.39 Parametric Correction for Serial Correlation

Suppose the state-level shock follows an AR(1) process:

$$v_{st} = \rho_v v_{s,t-1} + u_{st}.$$

Then

$$\text{Cov}(v_{st}, v_{s,t-\ell}) = \rho_v^{\ell} \, \text{Var}(v_{st}).$$

A parametric correction estimates ρ_v and adjusts the variance formula accordingly.

This can work well when the AR(1) structure is approximately correct. But if the serial correlation structure is misspecified, the correction can be misleading. Therefore,

$$\boxed{\text{parametric serial-correlation corrections trade robustness for efficiency.}}$$

8.40 Appendix: Matrix Setup for the Moulton Factor

Let

$$y_g = \begin{bmatrix} y_{1g} \\ y_{2g} \\ \vdots \\ y_{ng} \end{bmatrix}, \qquad e_g = \begin{bmatrix} e_{1g} \\ e_{2g} \\ \vdots \\ e_{ng} \end{bmatrix}.$$

Let ι_g be an $n_g \times 1$ vector of ones. If the group-level regressor is x_g, then the stacked regressor vector for group g is

$$\iota_g x_g.$$

The full error vector is

$$e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_G \end{bmatrix}.$$

The error covariance matrix is block diagonal:

$$\mathbb{E}[ee'] = \Psi = \begin{bmatrix} \Psi_1 & 0 & \cdots & 0 \\ 0 & \Psi_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Psi_G \end{bmatrix}.$$

Within group g,

$$\boxed{\Psi_g = \sigma_e^2 \left[(1-\rho) I_{n_g} + \rho \iota_g \iota_g' \right].}$$

8.41 Appendix: Deriving $X'\Psi X$ for Group-Level Regressors

For group g, the group-level regressor matrix is

$$X_g = \iota_g x_g'.$$

Then

$$X_g' \Psi_g X_g = x_g \iota_g' \Psi_g \iota_g x_g'.$$

Now

$$\Psi_g = \sigma_e^2[(1-\rho)I_{n_g} + \rho\iota_g\iota_g'].$$

Therefore,

$$\iota_g'\Psi_g\iota_g = \sigma_e^2\left[(1-\rho)\iota_g' I_{n_g}\iota_g + \rho\iota_g'\iota_g\iota_g'\iota_g\right].$$

Since

$$\iota_g'\iota_g = n_g,$$

we get

$$\iota_g'\Psi_g\iota_g = \sigma_e^2\left[(1-\rho)n_g + \rho n_g^2\right].$$

Factor out n_g:

$$= \sigma_e^2 n_g[1-\rho+\rho n_g] = \sigma_e^2 n_g[1+(n_g-1)\rho].$$

Define

$$\tau_g = 1 + (n_g-1)\rho.$$

Then

$$\boxed{X_g'\Psi_g X_g = \sigma_e^2 n_g\tau_g x_g x_g'.}$$

8.42 Appendix: Deriving the Simple Moulton Factor

The true variance of $\widehat{\beta}$ is

$$V(\widehat{\beta}) = (X'X)^{-1}X'\Psi X(X'X)^{-1}.$$

Because

$$X'X = \sum_g n_g x_g x_g',$$

and

$$X'\Psi X = \sigma_e^2\sum_g n_g\tau_g x_g x_g',$$

we have

$$V(\widehat{\beta}) = \sigma_e^2\left(\sum_g n_g x_g x_g'\right)^{-1}\left(\sum_g n_g\tau_g x_g x_g'\right)\left(\sum_g n_g x_g x_g'\right)^{-1}.$$

The conventional variance that ignores clustering is

$$V_c(\widehat{\beta}) = \sigma_e^2\left(\sum_g n_g x_g x_g'\right)^{-1}.$$

If all groups have equal size $n_g = n$, then

$$\tau_g = \tau = 1 + (n-1)\rho.$$

Therefore,

$$V(\widehat{\beta}) = \sigma_e^2 \left(\sum_g n x_g x_g'\right)^{-1} \left[\tau \sum_g n x_g x_g'\right] \left(\sum_g n x_g x_g'\right)^{-1}.$$

Cancel the middle matrix:

$$V(\widehat{\beta}) = \tau \sigma_e^2 \left(\sum_g n x_g x_g'\right)^{-1}.$$

Since

$$V_c(\widehat{\beta}) = \sigma_e^2 \left(\sum_g n x_g x_g'\right)^{-1},$$

we get

$$\boxed{V(\widehat{\beta}) = \tau V_c(\widehat{\beta}).}$$

Substitute

$$\tau = 1 + (n-1)\rho.$$

Hence,

$$\boxed{\frac{V(\widehat{\beta})}{V_c(\widehat{\beta})} = 1 + (n-1)\rho.}$$

The standard-error inflation factor is

$$\boxed{\sqrt{1 + (n-1)\rho}.}$$

8.43 Summary of Chapter 8

The main message of this chapter is that standard errors are part of the research design. Robust standard errors solve heteroskedasticity asymptotically, but they can be biased in small samples. Clustered data require cluster-level inference, and the relevant asymptotics depend on the number of clusters, not the number of individuals. Difference-in-differences designs add serial correlation concerns because group-level shocks are often persistent over time.

The practical implications are:

- Use HC corrections carefully, since HC standard errors can be too small in finite samples.
- If treatment varies at the group level, account for within-group correlation.
- In DD panels, cluster at the level where treatment varies over time, often the state or group level.
- With few clusters, use conservative methods such as t_{G-k} critical values, group means, BRL, or block bootstrap.